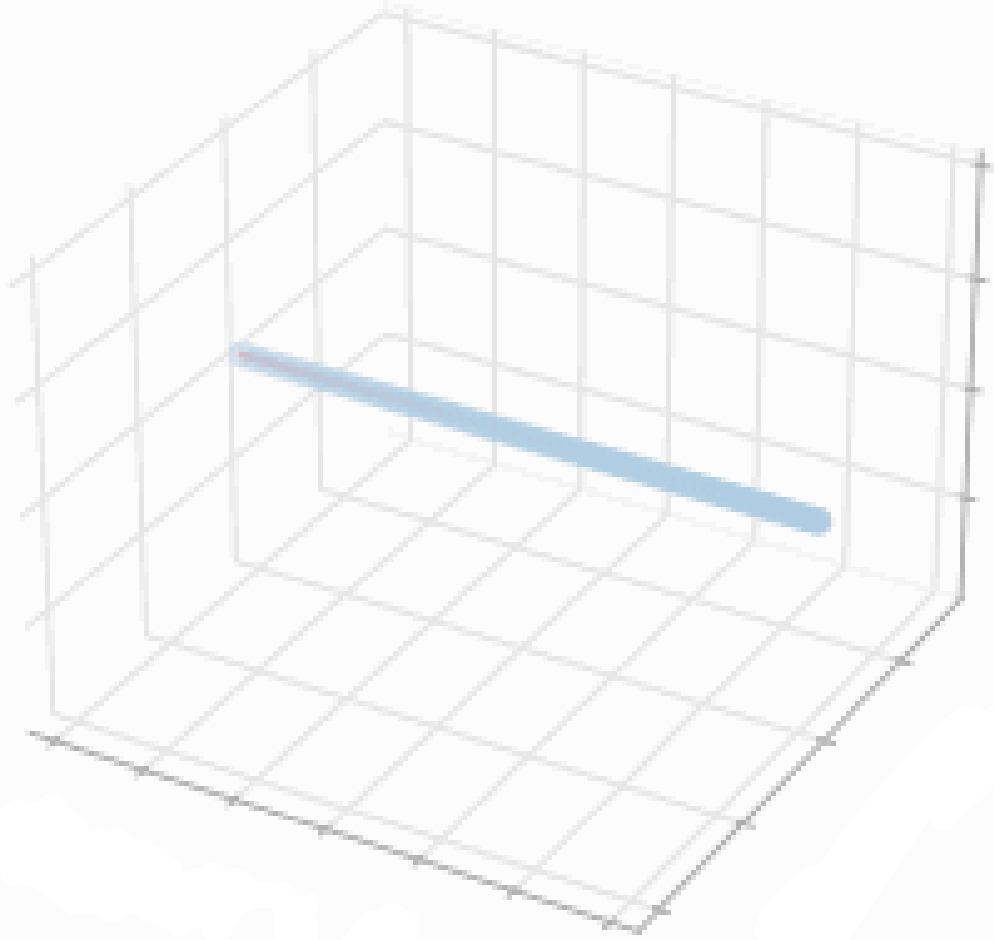


SIMULACIÓN MOLECULAR

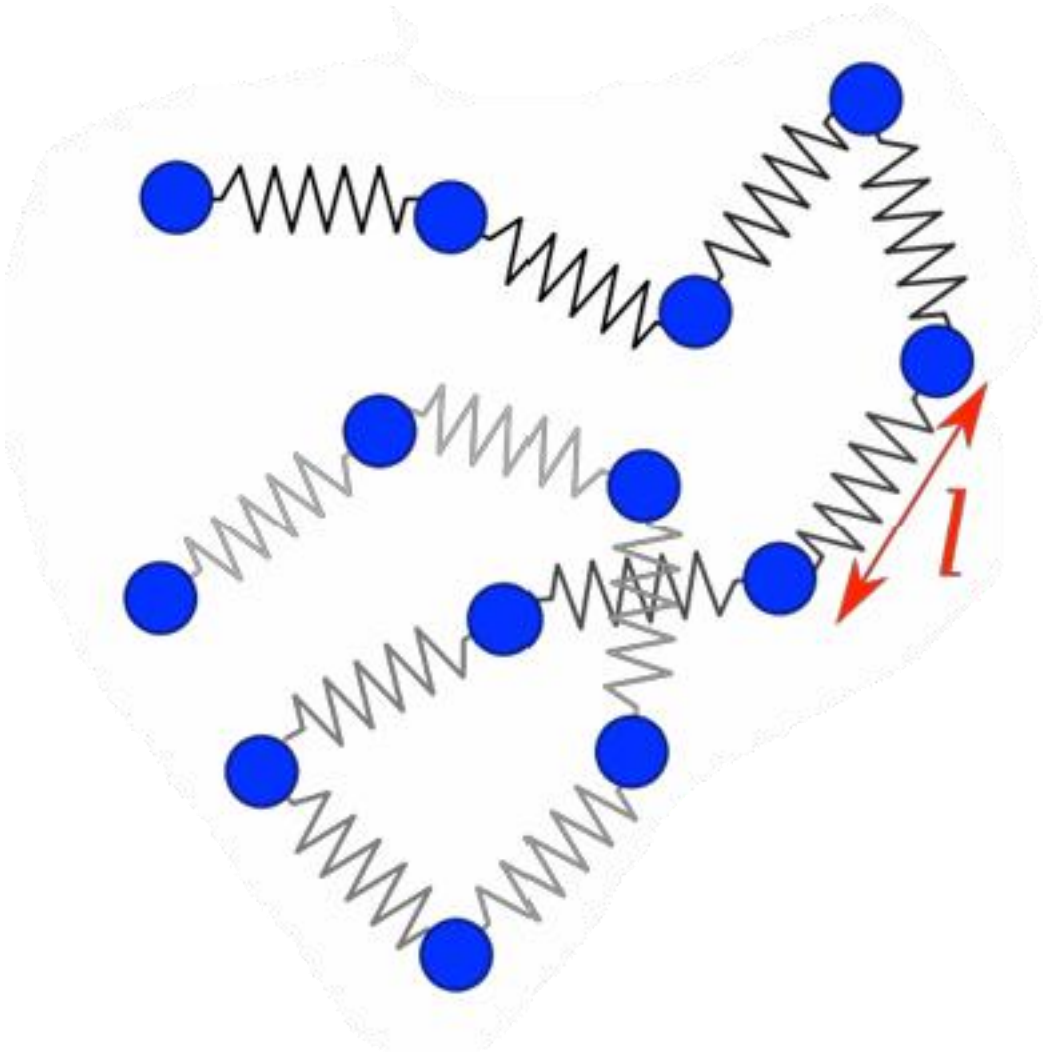
Manuel Morato, Paula Otal y Ricardo Sáez

INTRODUCCIÓN

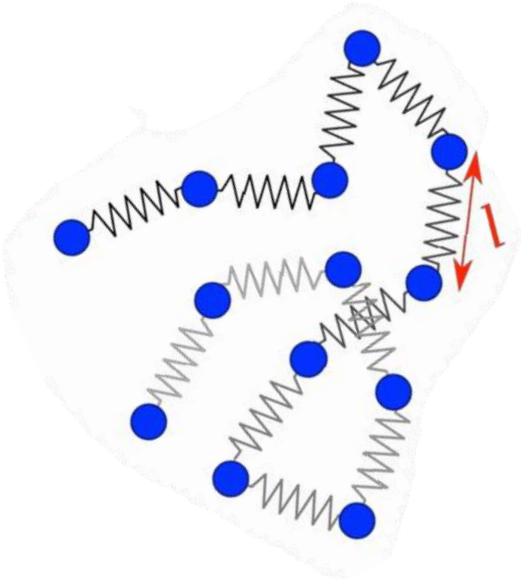
Marco teórico

Implementación

Sistemas físicos



INTRODUCCIÓN

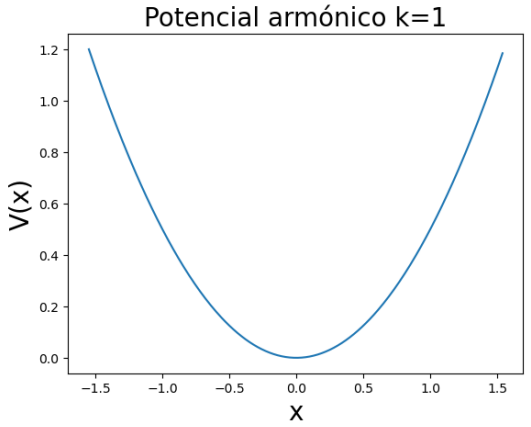


Marco teórico

Ecuación de Langevin:

$$m\frac{d^2\vec{r}}{dt^2} = -\vec{\nabla}V(\vec{r}) - \eta\frac{d\vec{r}}{dt} + \xi(t)$$

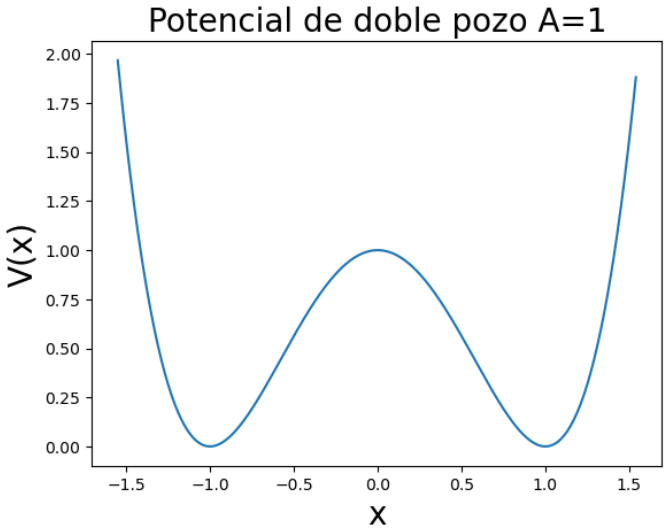
INTRODUCCIÓN



Marco teórico

Ecuación de Langevin:

$$m\frac{d^2\vec{r}}{dt^2} = \underbrace{-\vec{\nabla}V(\vec{r})}_{\text{Término de potencial}} - \eta\frac{d\vec{r}}{dt} + \xi(t)$$



$$V(\overrightarrow{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2}k_e(|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

INTRODUCCIÓN

Marco teórico

Ecuación de Langevin:

$$m\frac{d^2\vec{r}}{dt^2} = -\vec{\nabla}V(\vec{r}) - \eta\frac{d\vec{r}}{dt} + \xi(t)$$

Rozamiento viscoso

INTRODUCCIÓN

Marco teórico

$$\langle \xi(t) \rangle = 0$$

$$\langle \xi(t)\xi(t') \rangle = 2\eta k_B T \delta(t - t')$$

Ecuación de Langevin:

$$m\frac{d^2\vec{r}}{dt^2} = -\vec{\nabla}V(\vec{r}) - \eta\frac{d\vec{r}}{dt} + \xi(t)$$

Término estocástico

INTRODUCCIÓN

Implementación

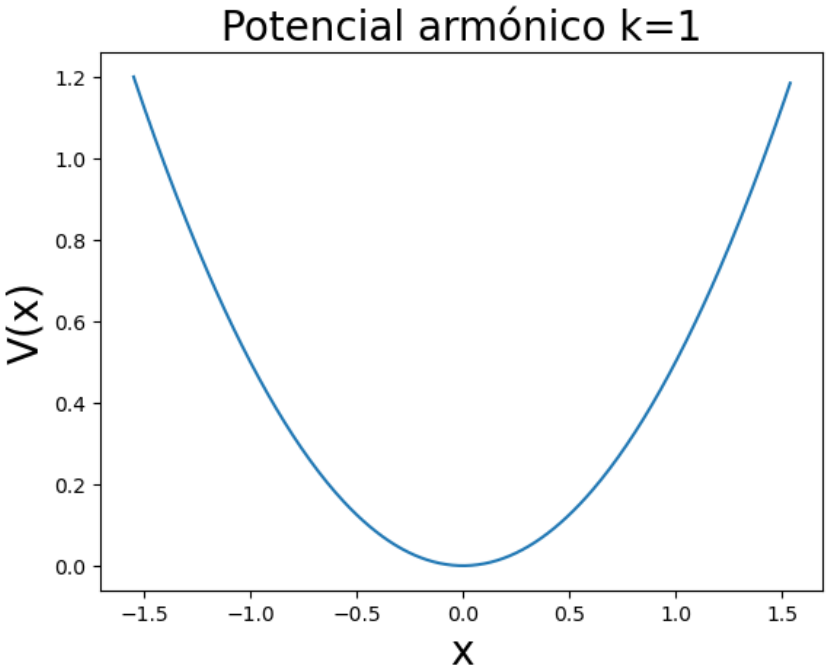
Métodos de integración:

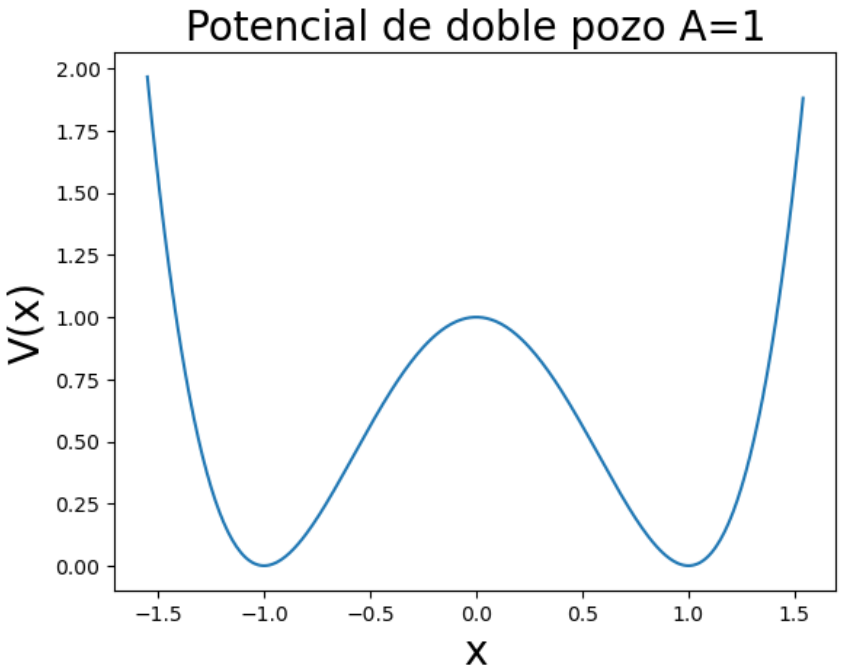
$h \equiv Paso\ de\ integración$

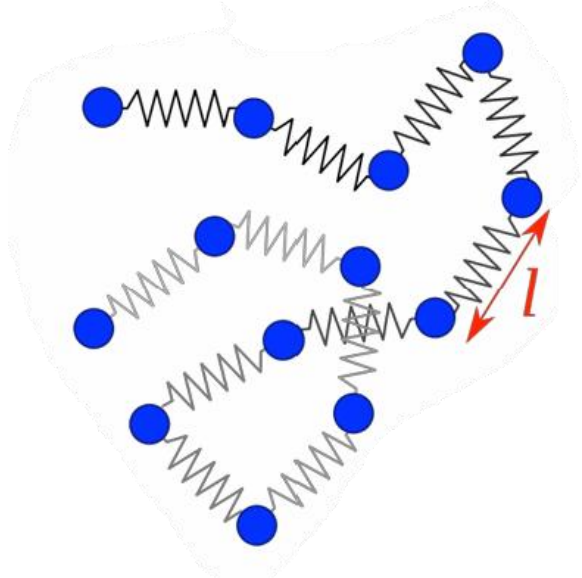
Euler	Runge-Kutta orden 2	Verlet explícito
- Primer orden - Simple y rápido	- Segundo orden	- Segundo orden
$error\ en\ x \sim \mathcal{O}(h^2)$	$error\ en\ x \sim \mathcal{O}(h^3)$	$error\ en\ x \sim \mathcal{O}(h^3)$

INTRODUCCIÓN

Sistemas físicos



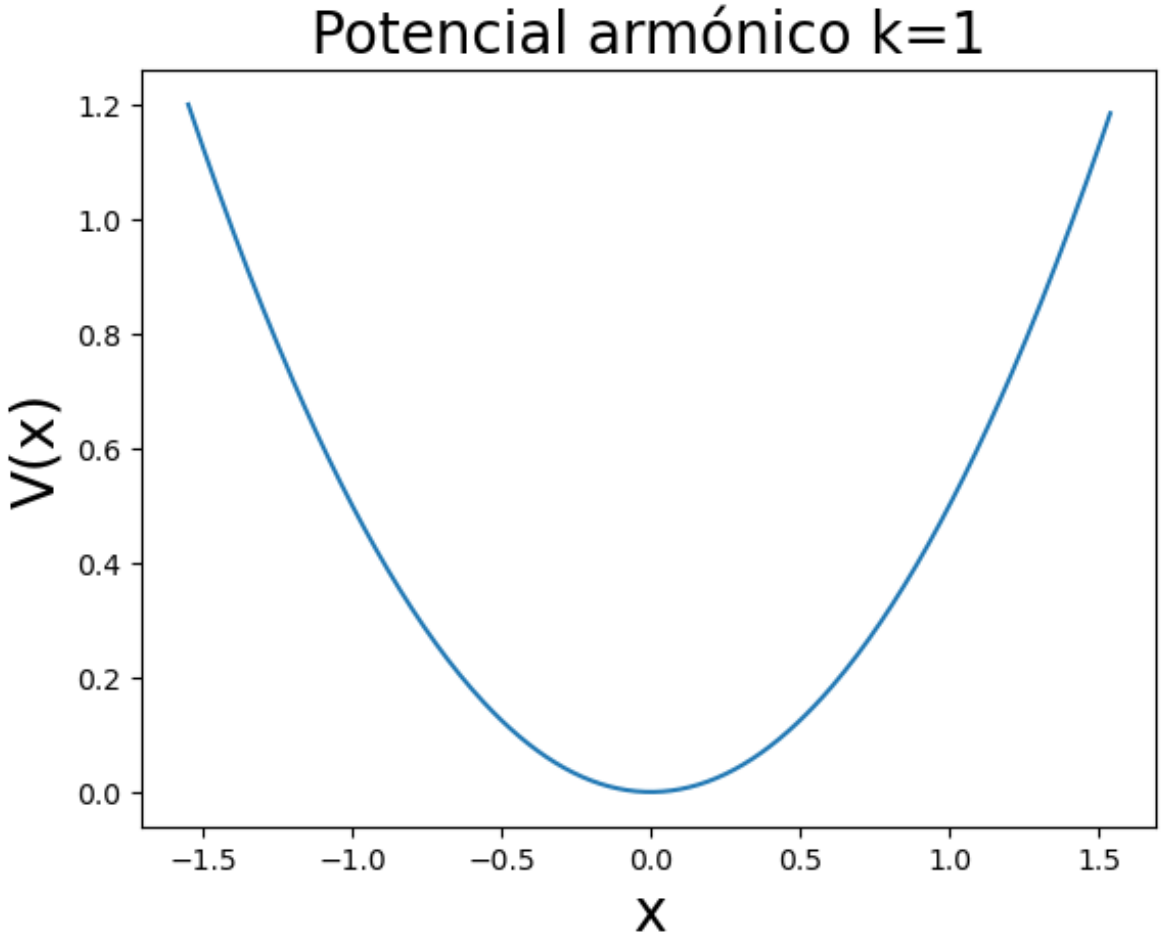




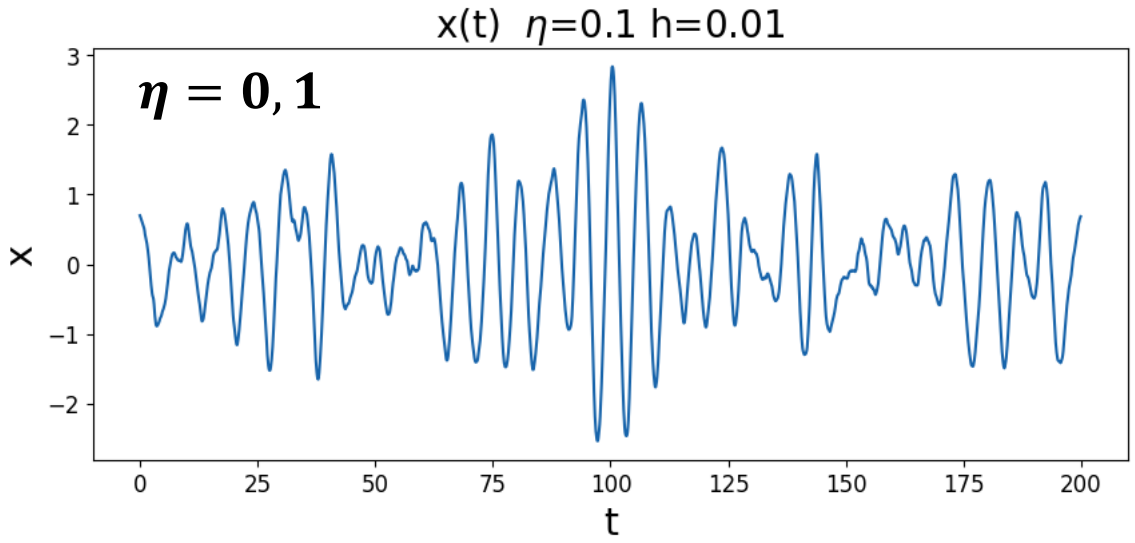
$$V(\vec{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2} k_e (|\vec{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

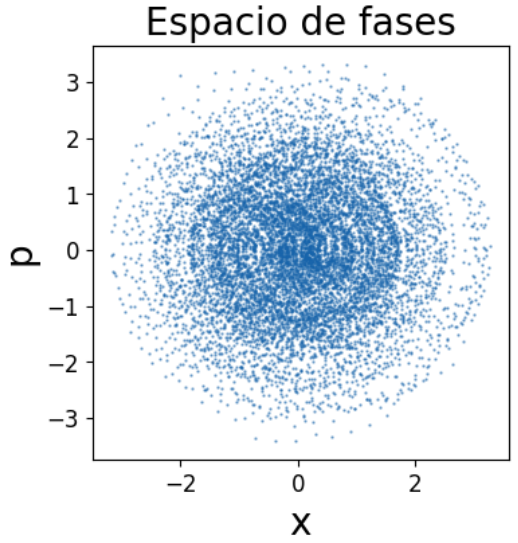
OSCILADOR ARMÓNICO

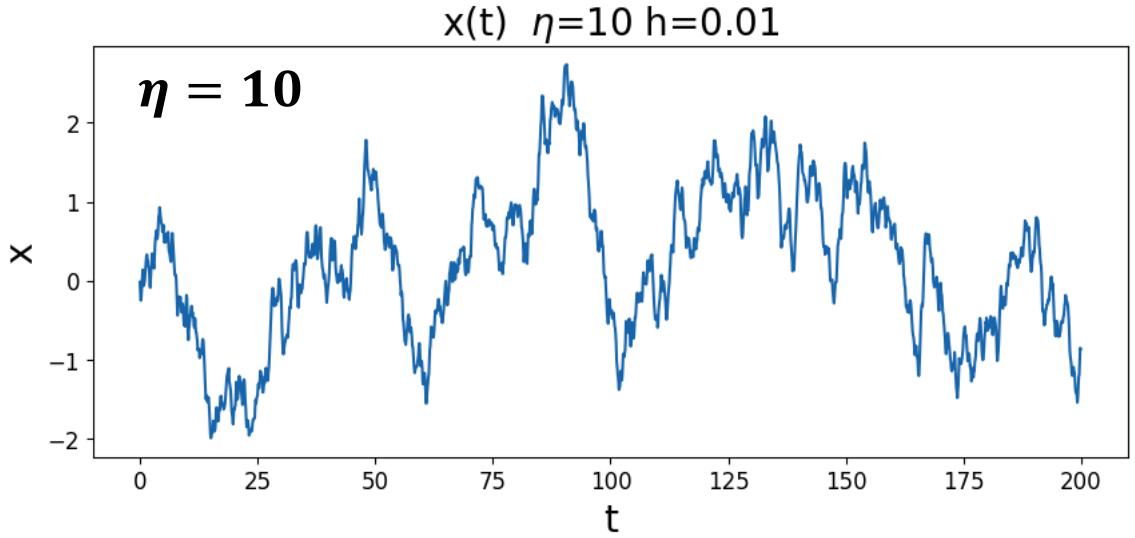
$$V(x) = \frac{1}{2}kx^2$$

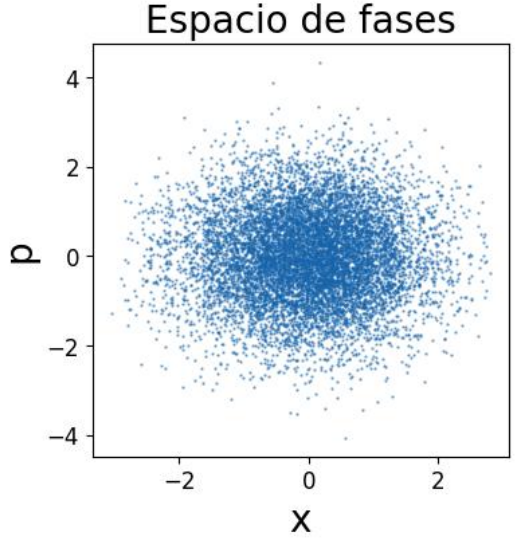


OSCILADOR ARMÓNICO: TRAYECTORIAS









Algoritmo: Verlet

OSCILADOR ARMÓNICO: EQUIPARTICIÓN

- Términos cuadráticos $\Longrightarrow$ Teorema de la equipartición de la energía

$$V = \frac{1}{2}kx^2 \Rightarrow \langle V \rangle = \frac{1}{2}k_B T = 0{,}5$$

$$T = \frac{p^2}{2m} \Rightarrow \langle T \rangle = \frac{1}{2}k_B T = 0{,}5$$

OSCILADOR ARMÓNICO: EQUIPARTICIÓN

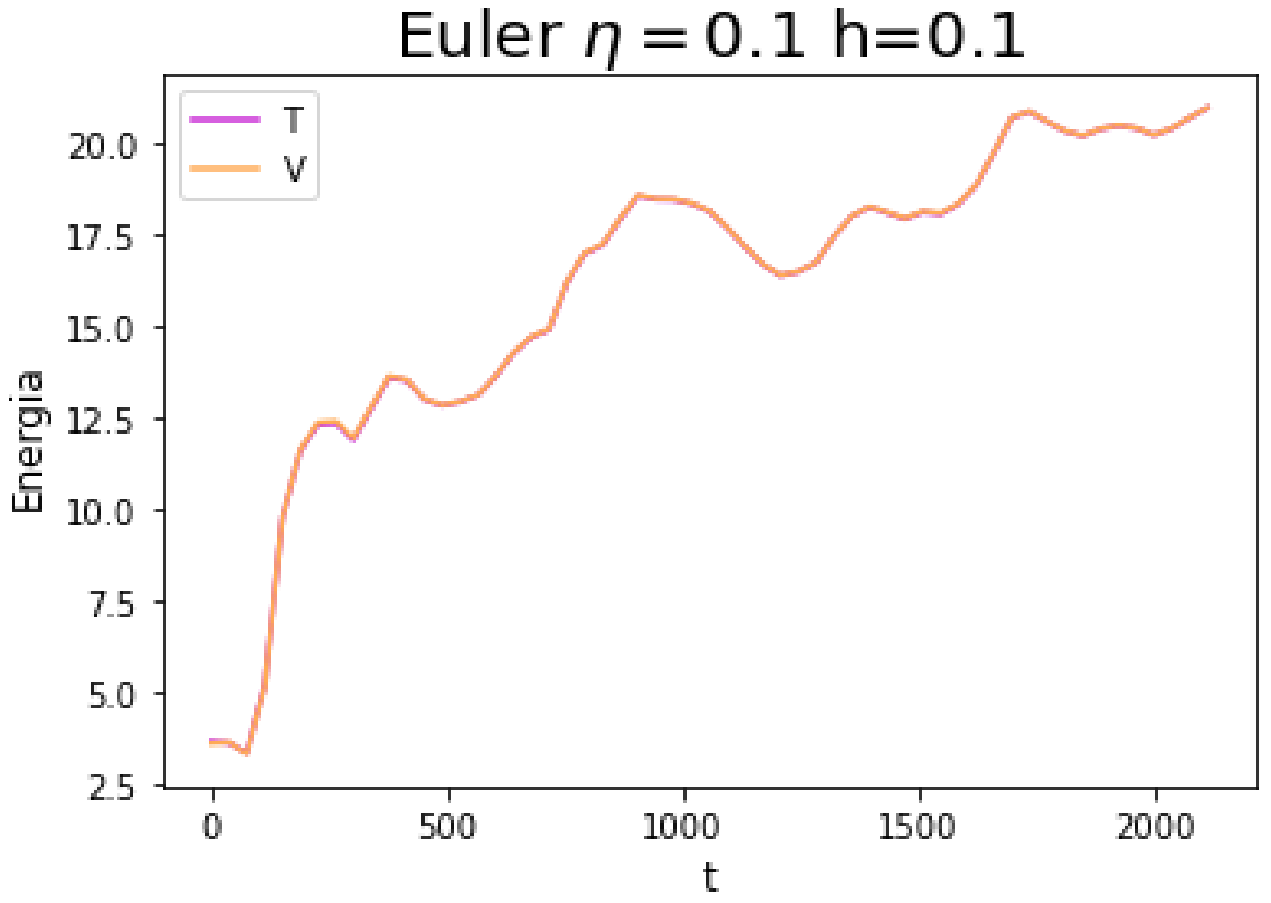
EULER													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	inf	245,88	0,56	0,48	0,57	0,52	0,53	0,52	0,50	0,50	0,51	0,50	0,50
<T>	inf	245,96	0,59	1,00	0,57	0,58	0,50	0,52	0,50	0,46	0,51	0,51	0,48
<E>	#¡VALOR!	491,84	1,15	1,48	1,15	1,10	1,03	1,03	1,00	0,96	1,02	1,01	0,98
RK													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,51	0,51	0,50	0,51	0,51	0,48	0,51	0,50	0,52	0,50	0,51	0,50
<T>	0,25	0,51	0,50	1,31	0,51	0,51	0,50	0,51	0,50	0,50	0,50	0,50	0,50
<E>	0,50	1,02	1,00	1,82	1,02	1,02	0,98	1,02	1,00	1,02	1,00	1,01	1,00
VERLET													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,50	0,50	0,47	0,52	0,51	0,50	0,51	0,50	0,46	0,51	0,51	0,48
<T>	0,25	0,50	0,51	0,50	0,51	0,51	0,50	0,51	0,50	0,50	0,51	0,50	0,50
<E>	0,50	1,01	1,01	0,97	1,03	1,02	1,00	1,02	1,00	0,96	1,02	1,01	0,98

OSCILADOR ARMÓNICO: EQUIPARTICIÓN

EULER													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	inf	245,88	0,56	0,48	0,57	0,52	0,53	0,52	0,50	0,50	0,51	0,50	0,50
<T>	inf	245,96	0,59	1,00	0,57	0,58	0,50	0,52	0,50	0,46	0,51	0,51	0,48
<E>	#¡VALOR!	491,84	1,15	1,48	1,15	1,10	1,03	1,03	1,00	0,96	1,02	1,01	0,98
RK													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,51	0,51	0,50	0,51	0,51	0,48	0,51	0,50	0,52	0,50	0,51	0,50
<T>	0,25	0,51	0,50	1,31	0,51	0,51	0,50	0,51	0,50	0,50	0,50	0,50	0,50
<E>	0,50	1,02	1,00	1,82	1,02	1,02	0,98	1,02	1,00	1,02	1,00	1,01	1,00
VERLET													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,50	0,50	0,47	0,52	0,51	0,50	0,51	0,50	0,46	0,51	0,51	0,48
<T>	0,25	0,50	0,51	0,50	0,51	0,51	0,50	0,51	0,50	0,50	0,51	0,50	0,50
<E>	0,50	1,01	1,01	0,97	1,03	1,02	1,00	1,02	1,00	0,96	1,02	1,01	0,98

OSCILADOR ARMÓNICO: EQUIPARTICIÓN

	h=0.1			
	η=0	η=0.1	η=1	η=10
<V>	inf	245,88	0,56	0,48
<T>	inf	245,96	0,59	1,00
<E>	#¡VALOR!	491,84	1,15	1,48

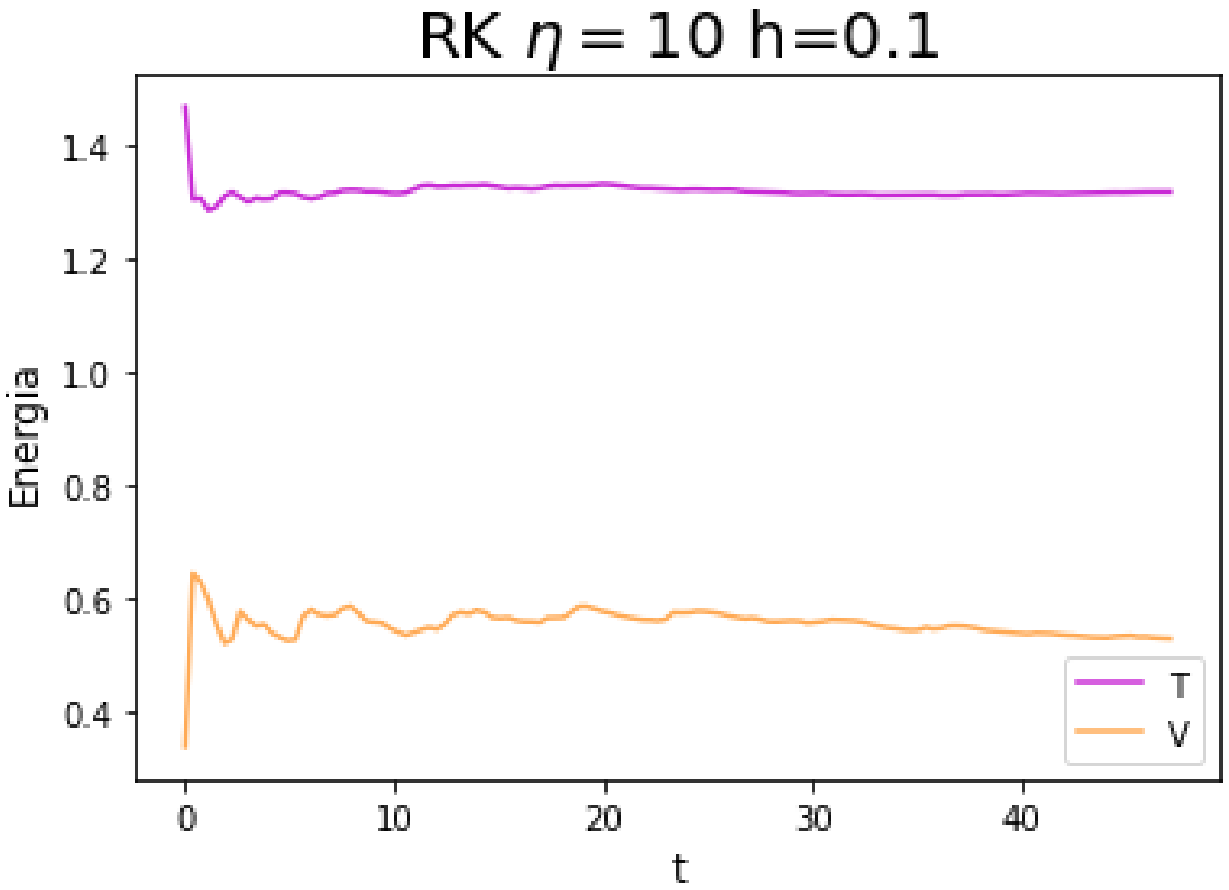


OSCILADOR ARMÓNICO: EQUIPARTICIÓN

EULER													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	inf	245,88	0,56	0,48	0,57	0,52	0,53	0,52	0,50	0,50	0,51	0,50	0,50
<T>	inf	245,96	0,59	1,00	0,57	0,58	0,50	0,52	0,50	0,46	0,51	0,51	0,48
<E>	#¡VALOR!	491,84	1,15	1,48	1,15	1,10	1,03	1,03	1,00	0,96	1,02	1,01	0,98
RK													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,51	0,51	0,50	0,51	0,51	0,48	0,51	0,50	0,52	0,50	0,51	0,50
<T>	0,25	0,51	0,50	1,31	0,51	0,51	0,50	0,51	0,50	0,50	0,50	0,50	0,50
<E>	0,50	1,02	1,00	1,82	1,02	1,02	0,98	1,02	1,00	1,02	1,00	1,01	1,00
VERLET													
	h=0.1				h=0.01			h=0.001			h=0.0001		
	η=0	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10	η=0.1	η=1	η=10
<V>	0,25	0,50	0,50	0,47	0,52	0,51	0,50	0,51	0,50	0,46	0,51	0,51	0,48
<T>	0,25	0,50	0,51	0,50	0,51	0,51	0,50	0,51	0,50	0,50	0,51	0,50	0,50
<E>	0,50	1,01	1,01	0,97	1,03	1,02	1,00	1,02	1,00	0,96	1,02	1,01	0,98

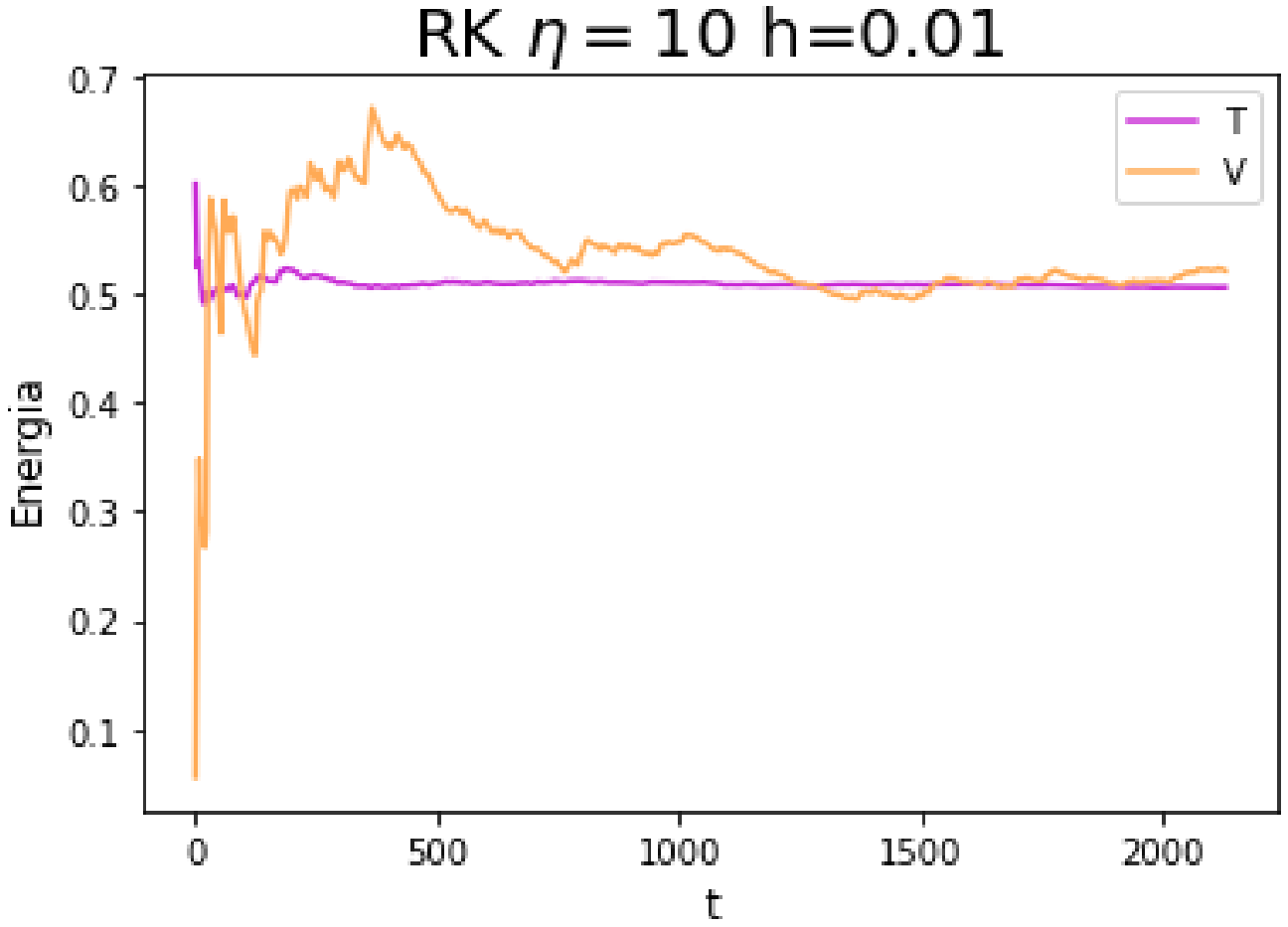
OSCILADOR ARMÓNICO: EQUIPARTICIÓN

h=0.1			
η=0	η=0.1	η=1	η=10
0,25	0,51	0,51	0,50
0,25	0,51	0,50	1,31
0,50	1,02	1,00	1,82



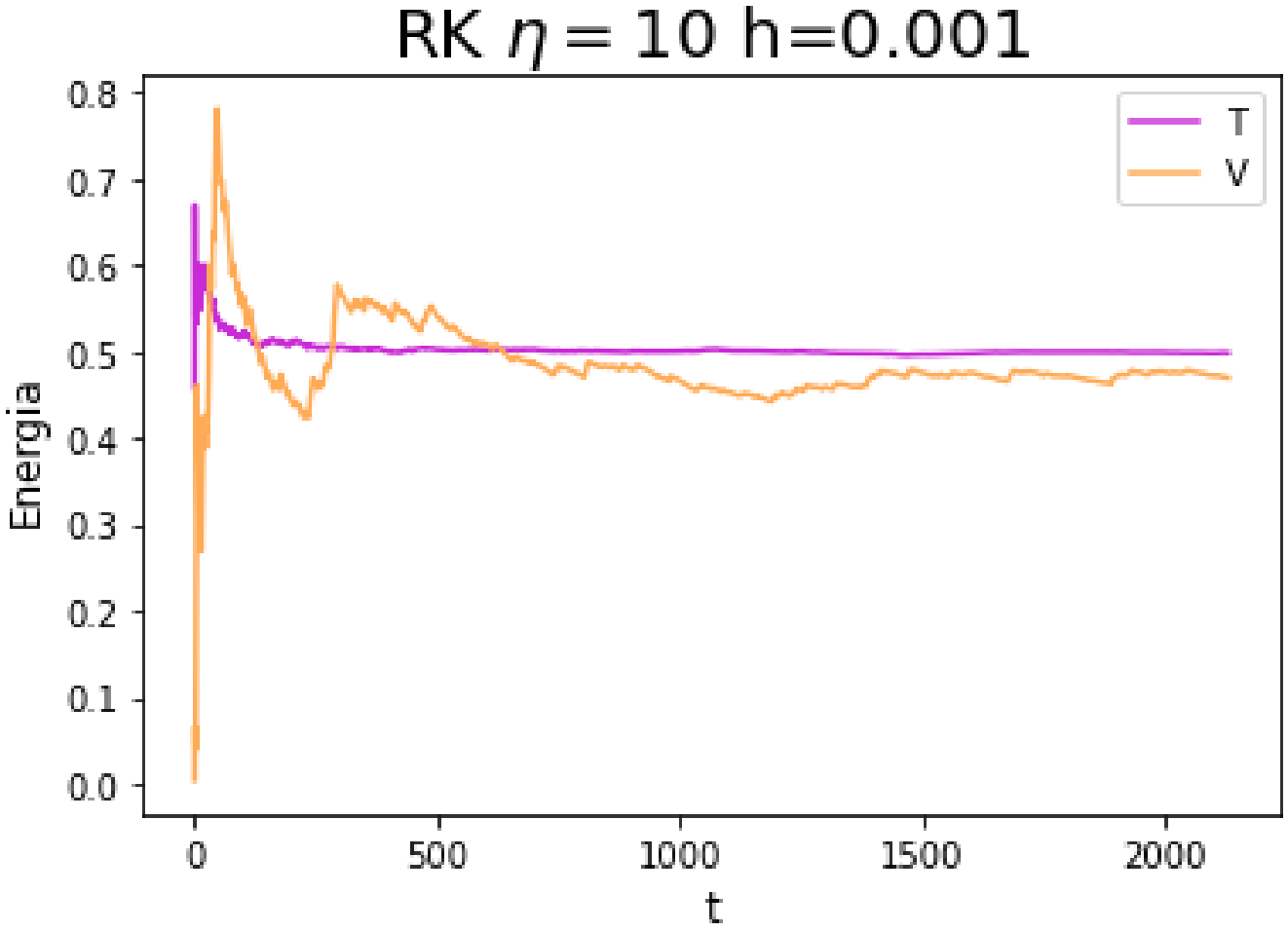
OSCILADOR ARMÓNICO: EQUIPARTICIÓN

h=0.01		
η=0.1	η=1	η=10
0,51	0,51	0,48
0,51	0,51	0,50
1,02	1,02	0,98



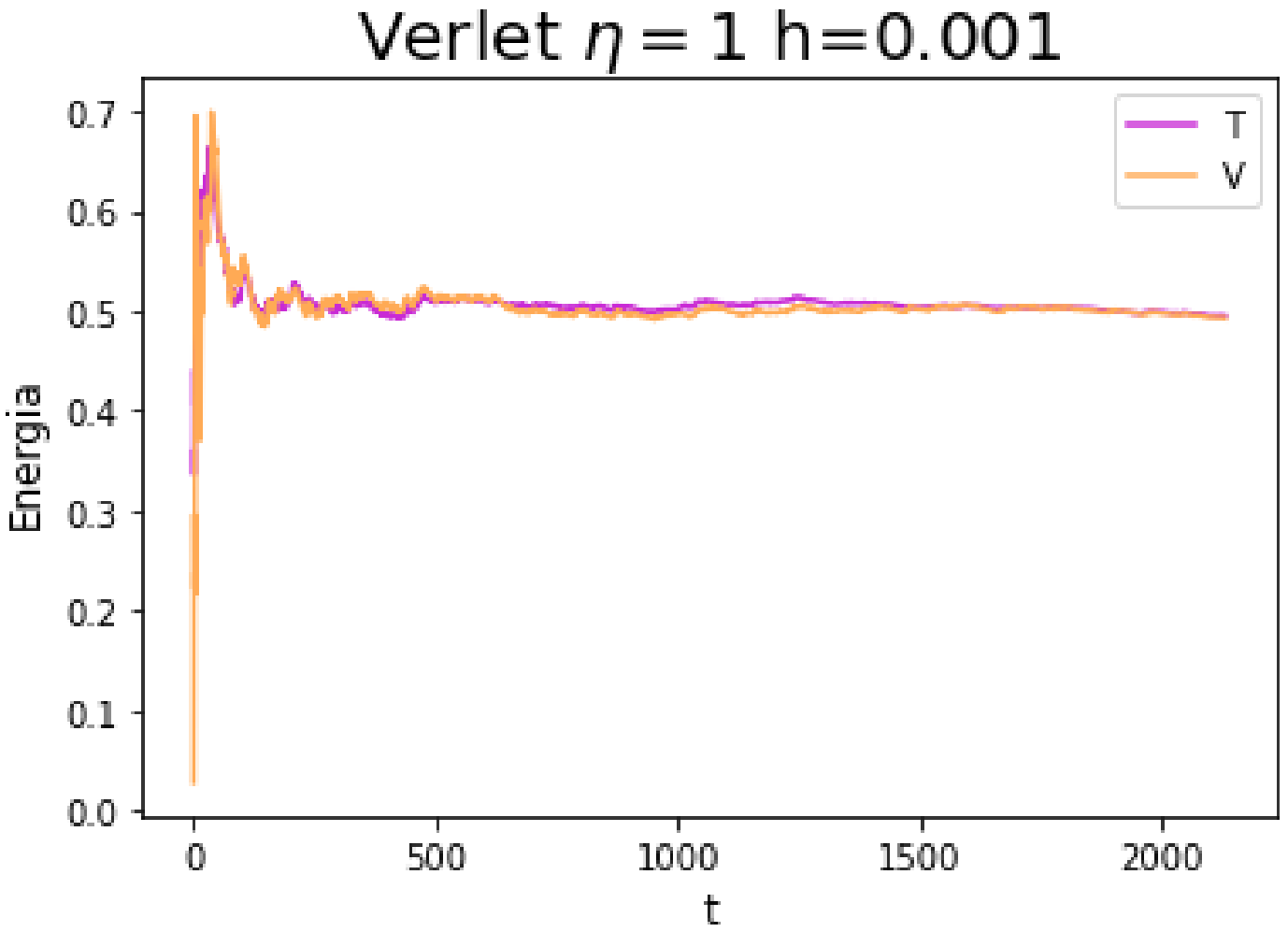
OSCILADOR ARMÓNICO: EQUIPARTICIÓN

h=0.001		
η=0.1	η=1	η=10
0,51	0,50	0,52
0,51	0,50	0,50
1,02	1,00	1,02



OSCILADOR ARMÓNICO: EQUIPARTICIÓN

h=0.001		
η=0.1	η=1	η=10
0,51	0,50	0,46
0,51	0,50	0,50
1,02	1,00	0,96



OSCILADOR ARMÓNICO: DISTRIBUCIONES

Formalismo canónico

$$P(x,p) = \frac{e^{-\frac{E(x,p)}{k_B T}}}{Z}, \qquad Z = \int_{x=-\infty}^{x=\infty} \int_{p=-\infty}^{p=\infty} e^{-\frac{E(x,p)}{k_B T}}$$

OSCILADOR ARMÓNICO: DISTRIBUCIONES

Formalismo canónico

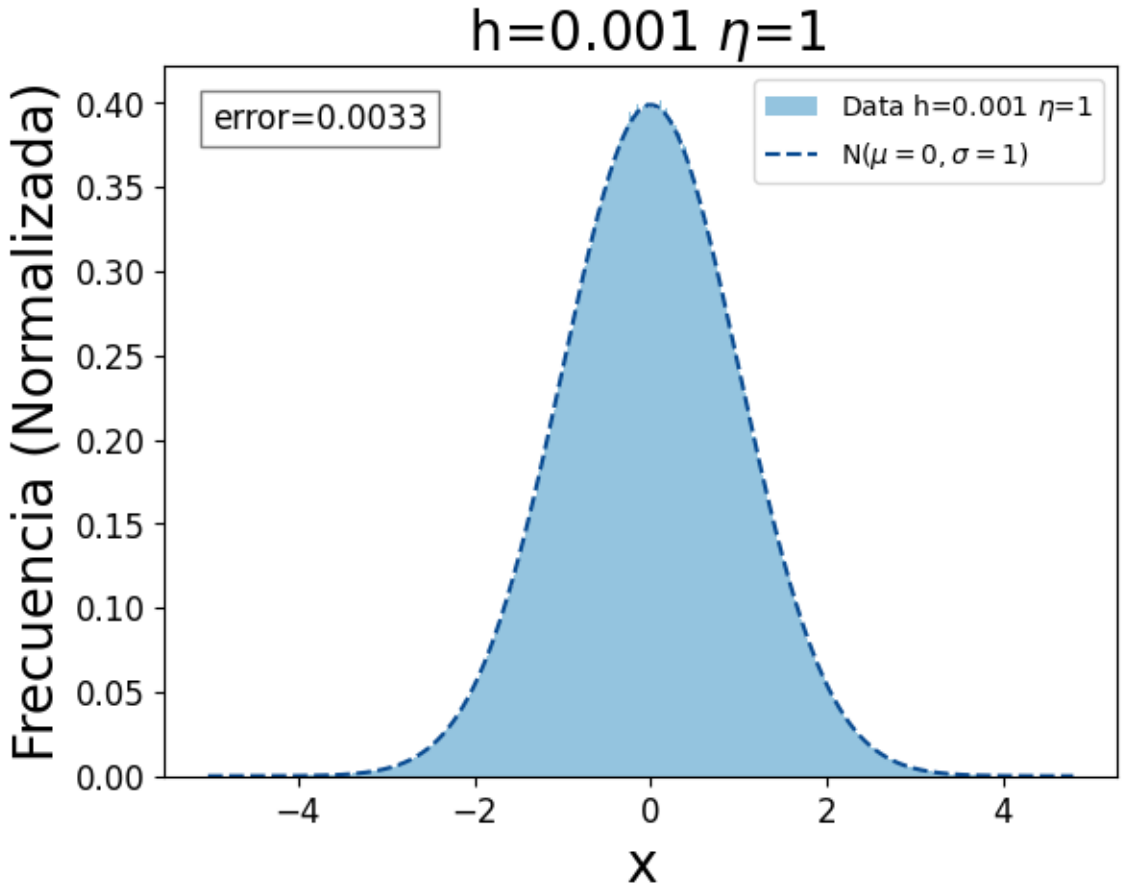
$$Z_p = \int_{p=-\infty}^{p=\infty} e^{-\frac{T(p)}{k_B T}} \qquad Z_x = \int_{x=-\infty}^{x=\infty} e^{-\frac{V(x)}{k_B T}}$$

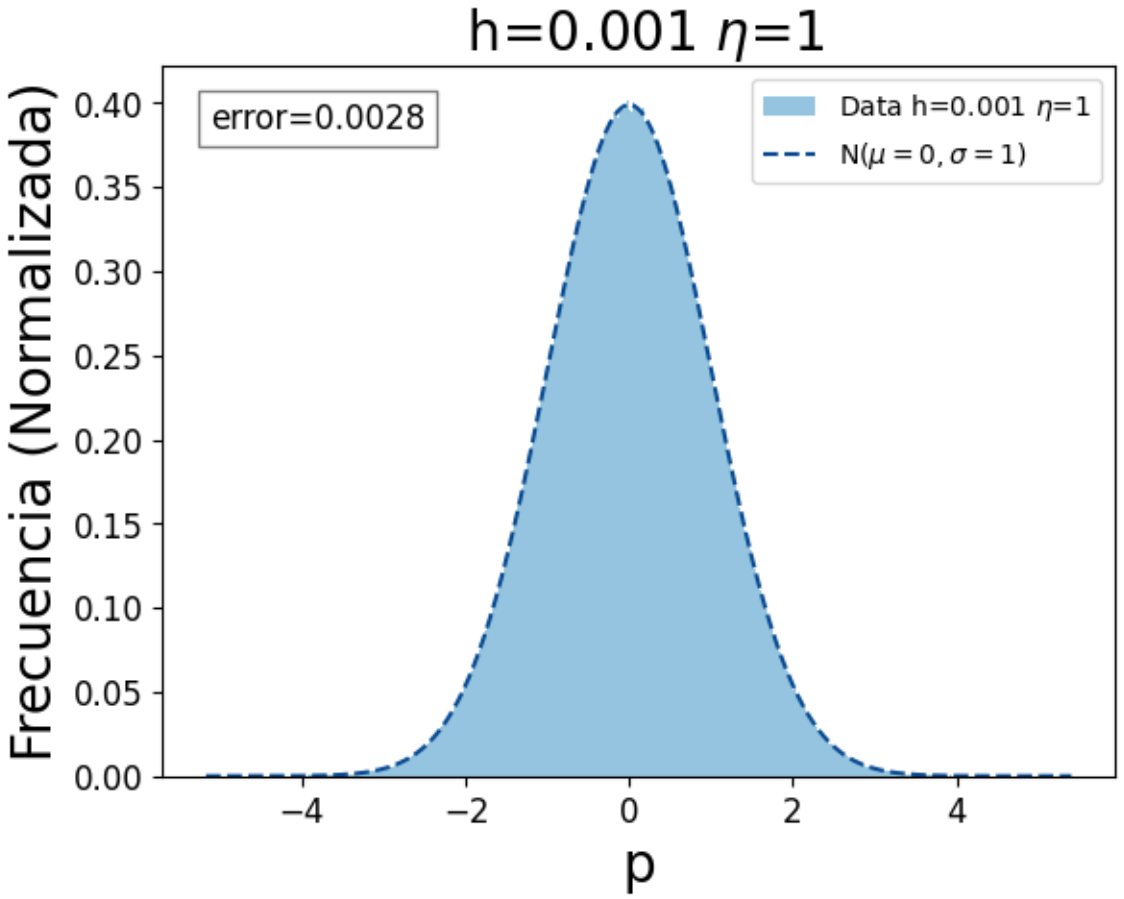
$\boxed{k = 1, m = 1, k_B T = 1}$

$$\begin{cases} V(x) = \frac{1}{2} k x^2 = \frac{x^2}{2} \\ T(p) = \frac{p^2}{2m} = \frac{p^2}{2} \end{cases} \Longrightarrow E(x,p) = \frac{x^2}{2} + \frac{p^2}{2} \Longrightarrow P(x,p) = \frac{1}{Z} e^{-\frac{x^2}{2}} e^{-\frac{p^2}{2}} \Longrightarrow \begin{cases} P(x) = \frac{1}{Z_x} e^{-\frac{x^2}{2}} \\ P(p) = \frac{1}{Z_p} e^{-\frac{p^2}{2}} \end{cases} \Longrightarrow$$

$$\boldsymbol{P(x), P(p) \propto N(\mu = 0, \sigma = 1)}$$

OSCILADOR ARMÓNICO: DISTRIBUCIONES



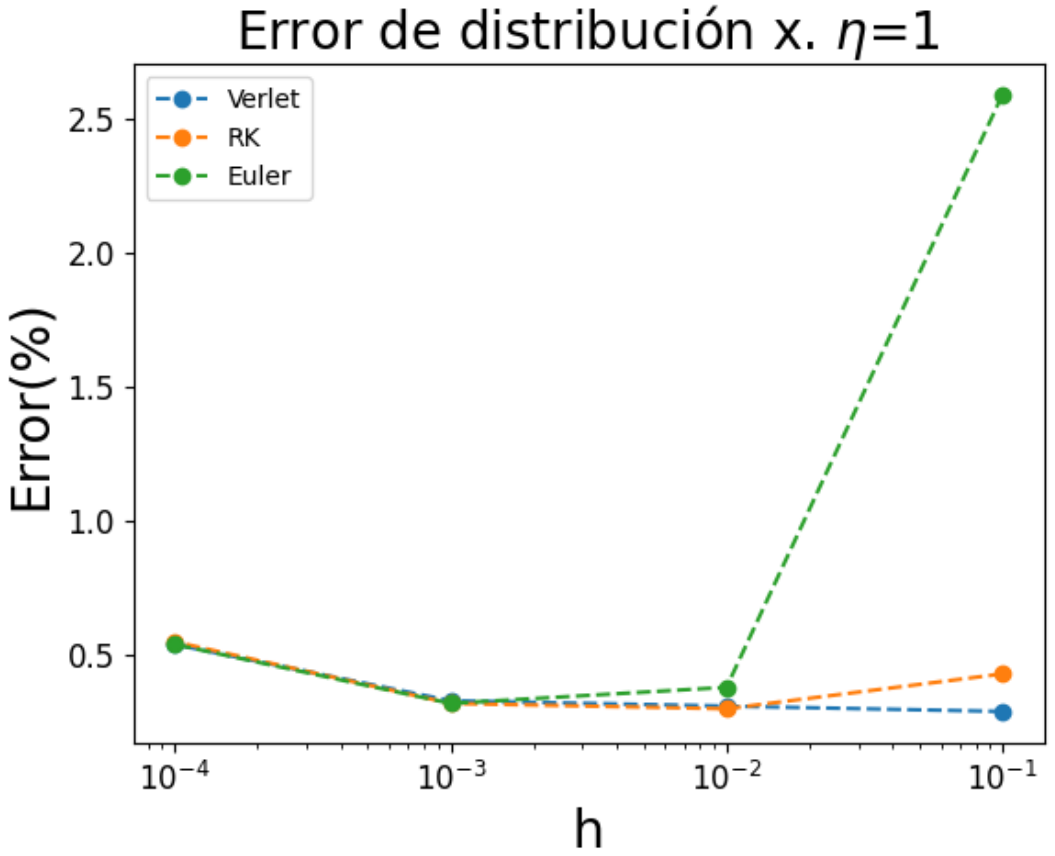


Algoritmo: Verlet, $t = 10^6$

OSCILADOR ARMÓNICO: DISTRIBUCIONES

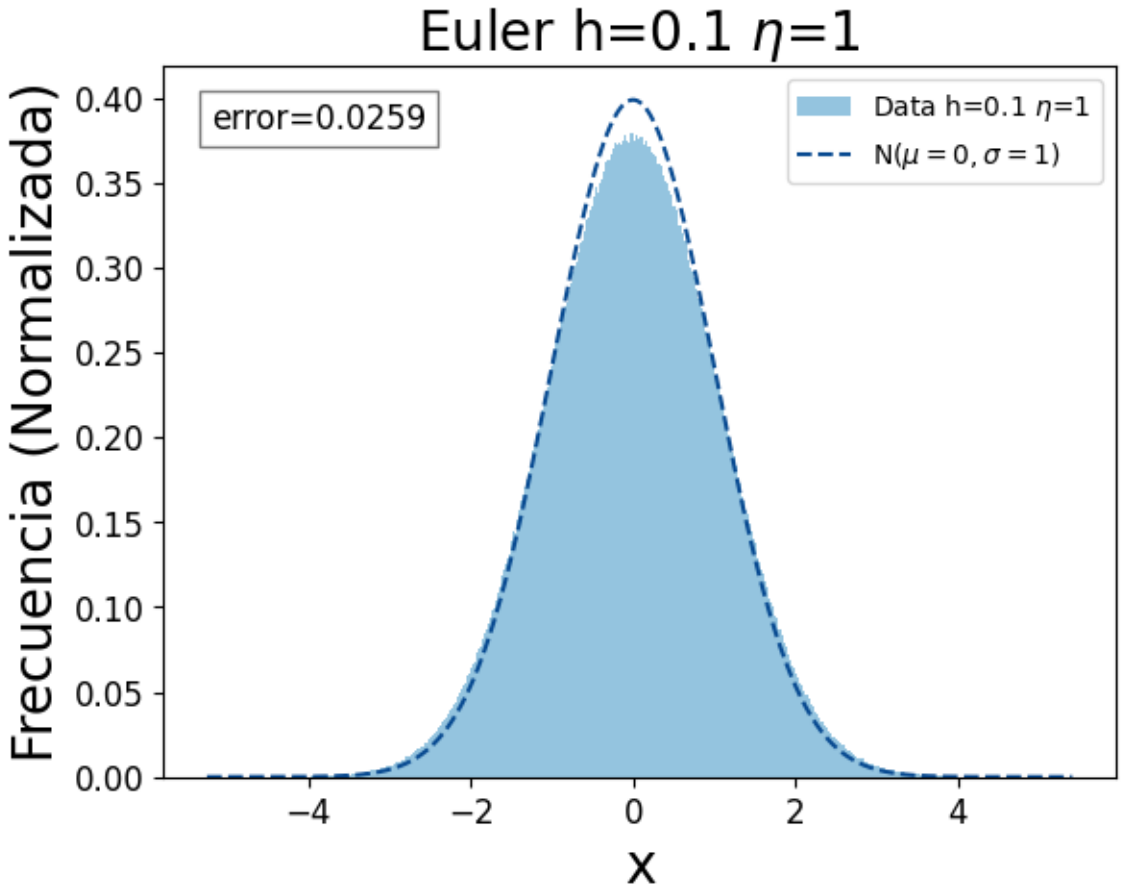
Efecto de h

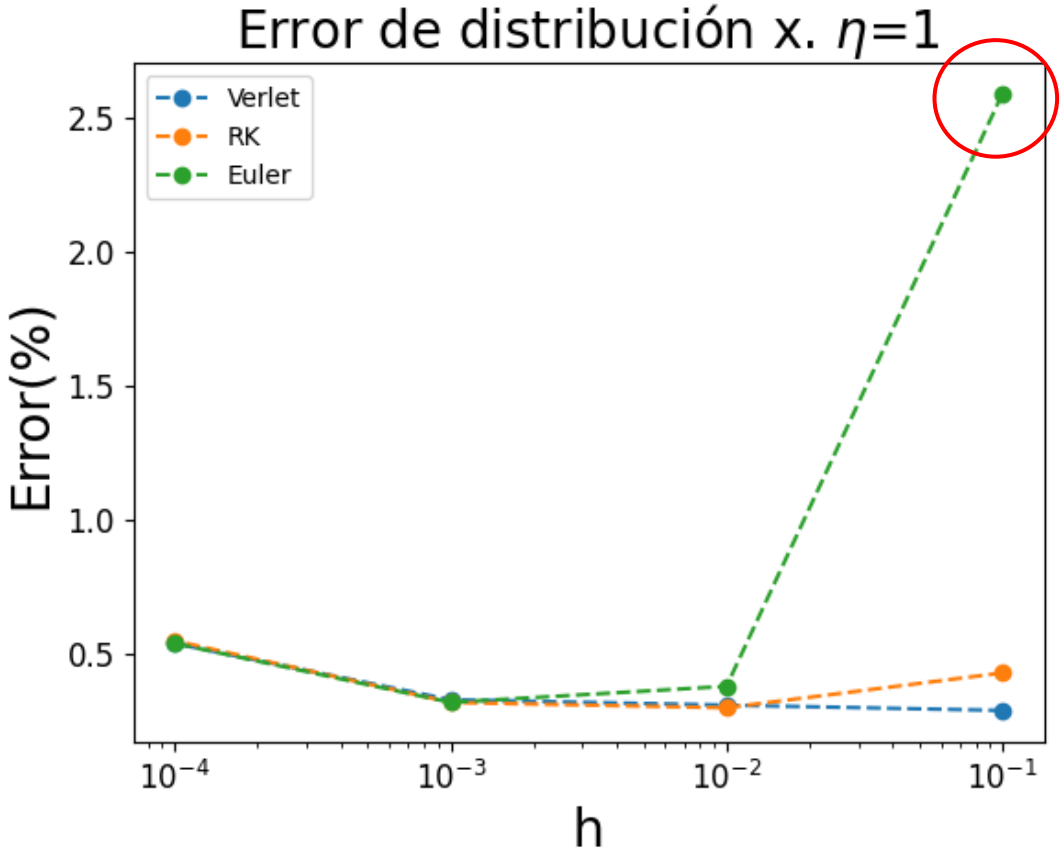
$$Error = \frac{1}{2}\int_{-\infty}^{\infty}\left(P_{data}(x) - P_{teórica}(x)\right)dx \in [0,1]$$



$t = 10^6$

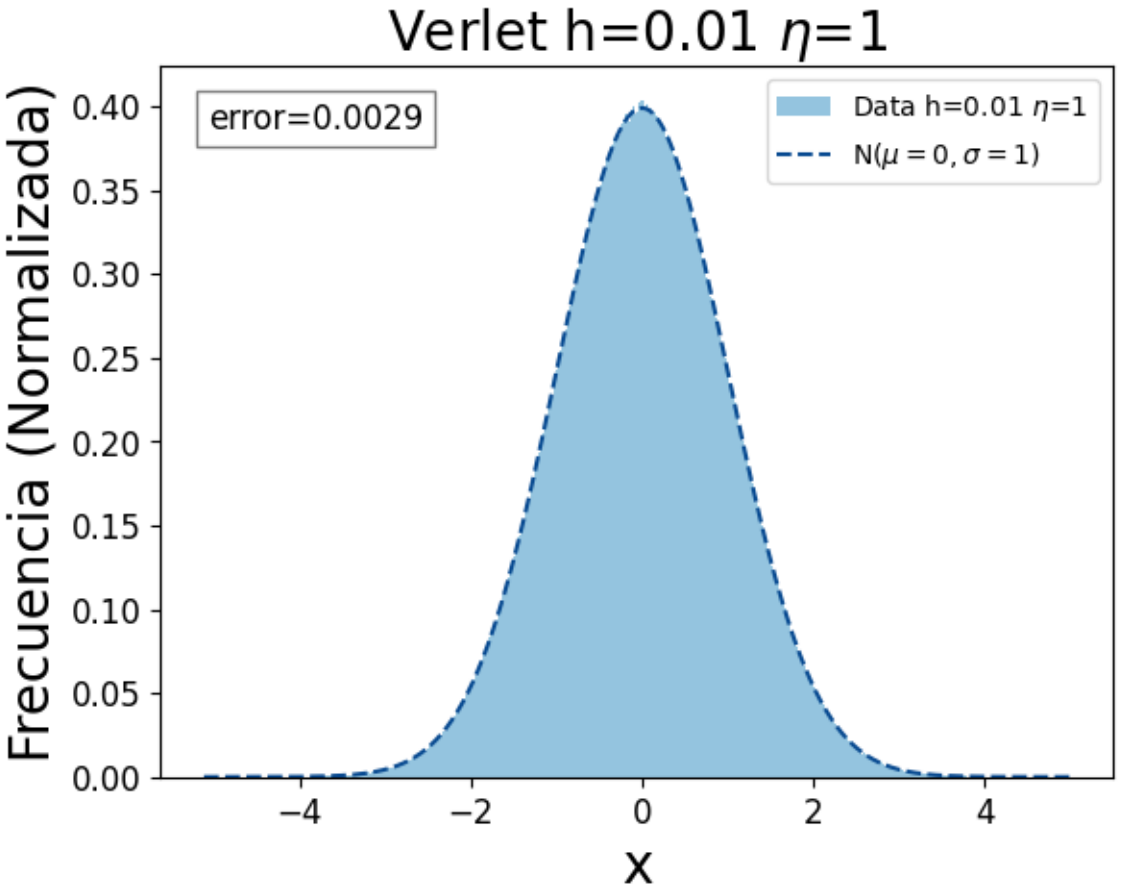
OSCILADOR ARMÓNICO: DISTRIBUCIONES

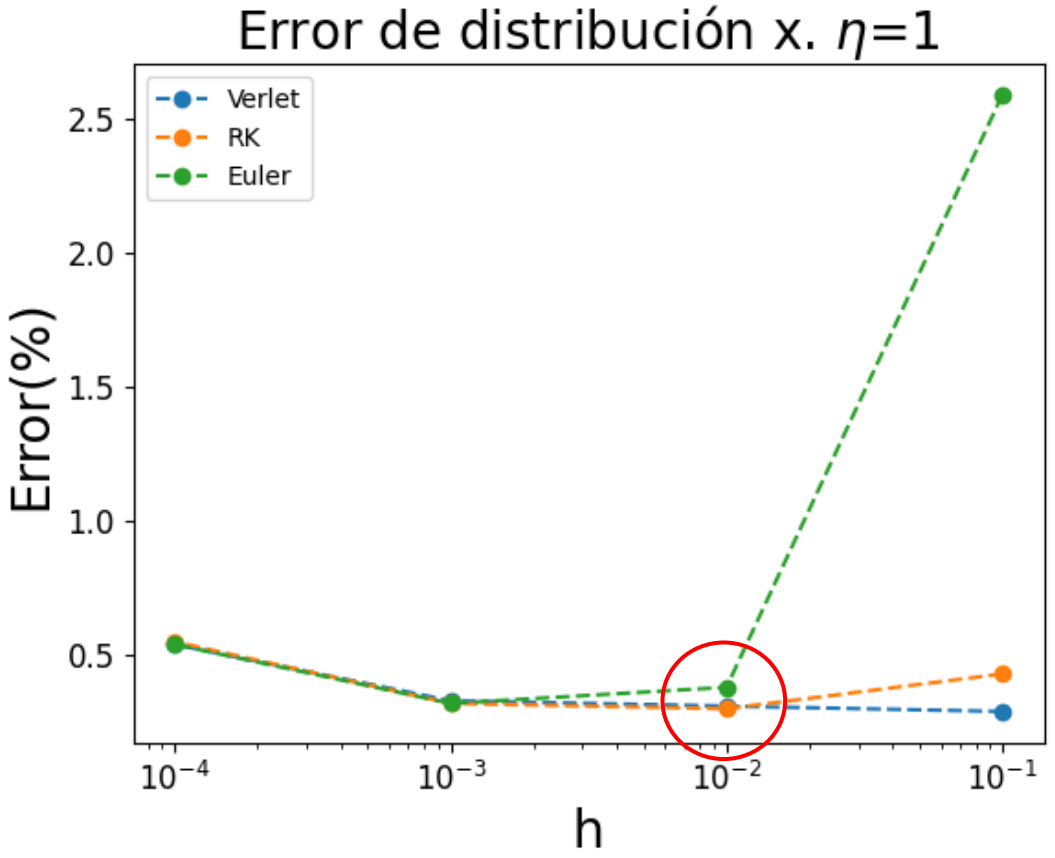




$t = 10^6$

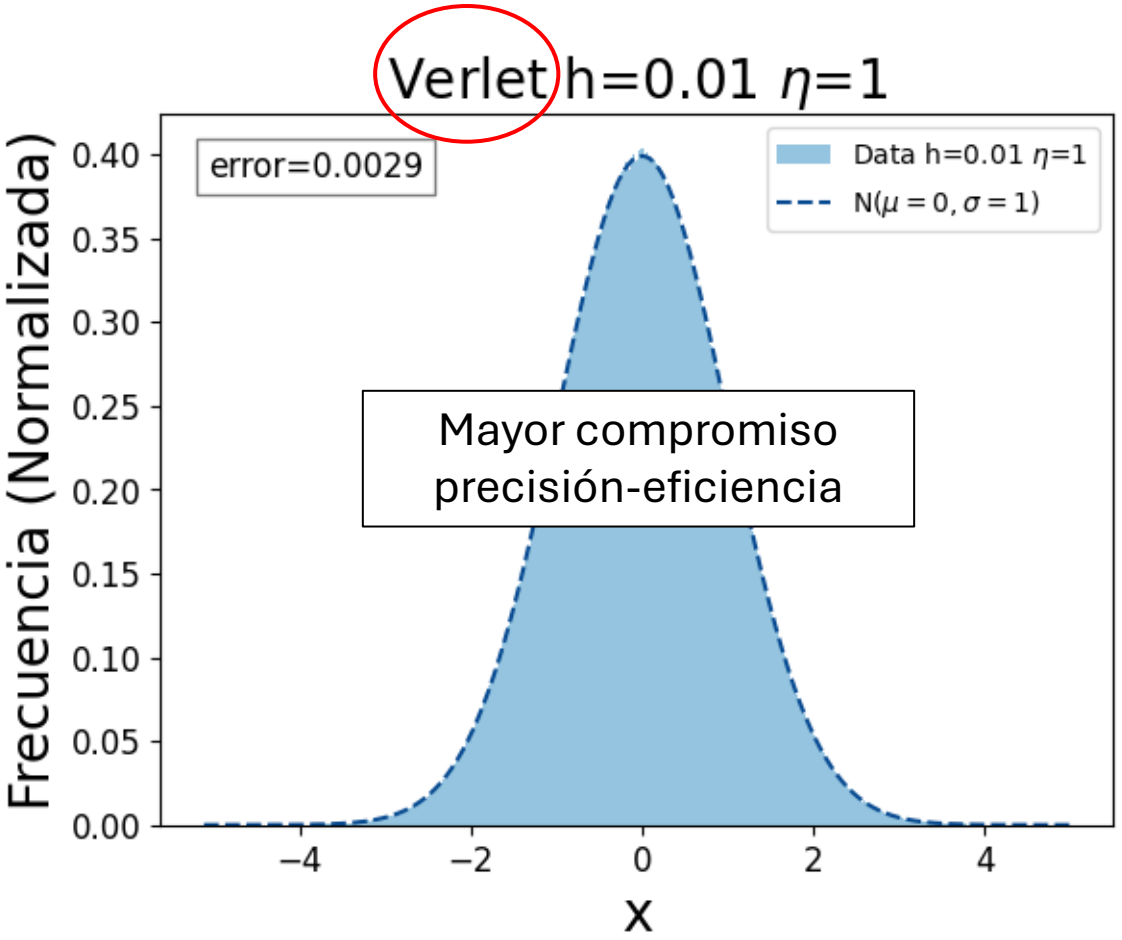
OSCILADOR ARMÓNICO: DISTRIBUCIONES

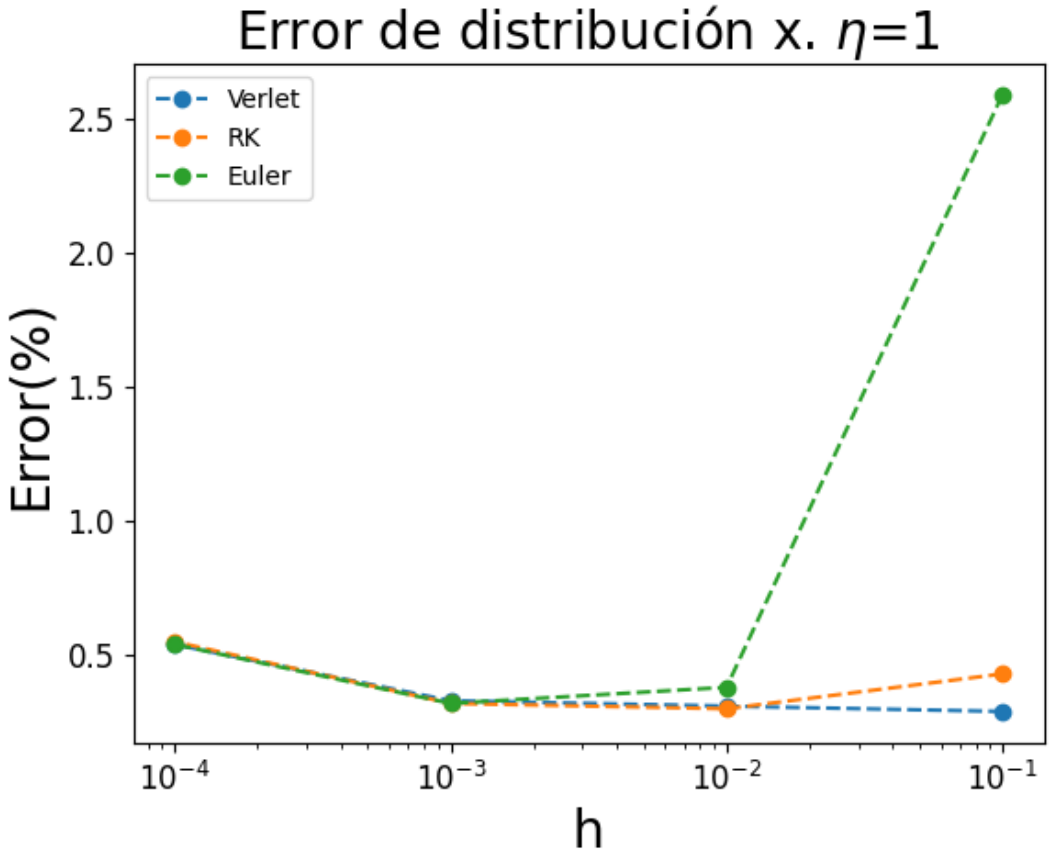




$t = 10^6$

OSCILADOR ARMÓNICO: DISTRIBUCIONES



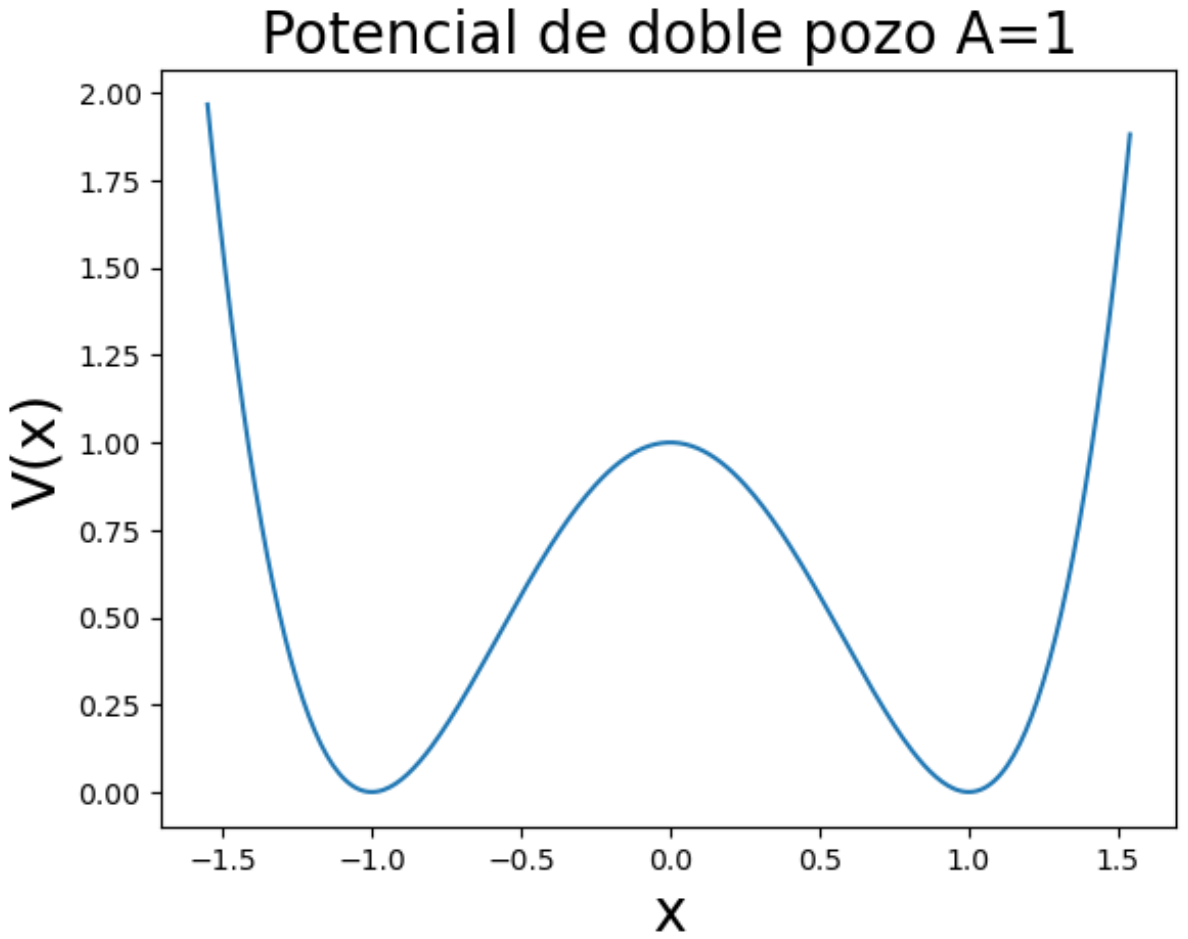


$t = 10^6$

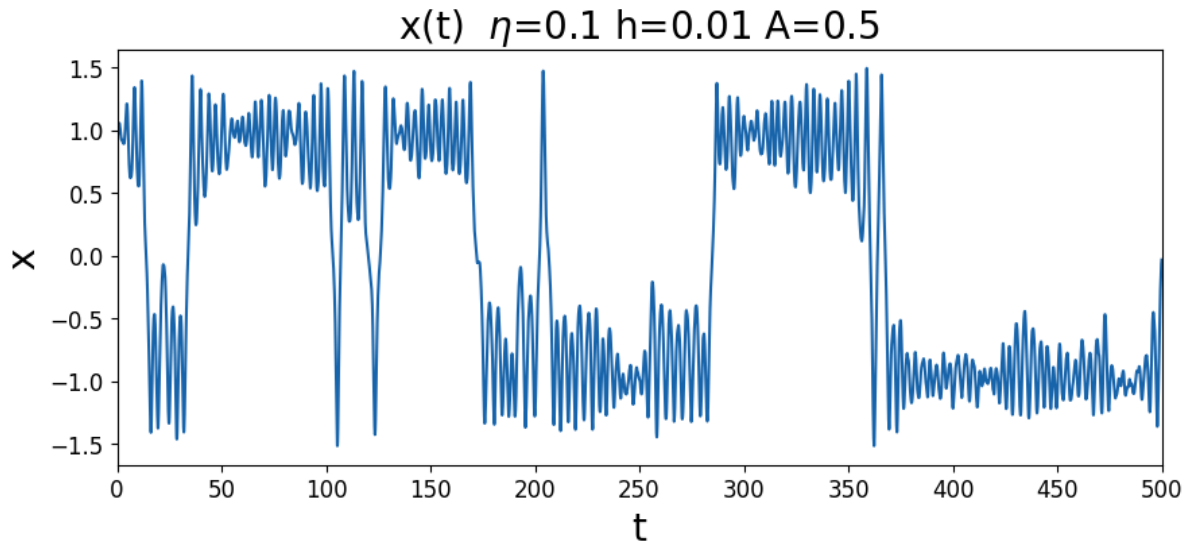
DOBLE POZO

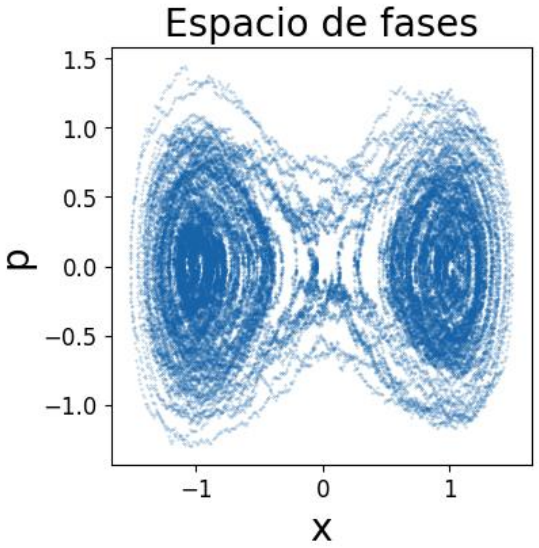
$$V(x) = A(x^2 - 1)^2$$

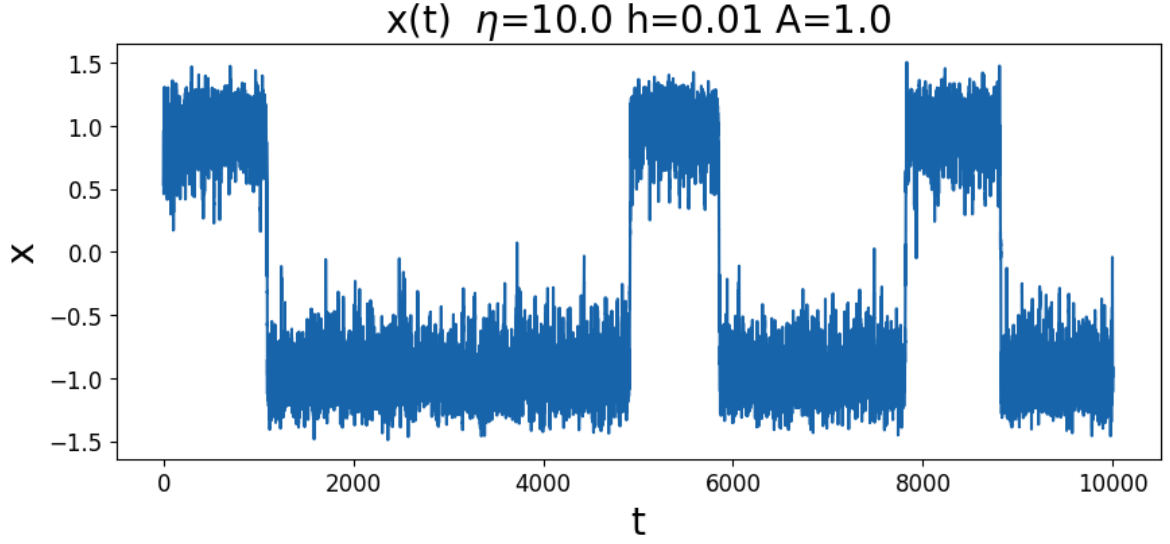
A ⟶ barrera de potencial

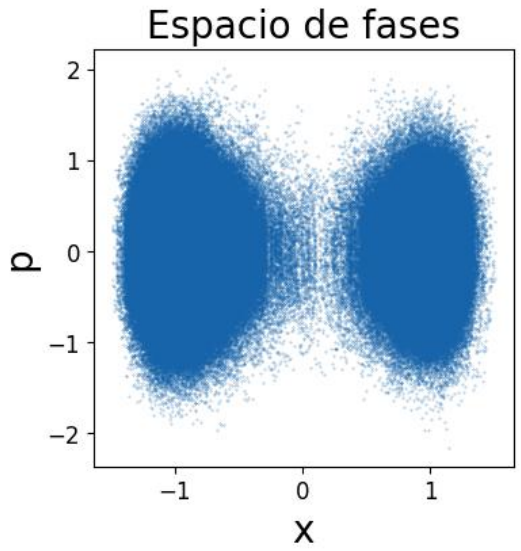


DOBLE POZO: TRAYECTORIAS









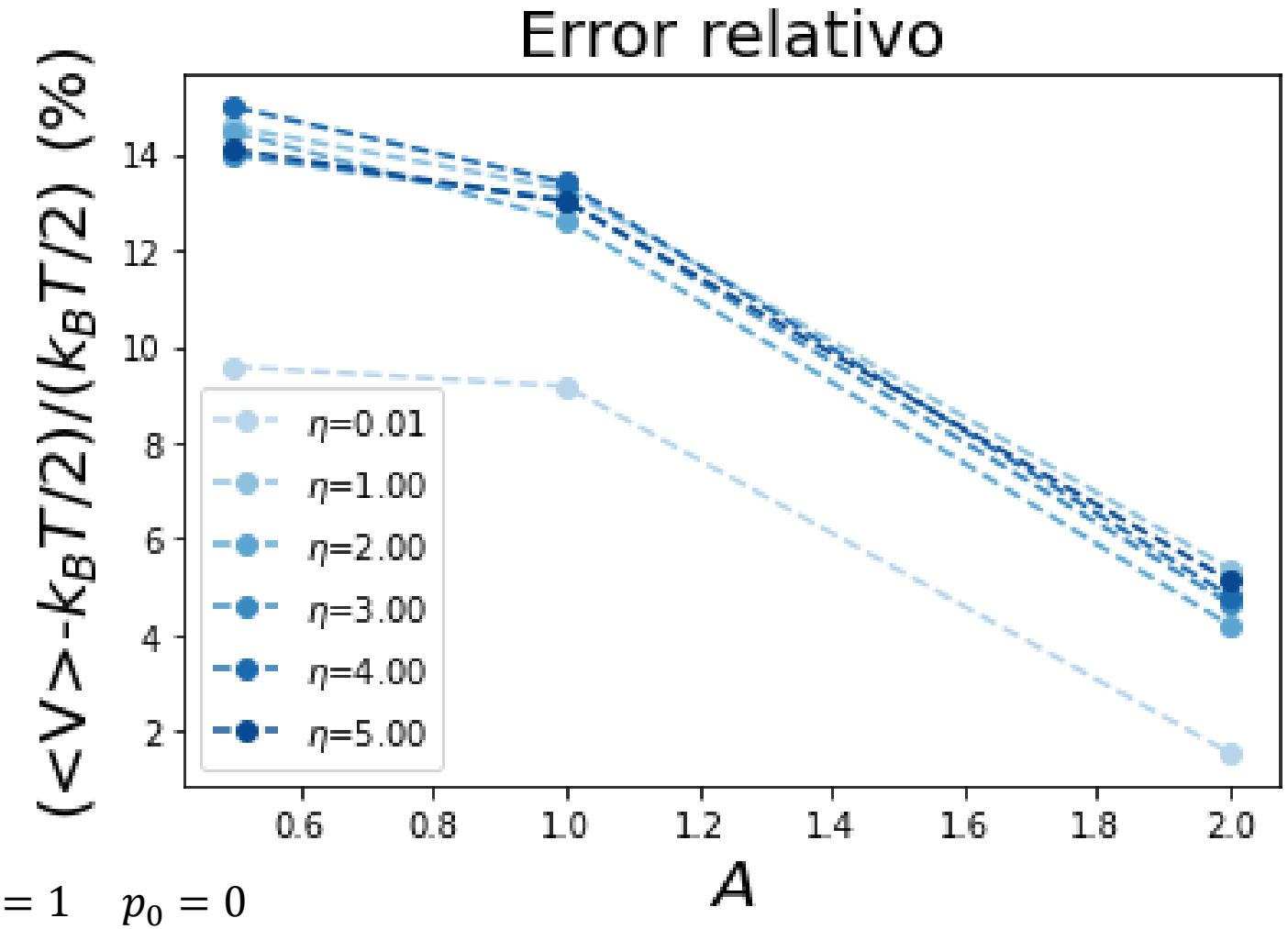
DOBLE POZO: EQUIPARTICIÓN

- Ya no se cumple el teorema de equipartición de la energía para el potencial, aunque sí para la cinética:

$$\langle V \rangle = \int_{-\infty}^{+\infty} V(x)P(x)dx = \int_{-\infty}^{+\infty} A(x^2-1)^2 e^{A\left(x^2-1\right)^2/k_B T} dx$$

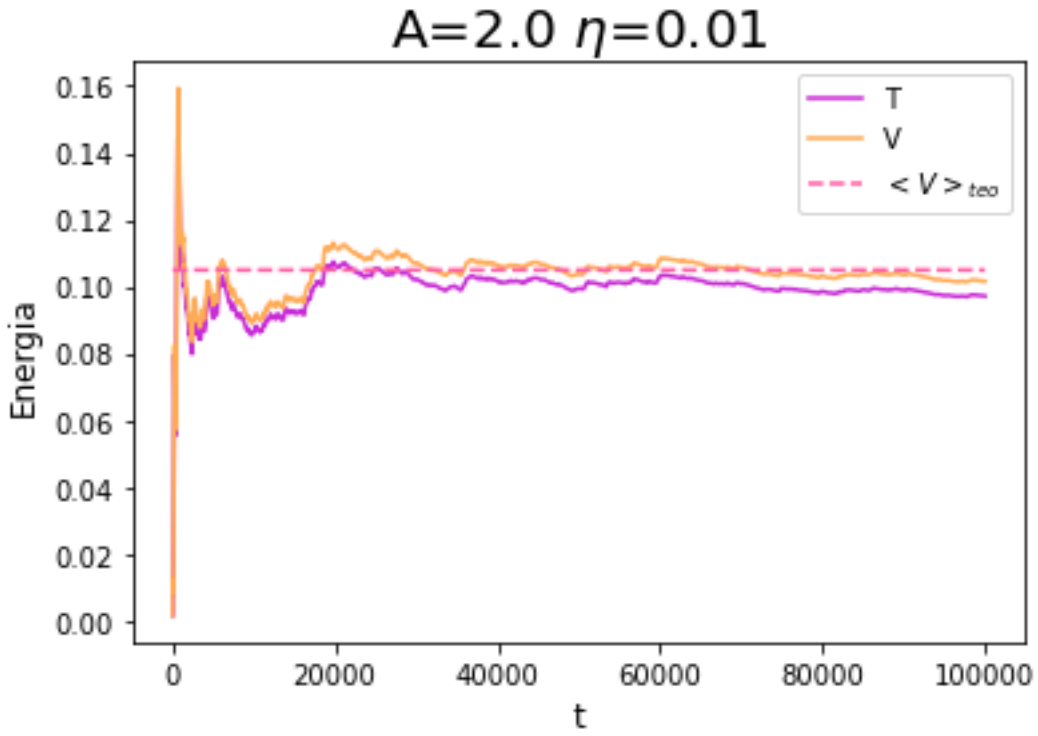
- No tiene solución analítica, pero se puede estimar numéricamente
- Para barreras altas, veremos que no se aleja mucho del oscilador

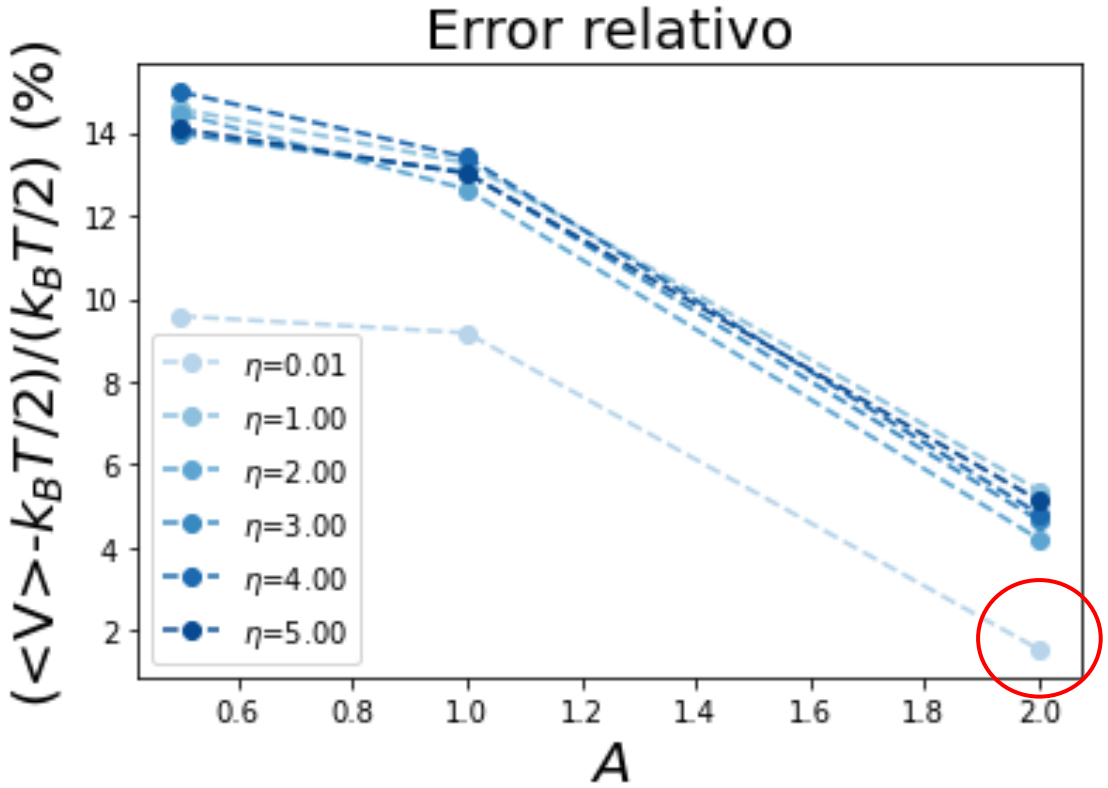
DOBLE POZO: EQUIP ENERGÍA



$k_BT = 0.2$ $x_0 = 1$ $p_0 = 0$

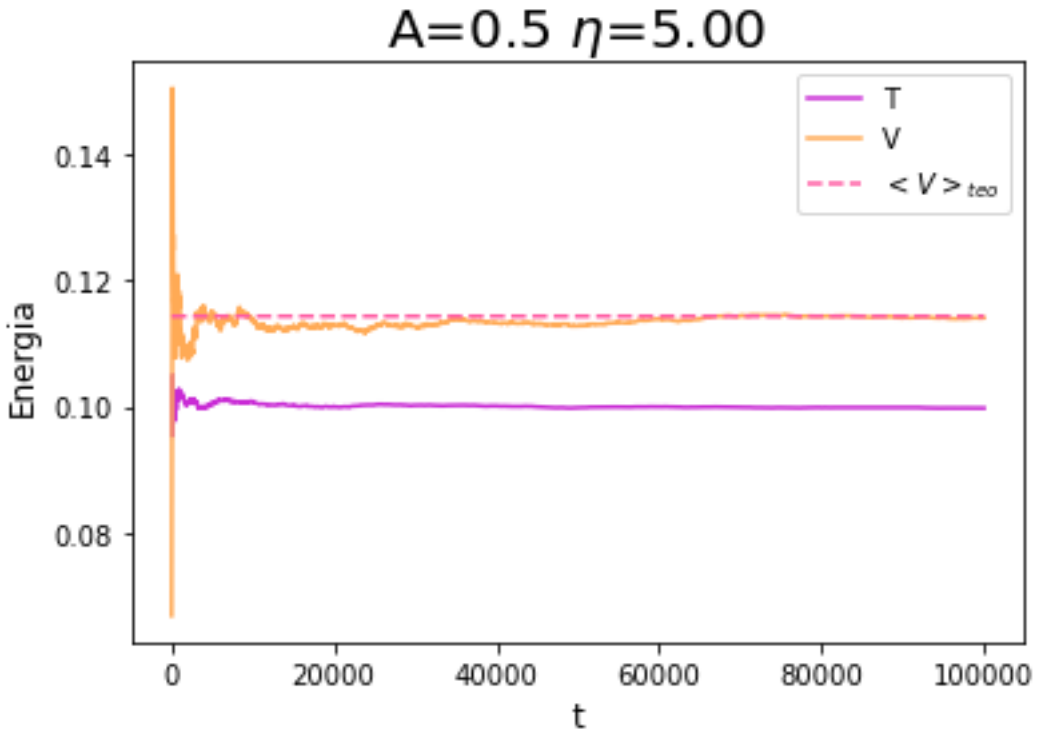
DOBLE POZO: EQUIP ENERGÍA

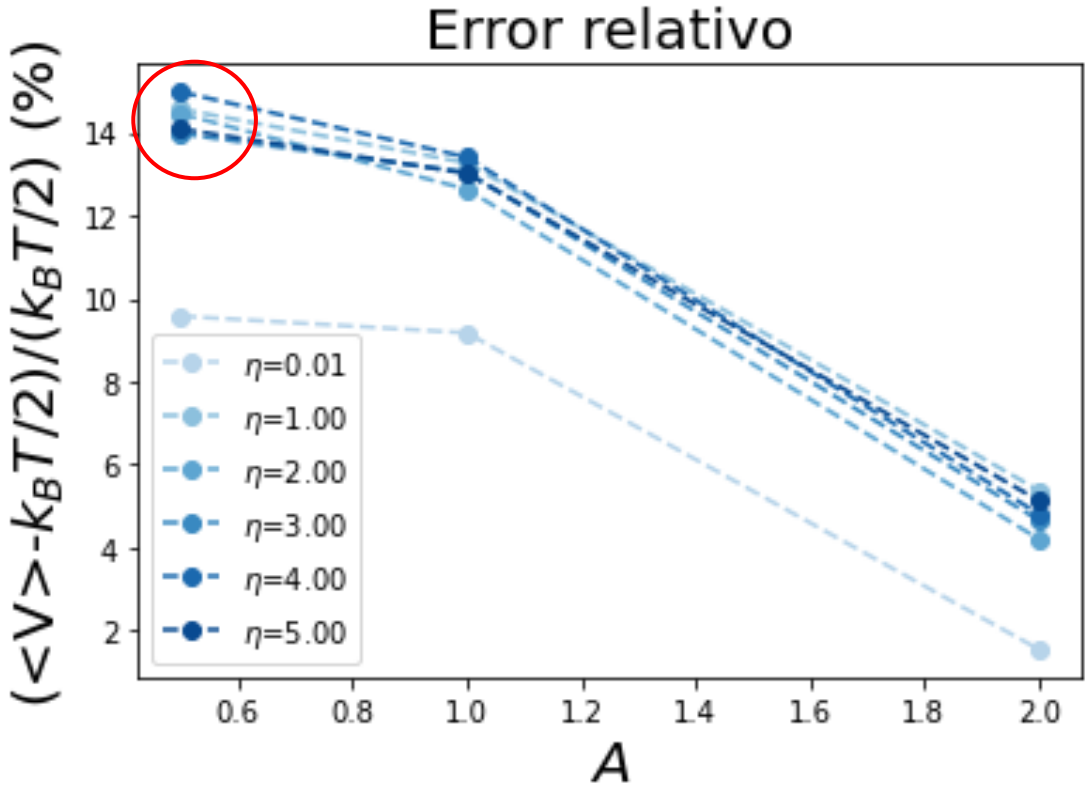




$k_B T = 0.2$ $x_0 = 1$ $p_0 = 0$

DOBLE POZO: EQUIP ENERGÍA





$k_B T = 0.2$ $x_0 = 1$ $p_0 = 0$

DOBLE POZO: DISTRIBUCIONES

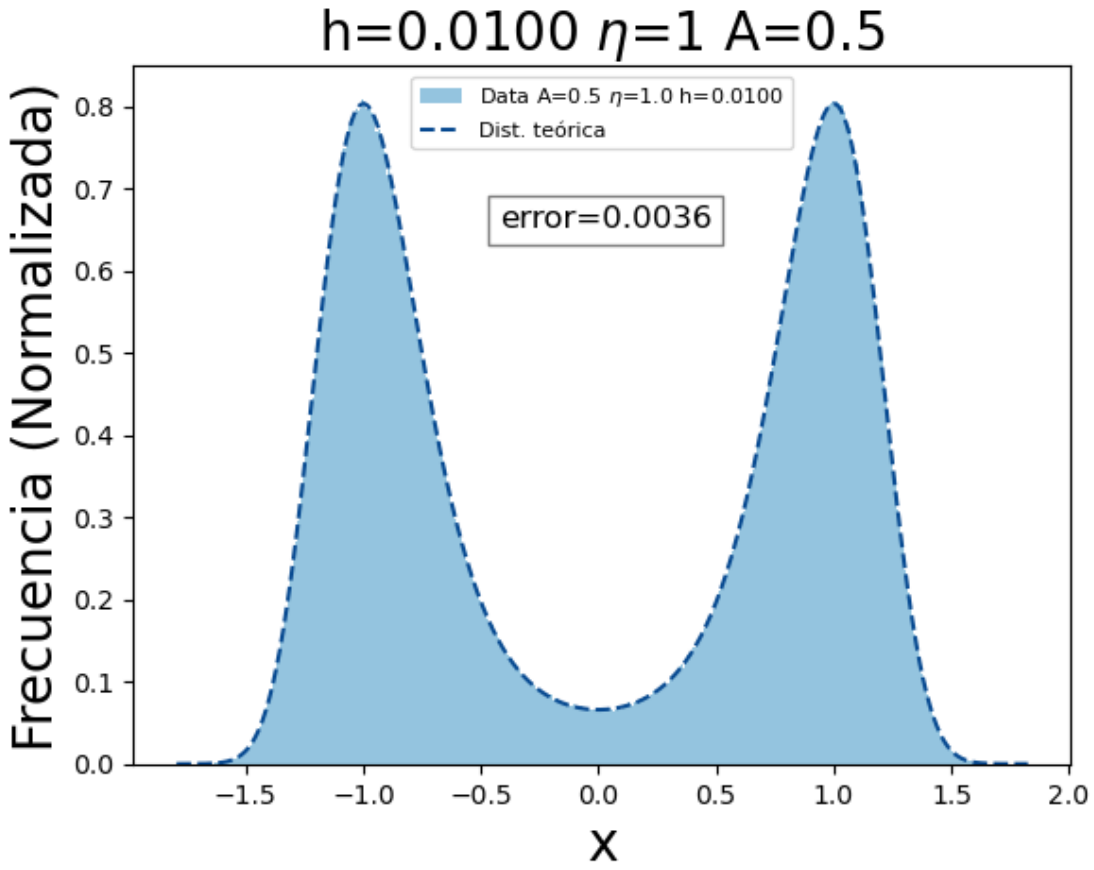
Formalismo canónico

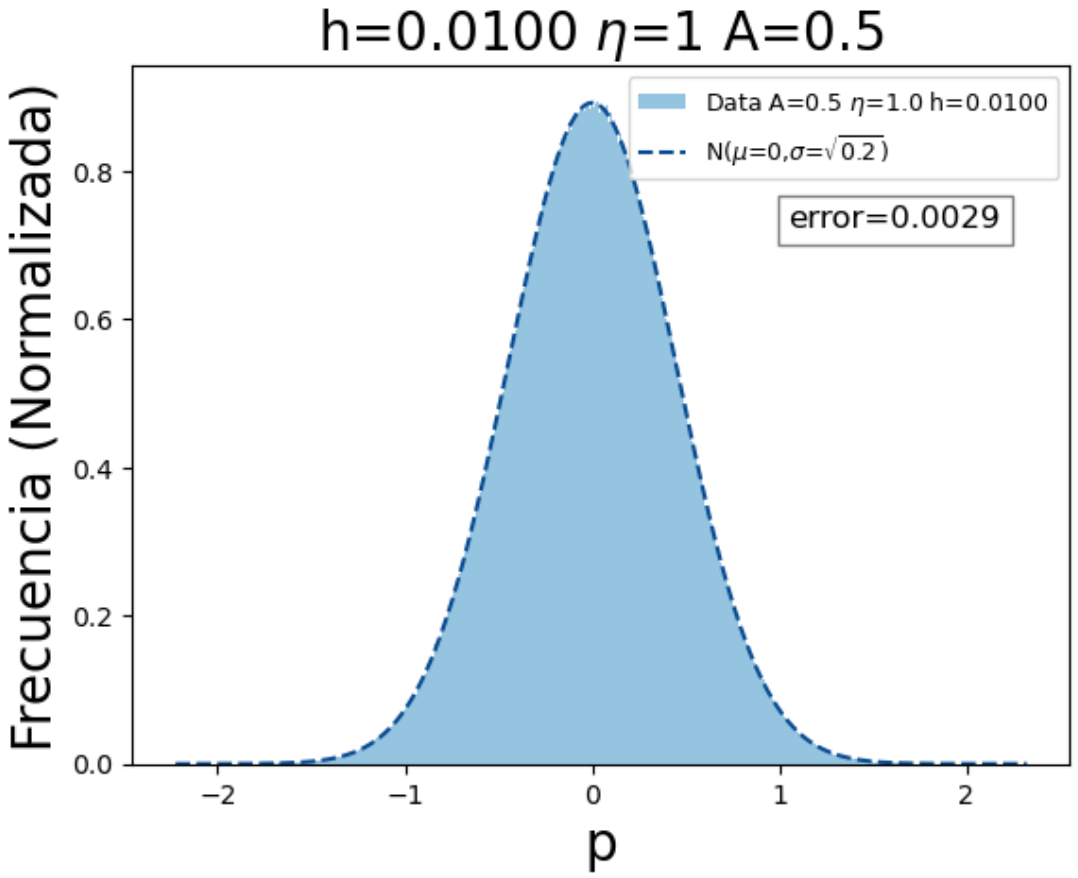
$\boxed{m = 1, k_B T = 0{,}2}$

$$\begin{cases} V(x) = A(x^2-1)^2 \\ T(p) = \dfrac{p^2}{2m} = \dfrac{p^2}{2} \end{cases} \Longrightarrow E(x,p) = A(x^2-1)^2 + \frac{p^2}{2} \Longrightarrow P(x,p) = \frac{1}{Z} e^{-\frac{A(x^2-1)^2}{k_B T}} e^{-\frac{p^2}{2k_B T}} \Longrightarrow \begin{cases} P(x) = \dfrac{1}{Z_x} e^{-\frac{A(x^2-1)^2}{k_B T}} \\ P(p) = \dfrac{1}{Z_p} e^{-\frac{p^2}{2k_B T}} \end{cases} \Longrightarrow$$

$$\boldsymbol{P(x) \propto \frac{1}{Z_p} e^{-\frac{A(x^2-1)^2}{k_B T}}} \qquad \boldsymbol{P(p) \propto N(\mu = 0, \sigma = \sqrt{0,2})}$$

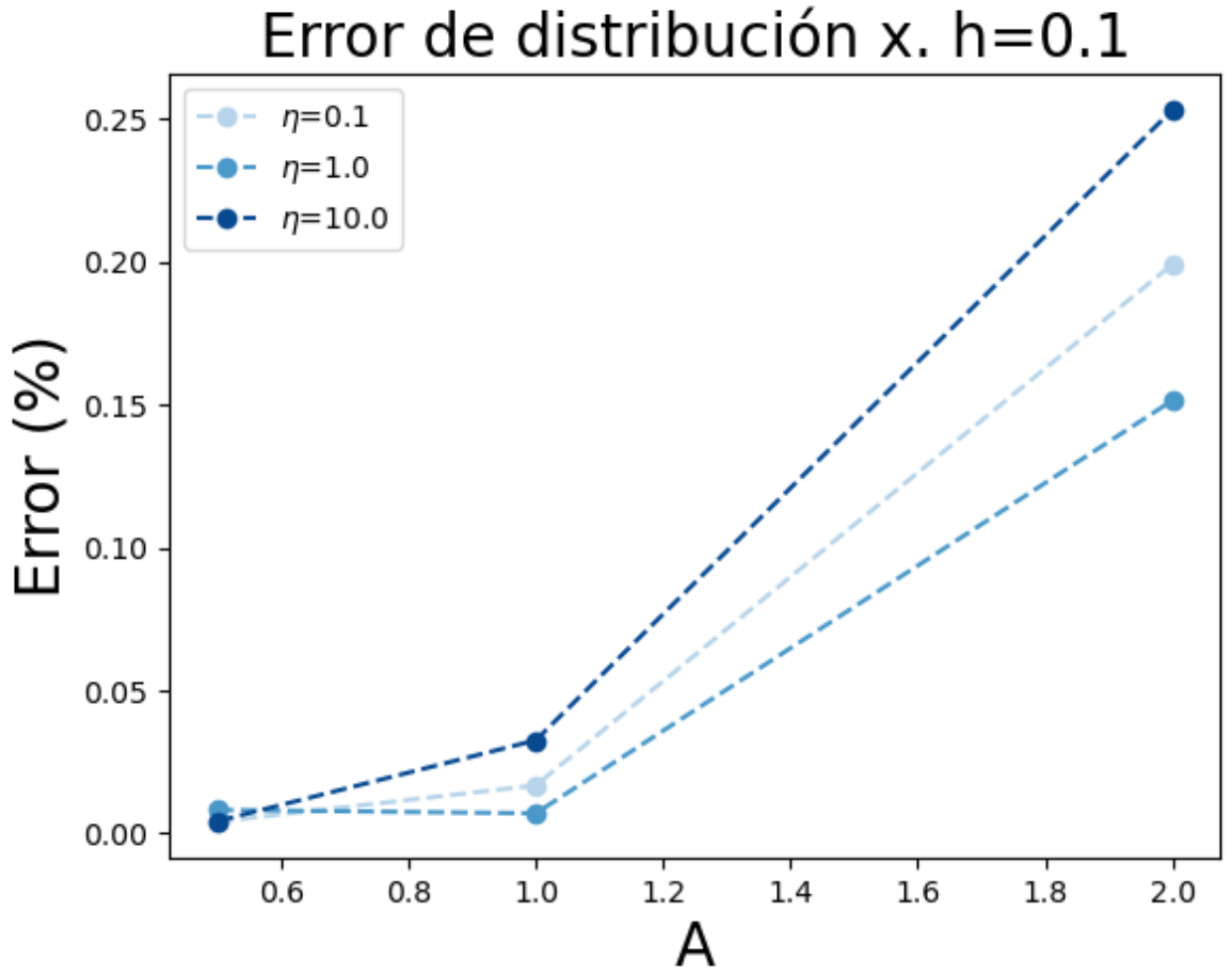
DOBLE POZO: DISTRIBUCIONES





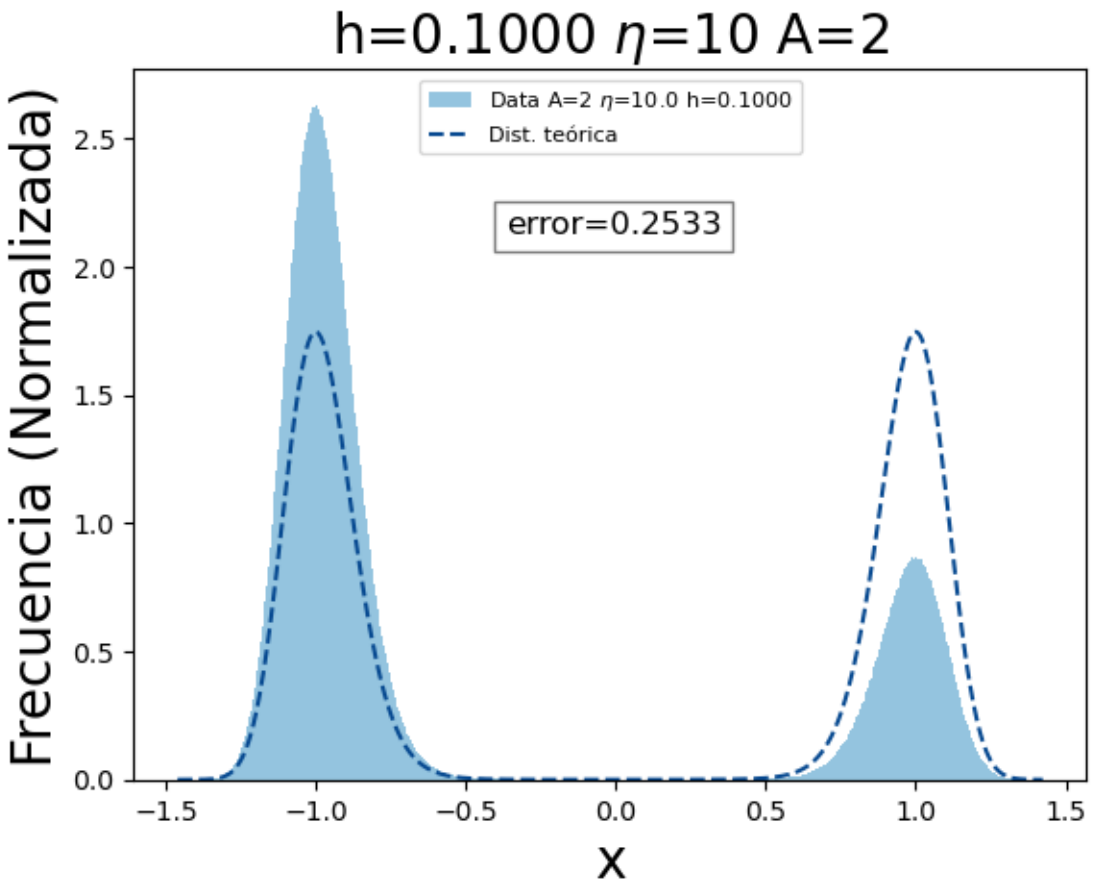
$t = 10^6$

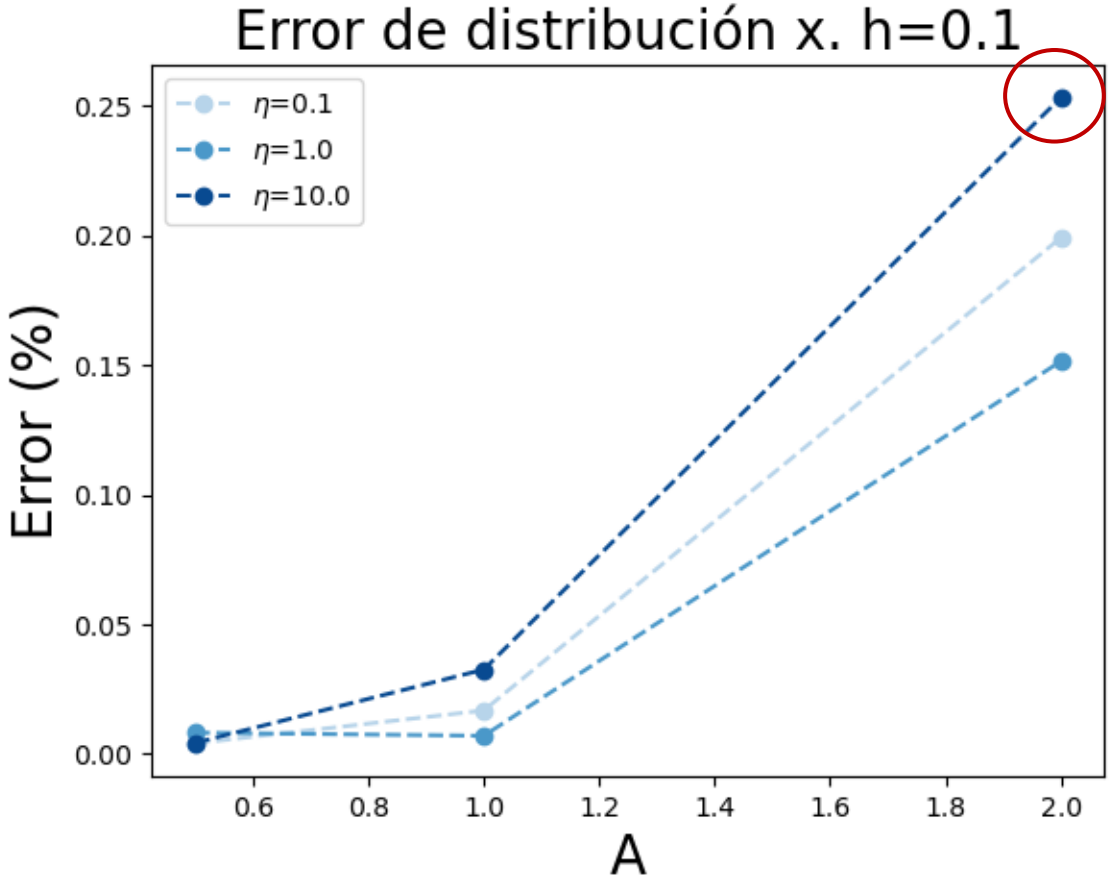
DOBLE POZO: DISTRIBUCIONES



$t = 10^6$

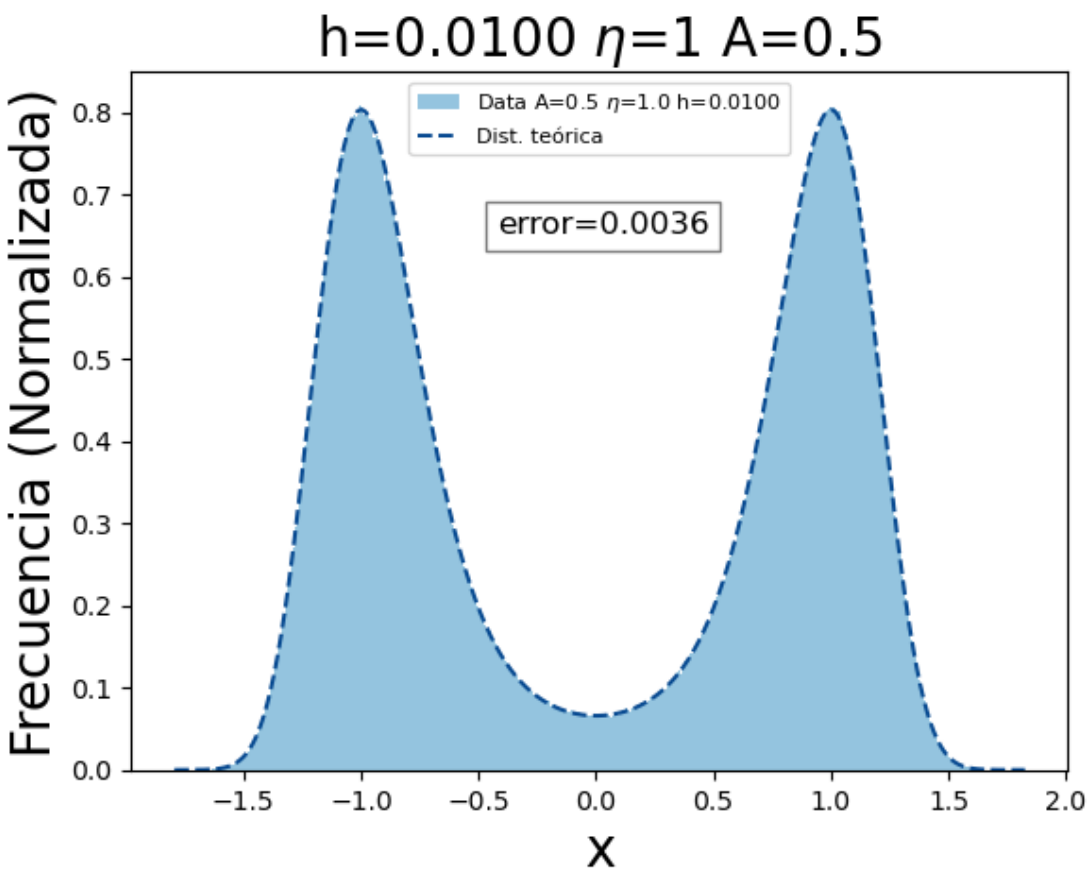
DOBLE POZO: DISTRIBUCIONES

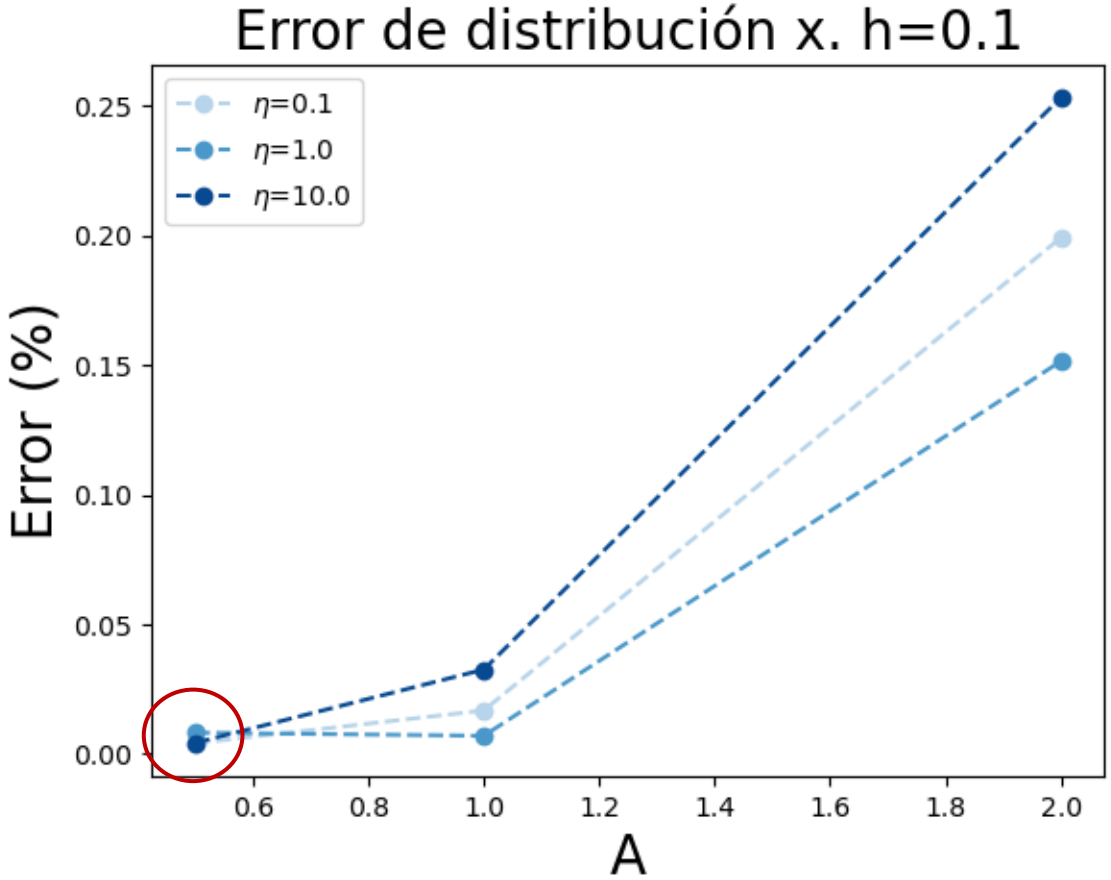




$t = 10^6$

DOBLE POZO: DISTRIBUCIONES



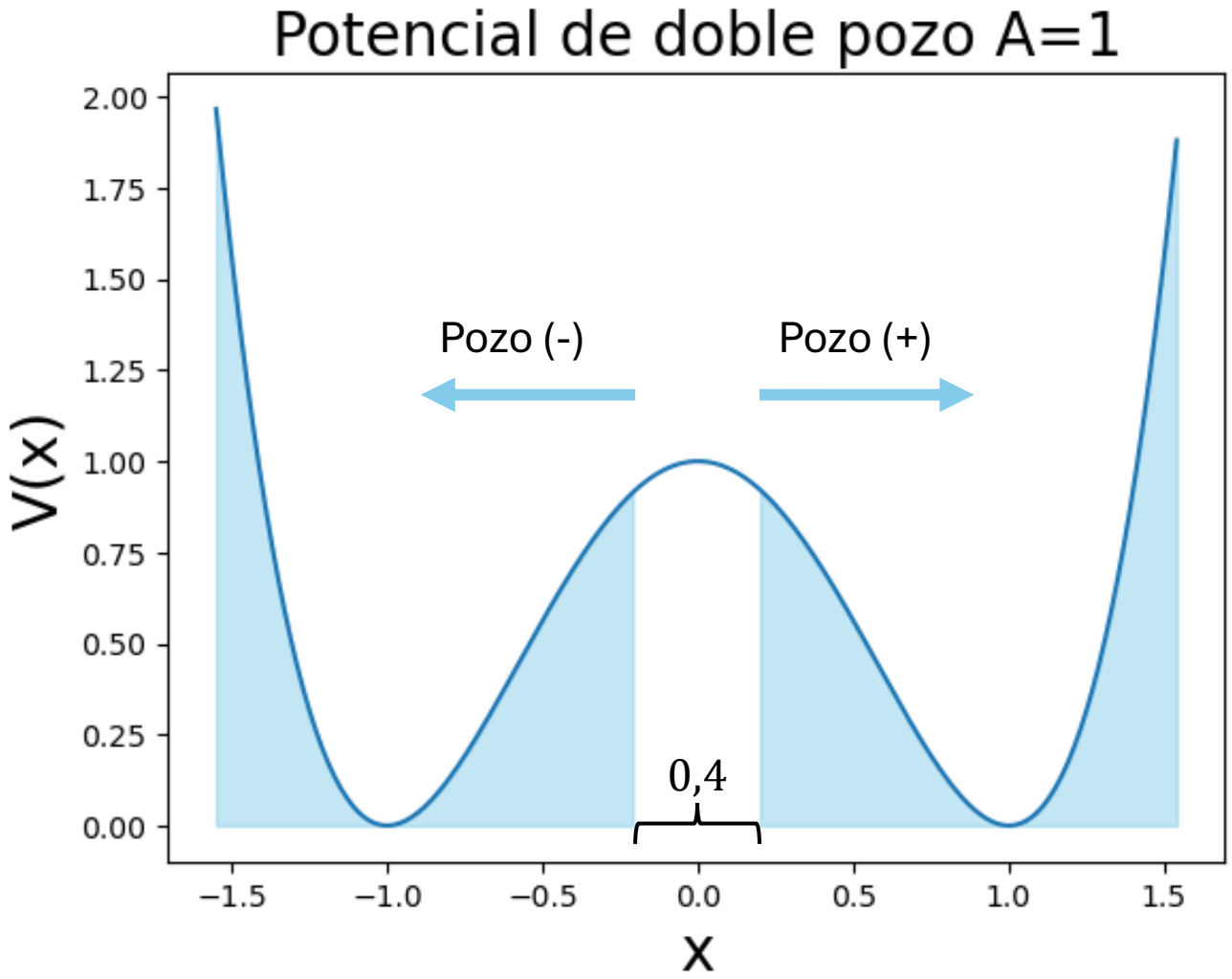


$t = 10^6$

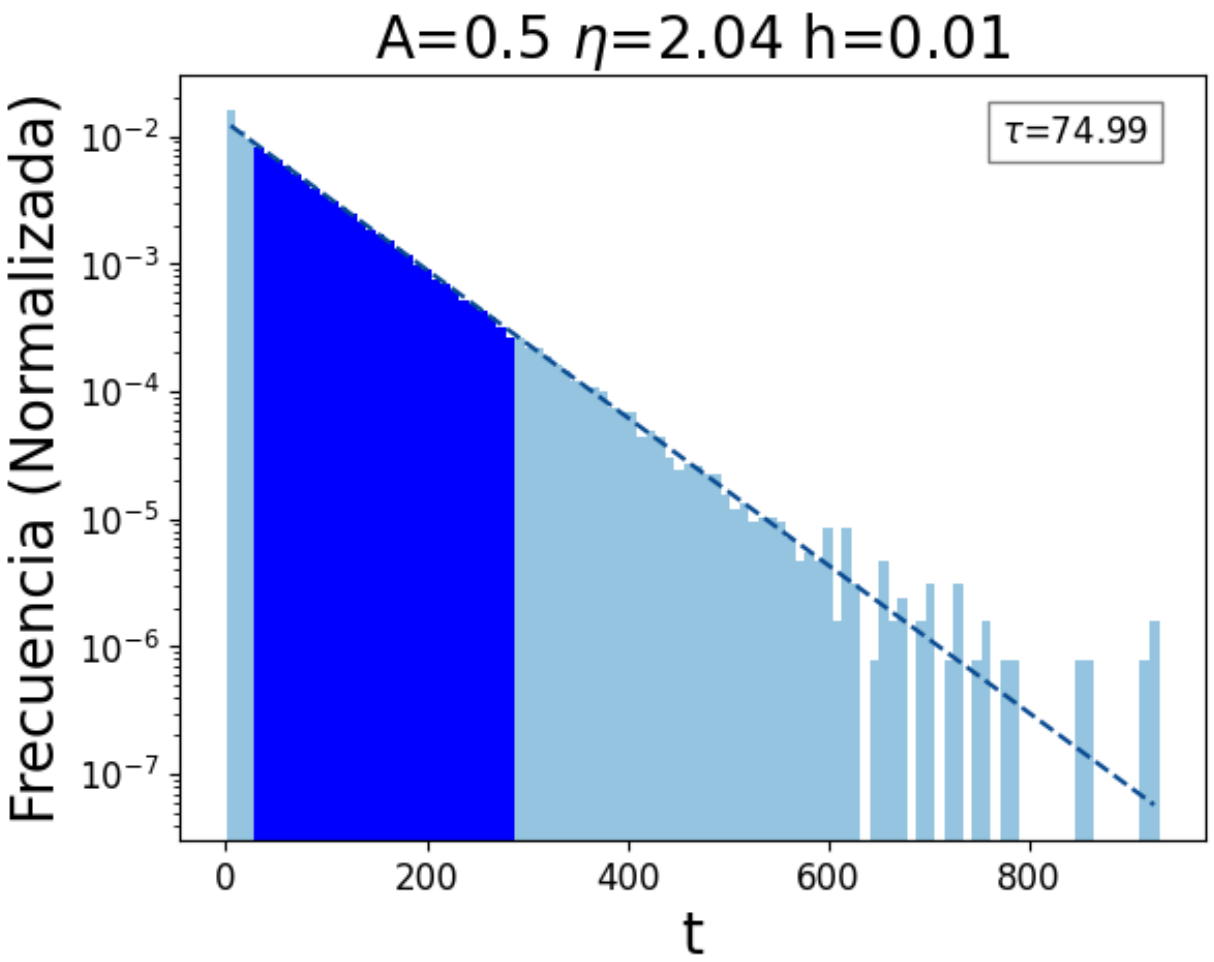
DOBLE POZO: TIEMPO DE OCUPACIÓN

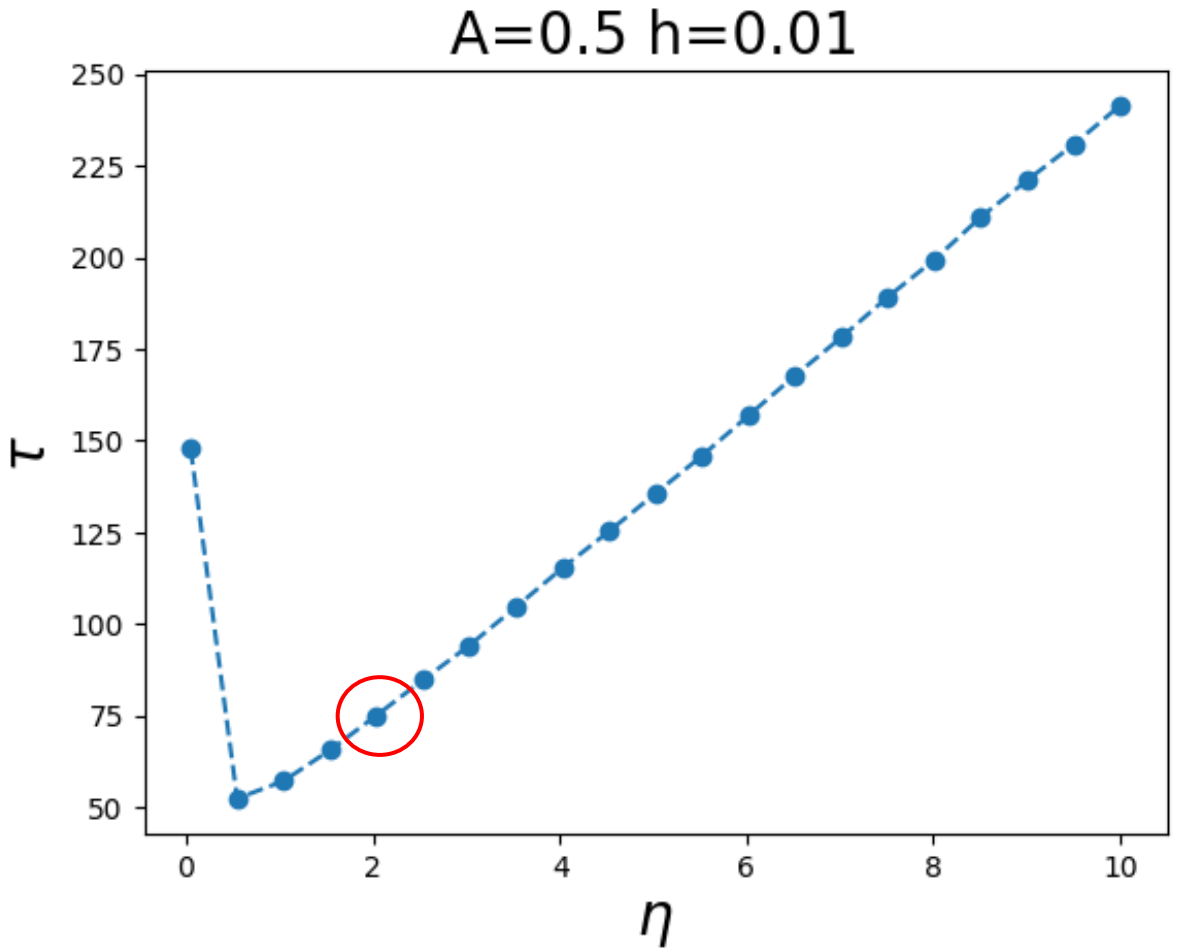
$$\boldsymbol{P(t) \propto e^{-\frac{t}{\tau}}}$$

$$\ln P(t) \propto -\frac{t}{\tau}$$



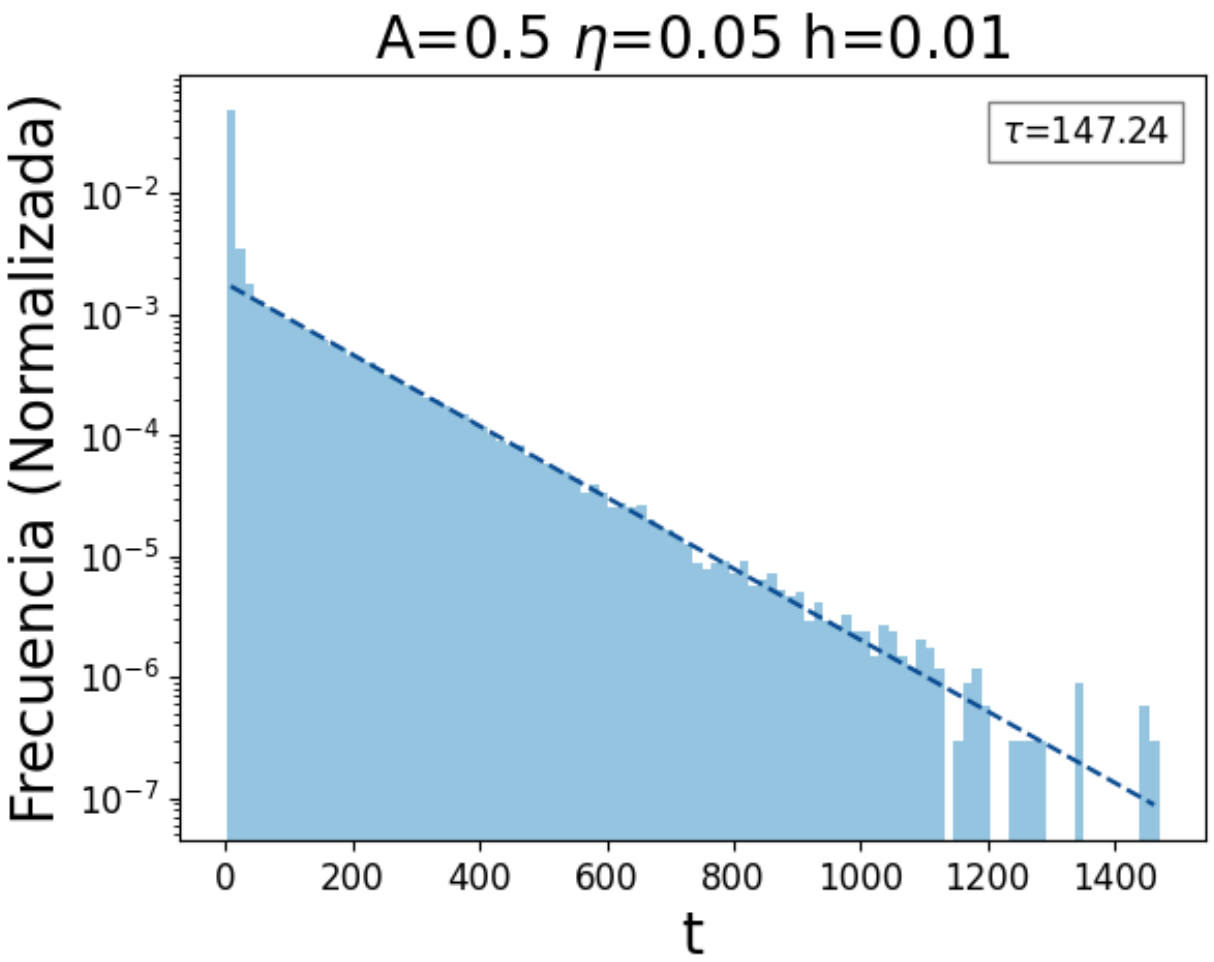
DOBLE POZO: TIEMPO DE OCUPACIÓN

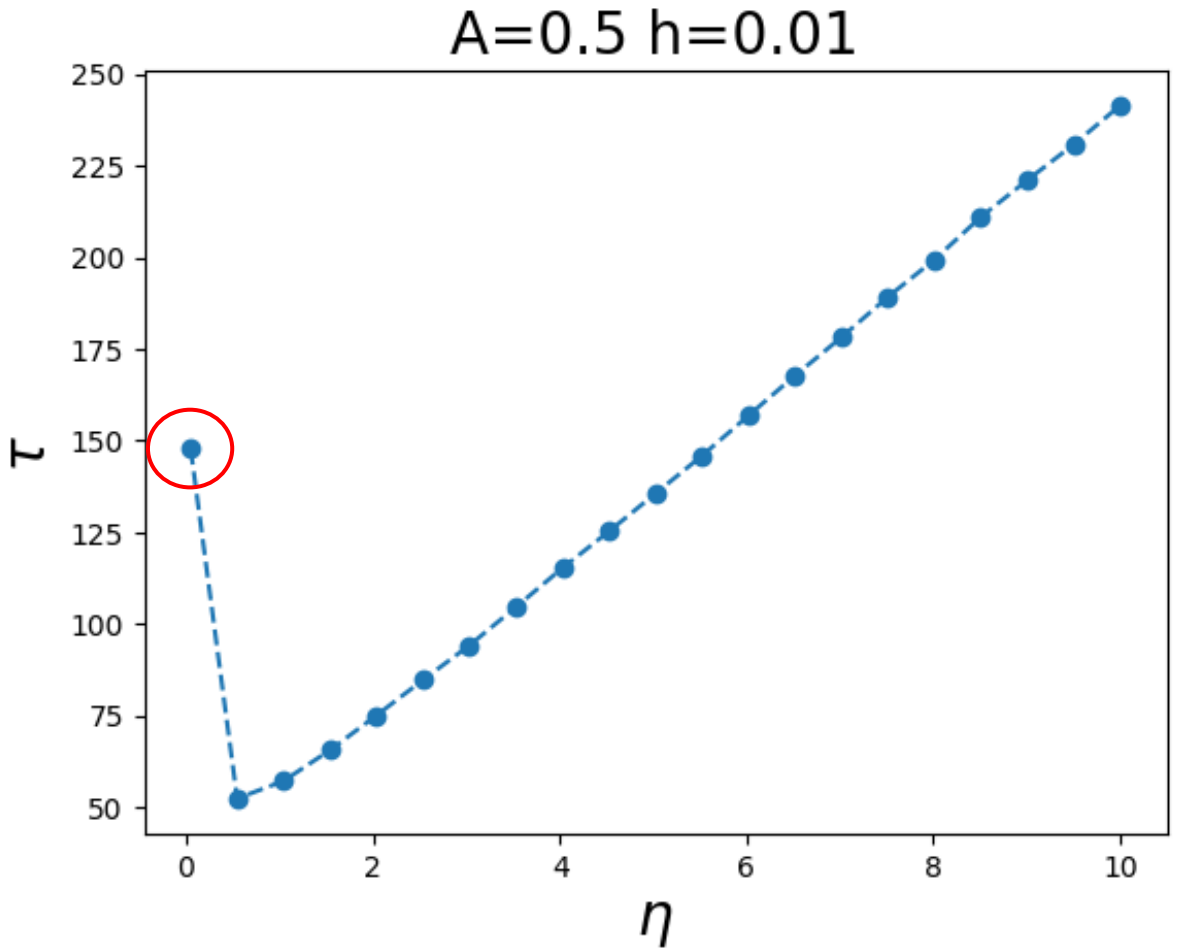




10^9 iteraciones

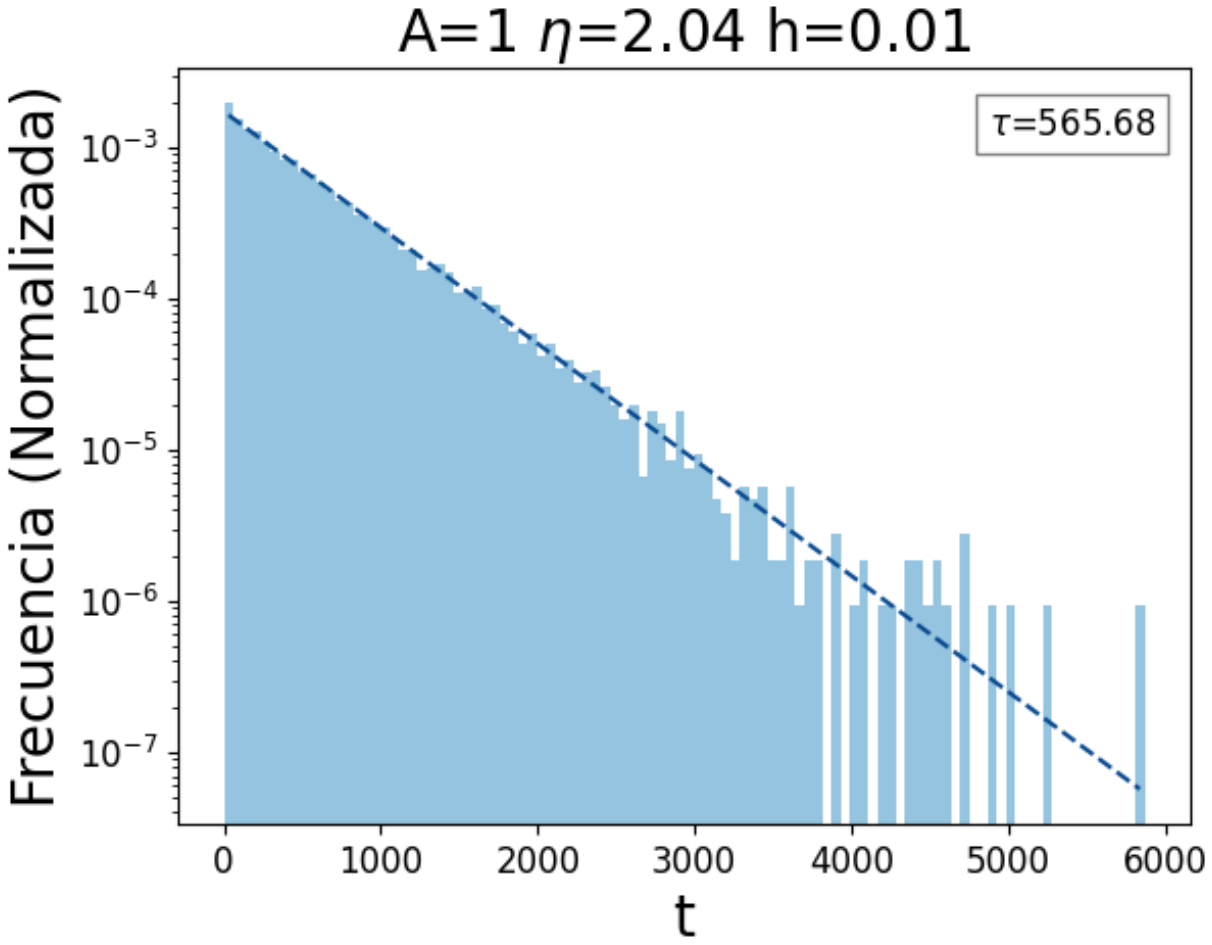
DOBLE POZO: TIEMPO DE OCUPACIÓN

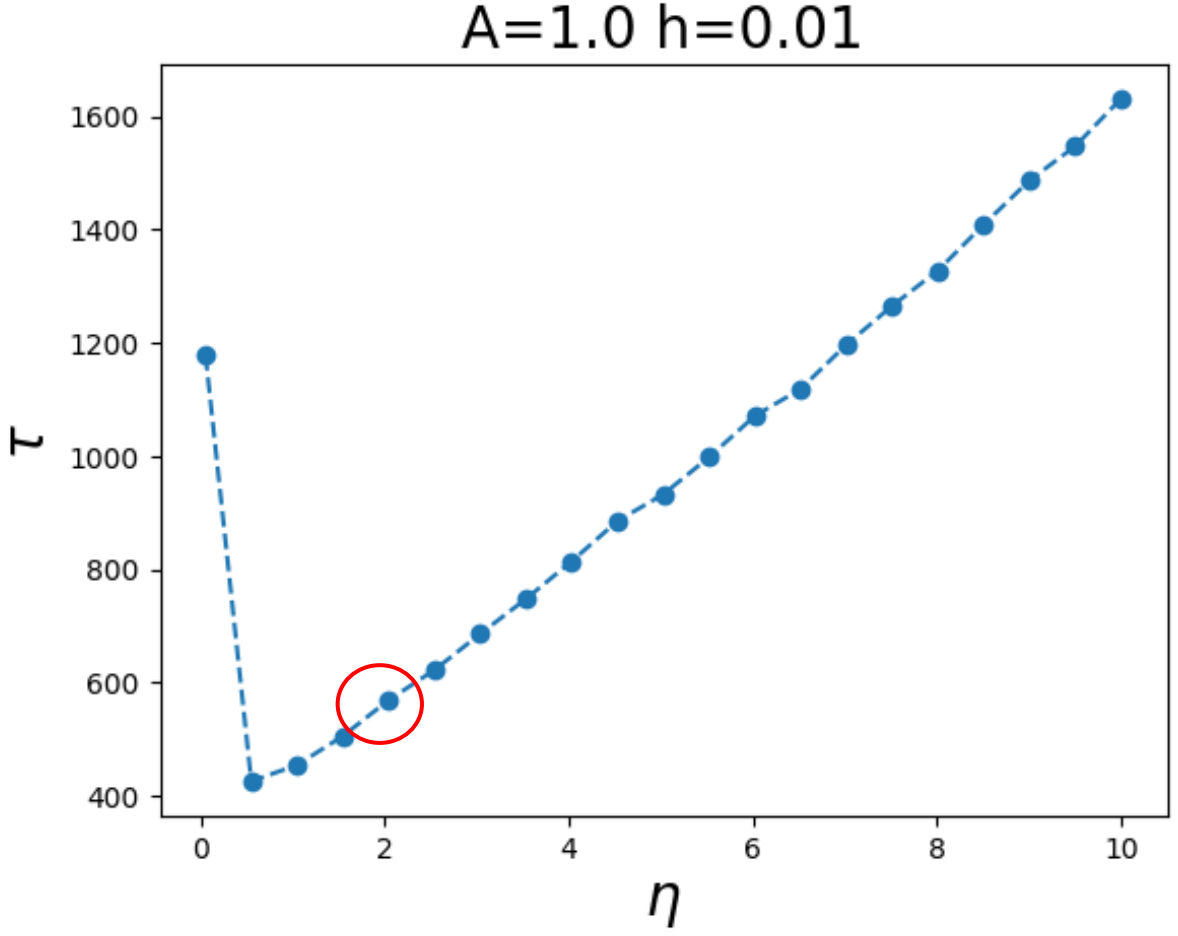




10^9 iteraciones

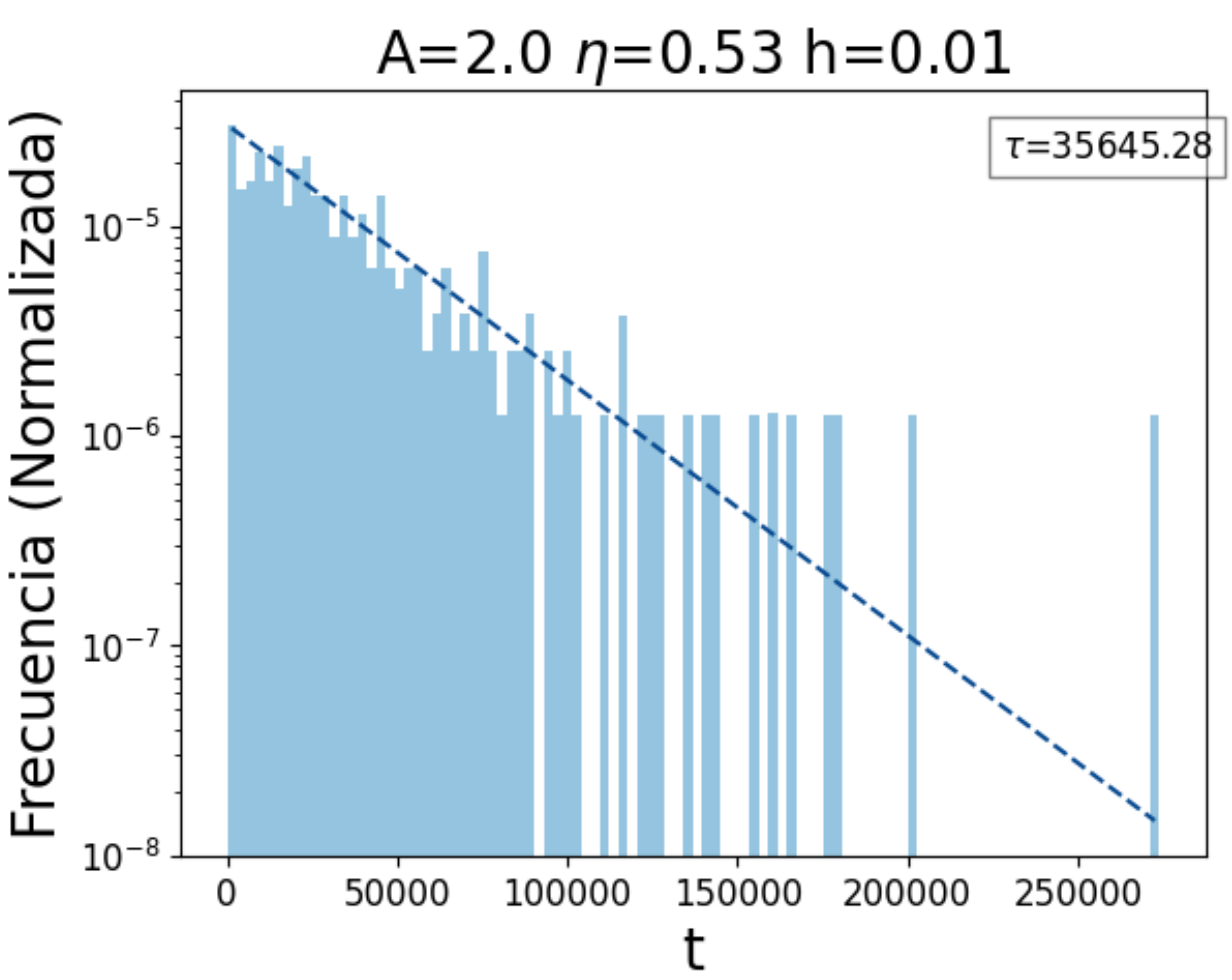
DOBLE POZO: TIEMPO DE OCUPACIÓN

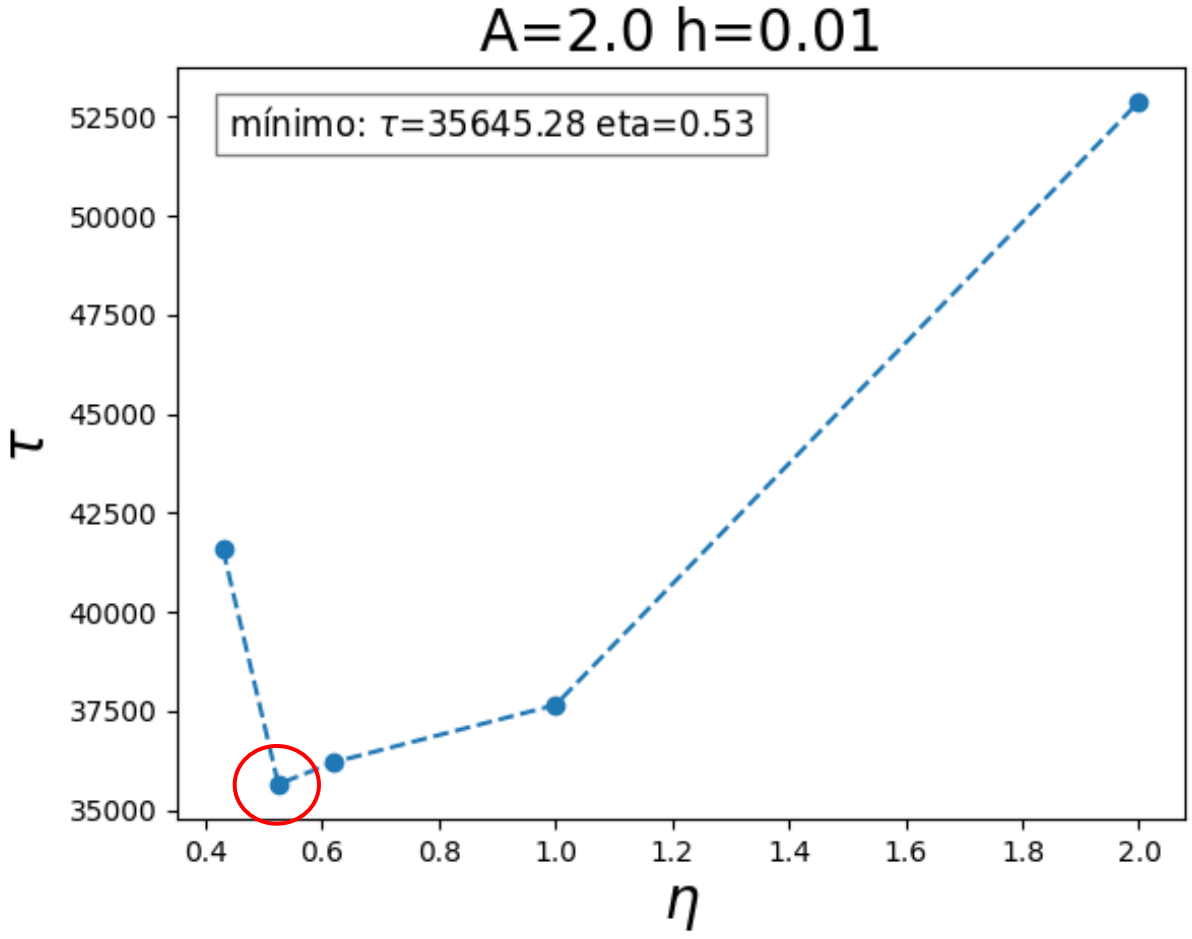




10^9 iteraciones

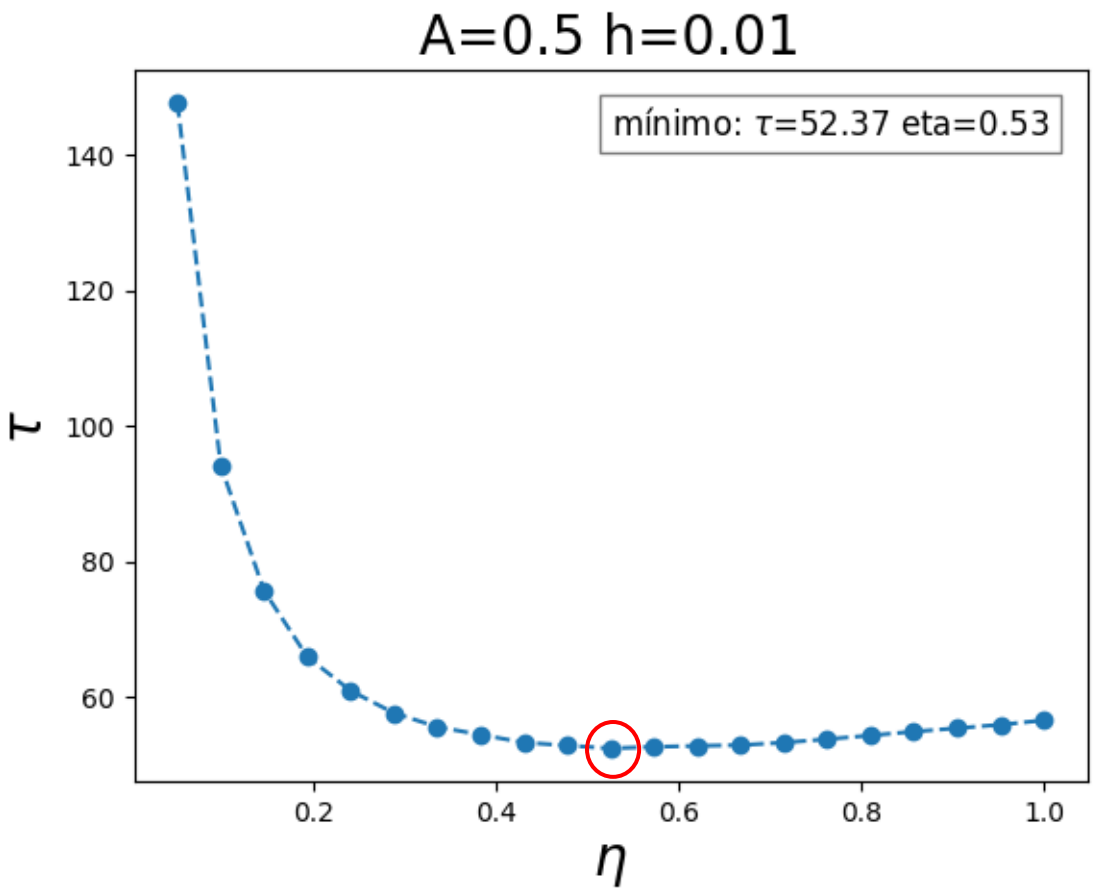
DOBLE POZO: TIEMPO DE OCUPACIÓN

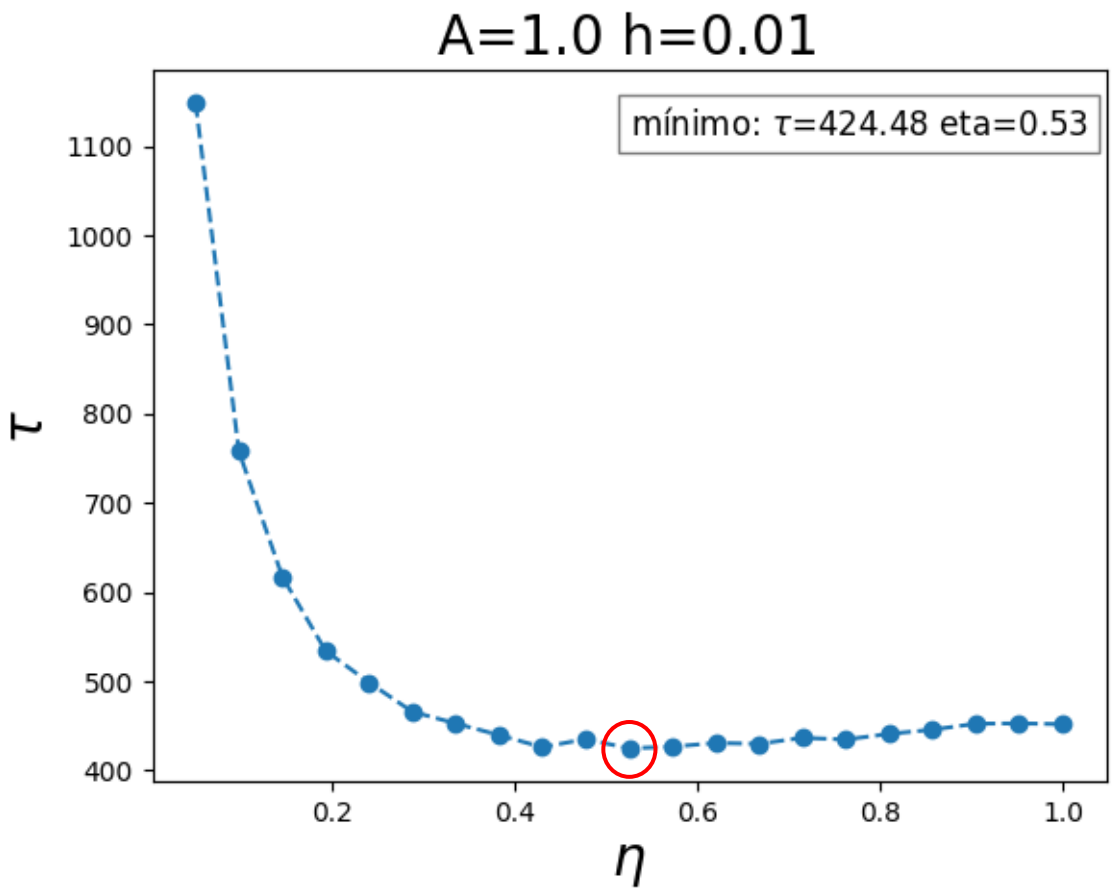




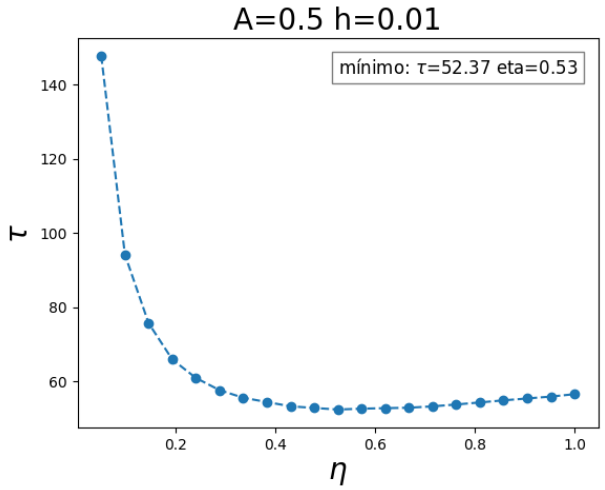
10^{11} iteraciones

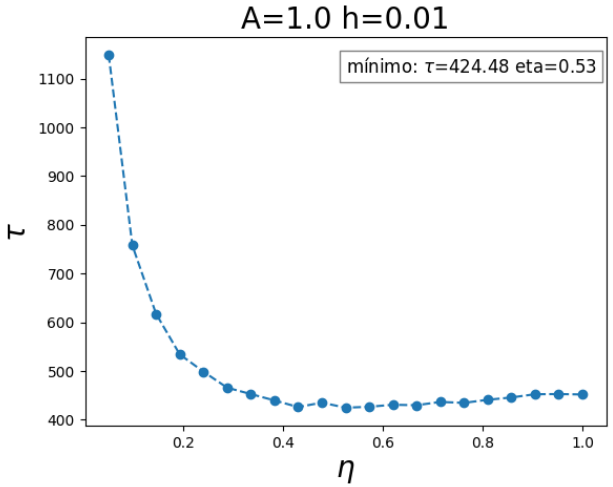
DOBLE POZO: TIEMPO DE OCUPACIÓN

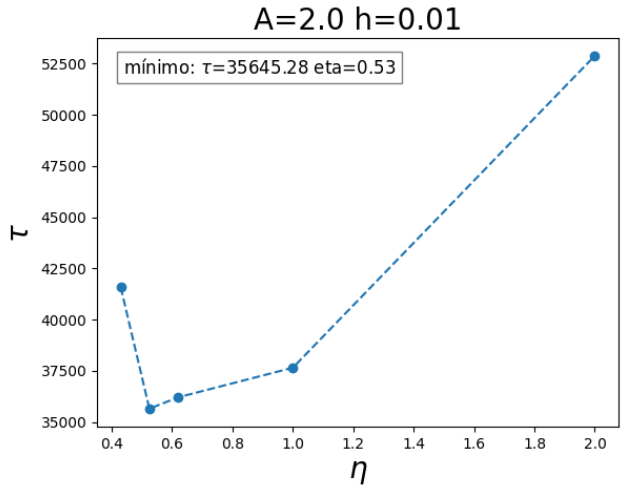




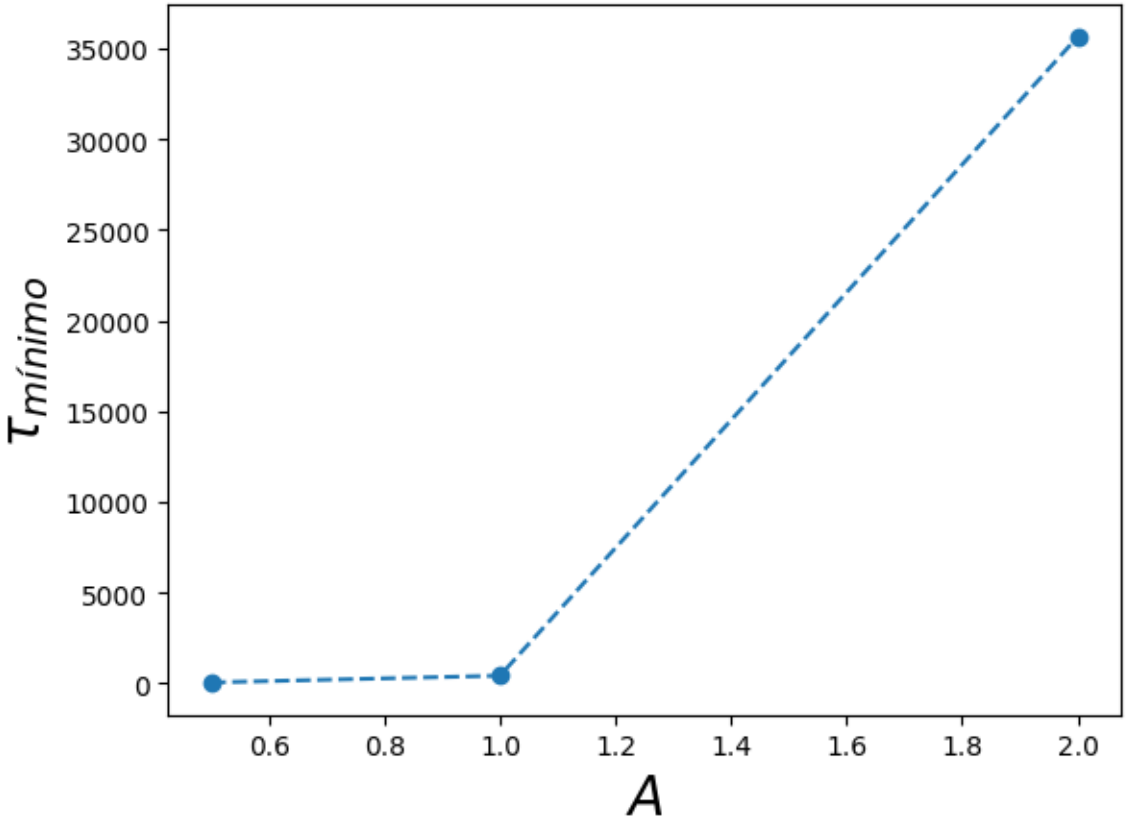
DOBLE POZO: TIEMPO DE OCUPACIÓN



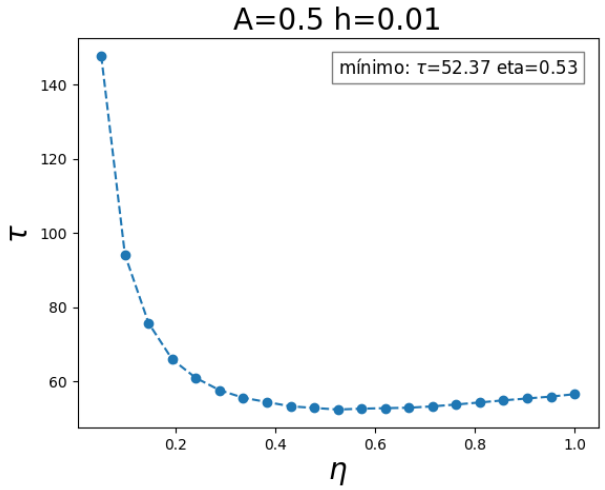


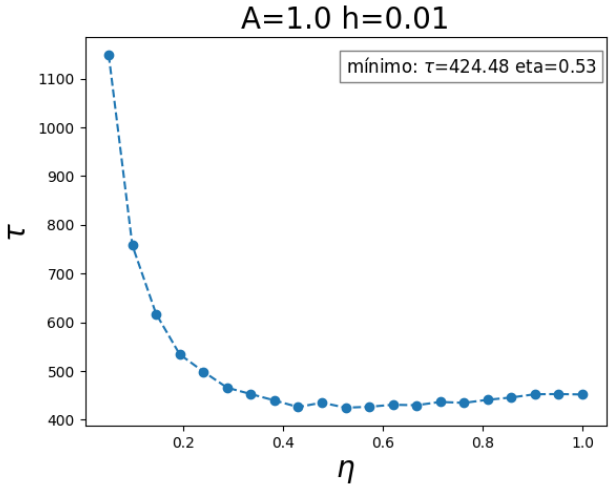


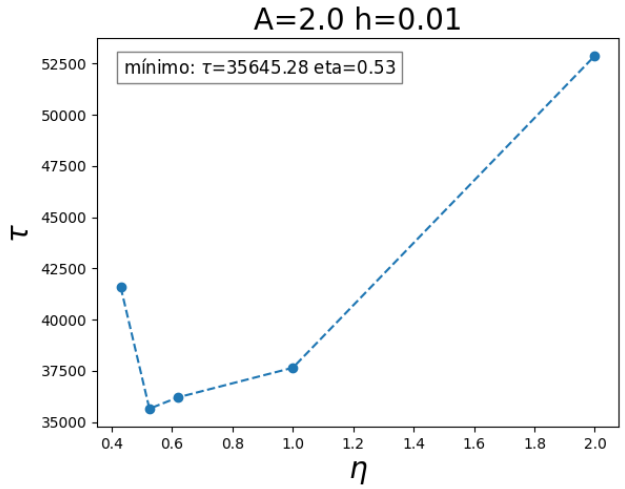
$$\eta_{min} \cong 0,53$$



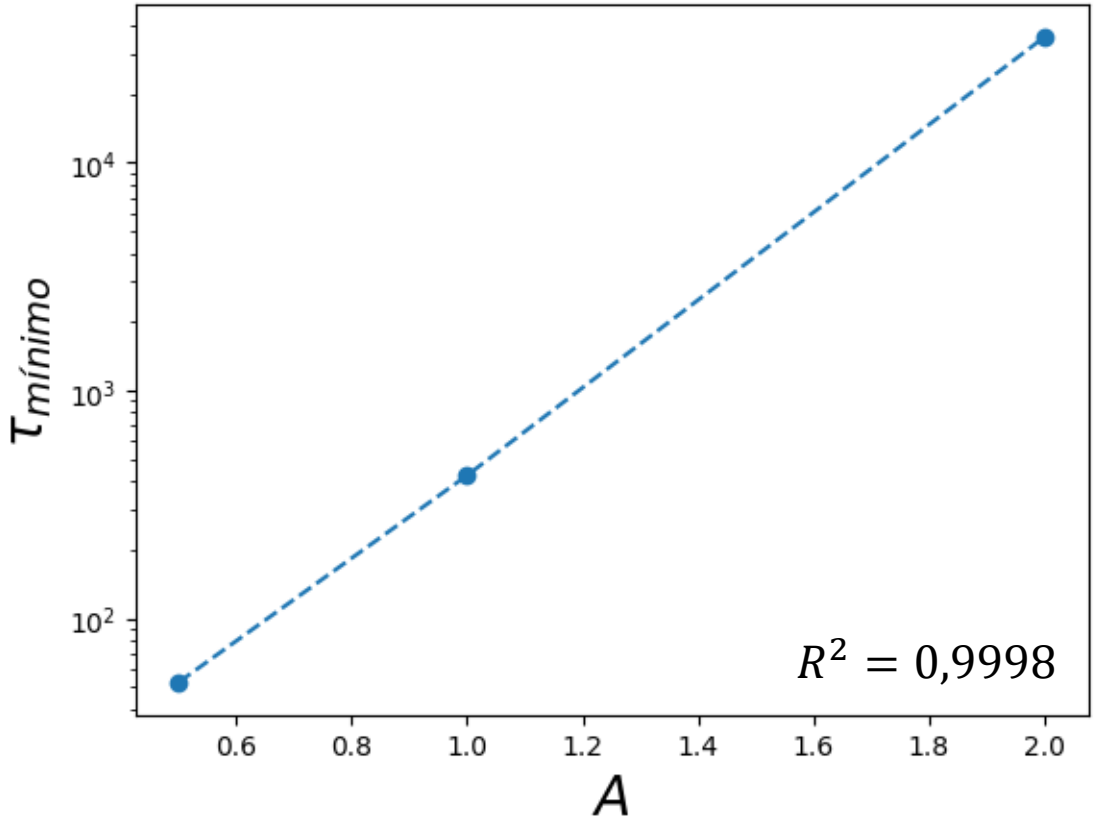
DOBLE POZO: TIEMPO DE OCUPACIÓN





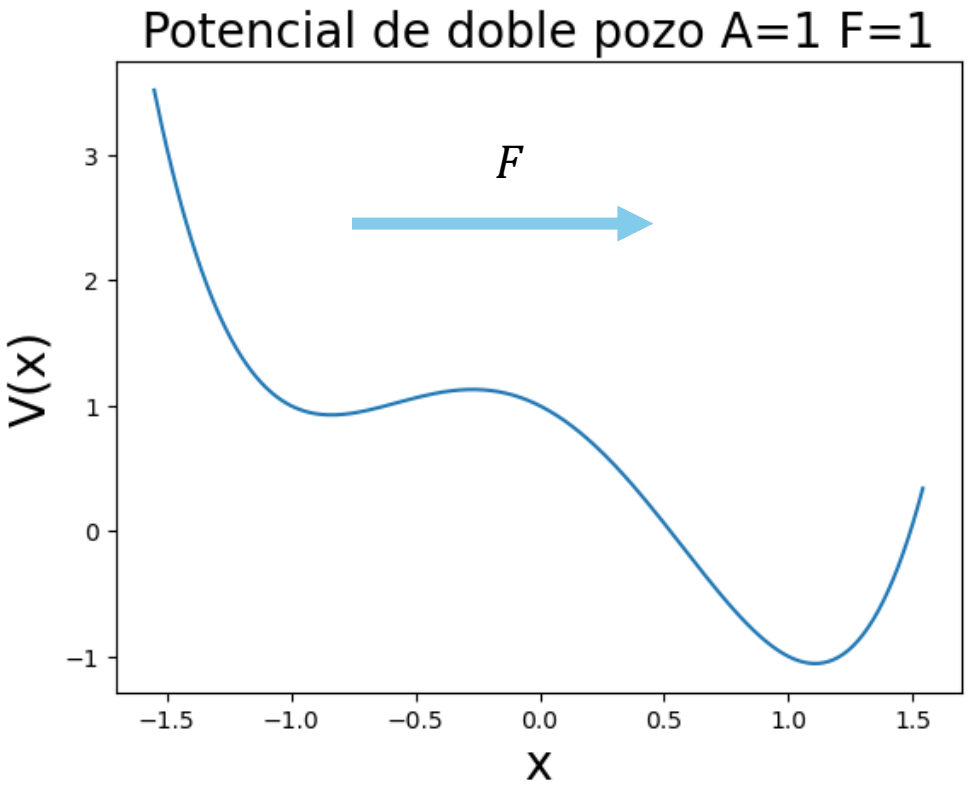


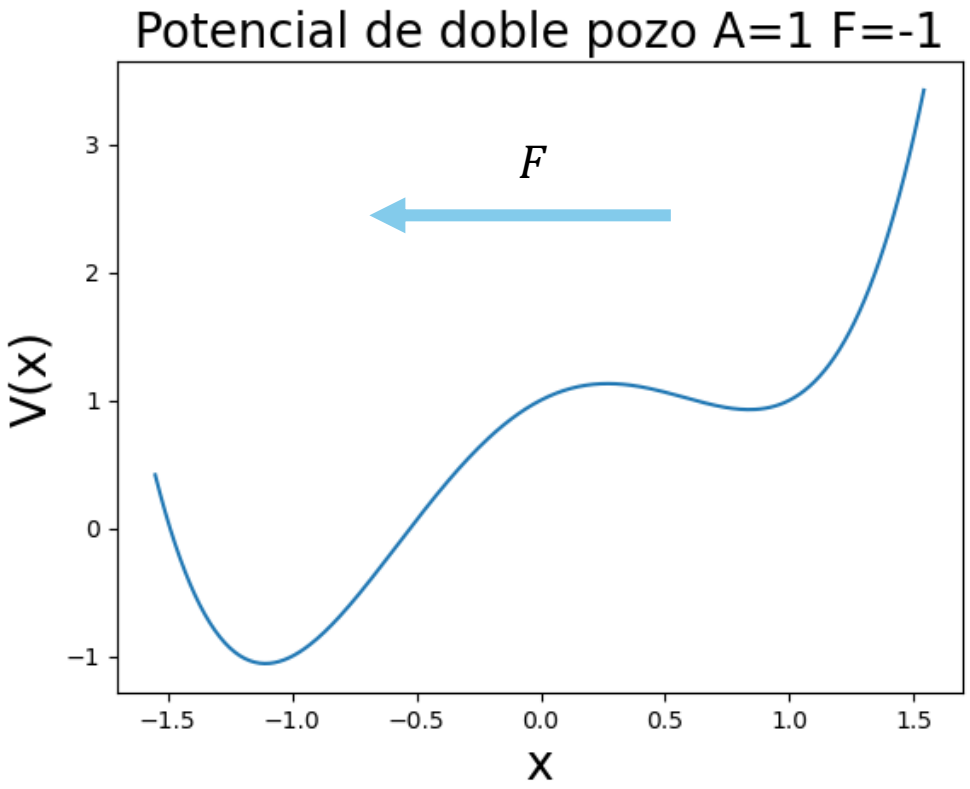
$$\eta_{min} \cong 0,53$$



DOBLE POZO: Fuerza constante

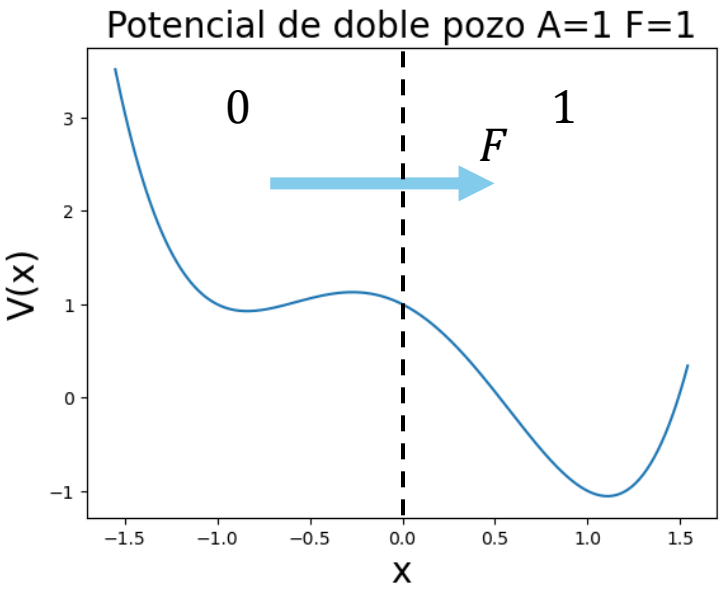
Potencial modificado: $V = A(x^2 - 1)^2 - \boldsymbol{Fx}$

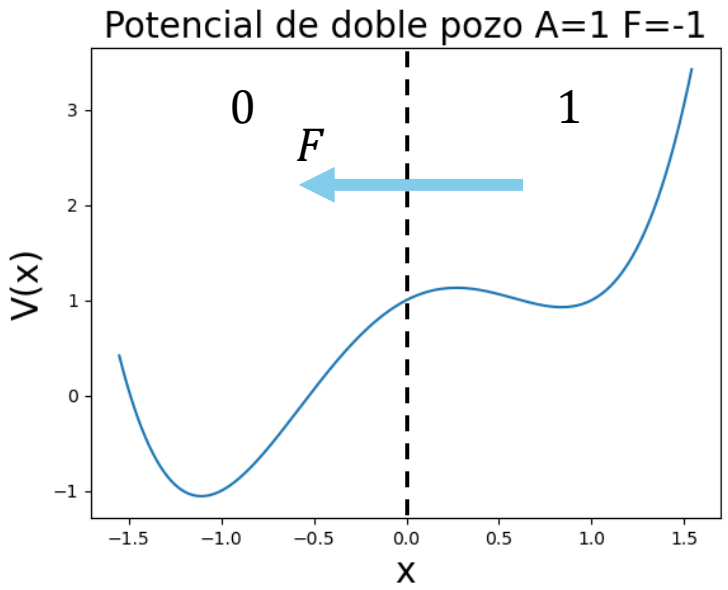




DOBLE POZO: Fuerza constante

Potencial modificado: $V = A(x^2 - 1)^2 - \boldsymbol{Fx}$





Solución exacta:

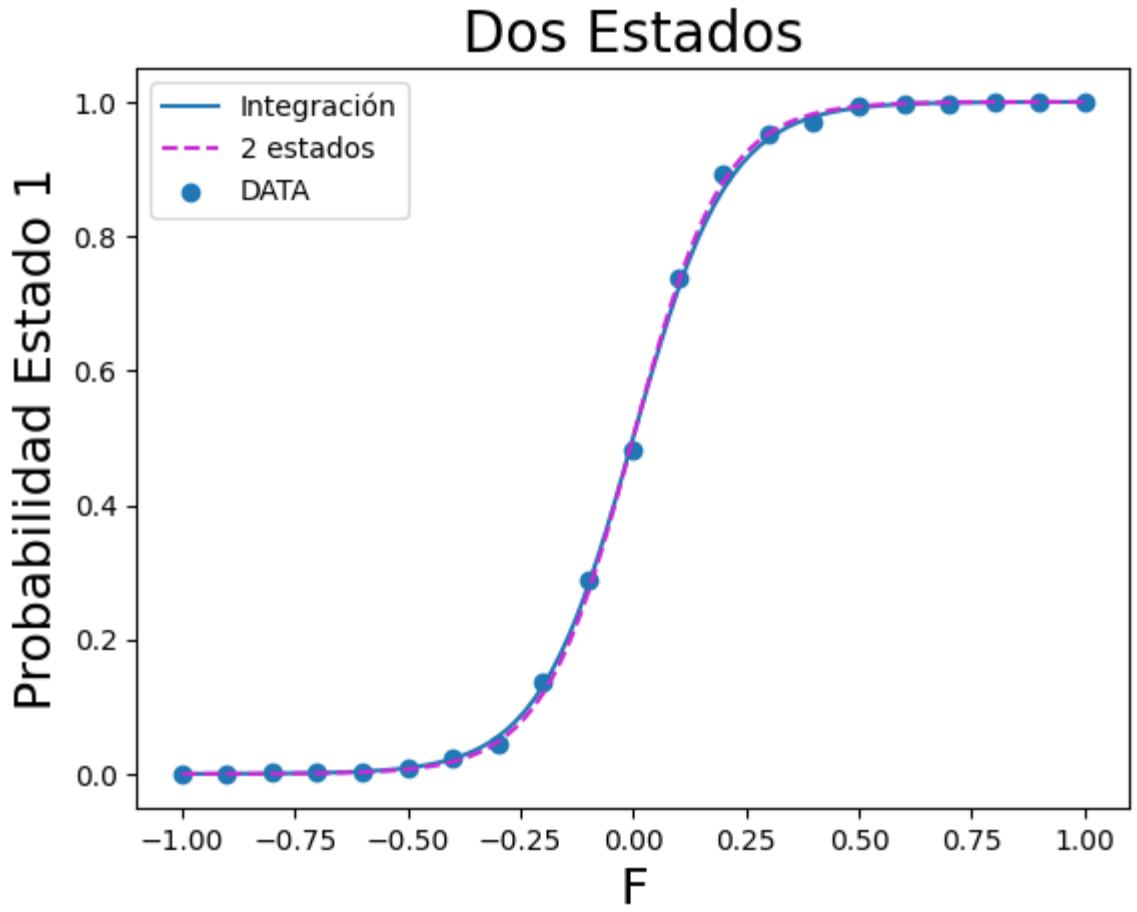
$$P(0) = \int_{-\infty}^{0} e^{-V(x)/k_B T} dx$$

$$P(1) = 1 - P(0) = \int_{0}^{+\infty} e^{-V(x)/k_B T} dx$$

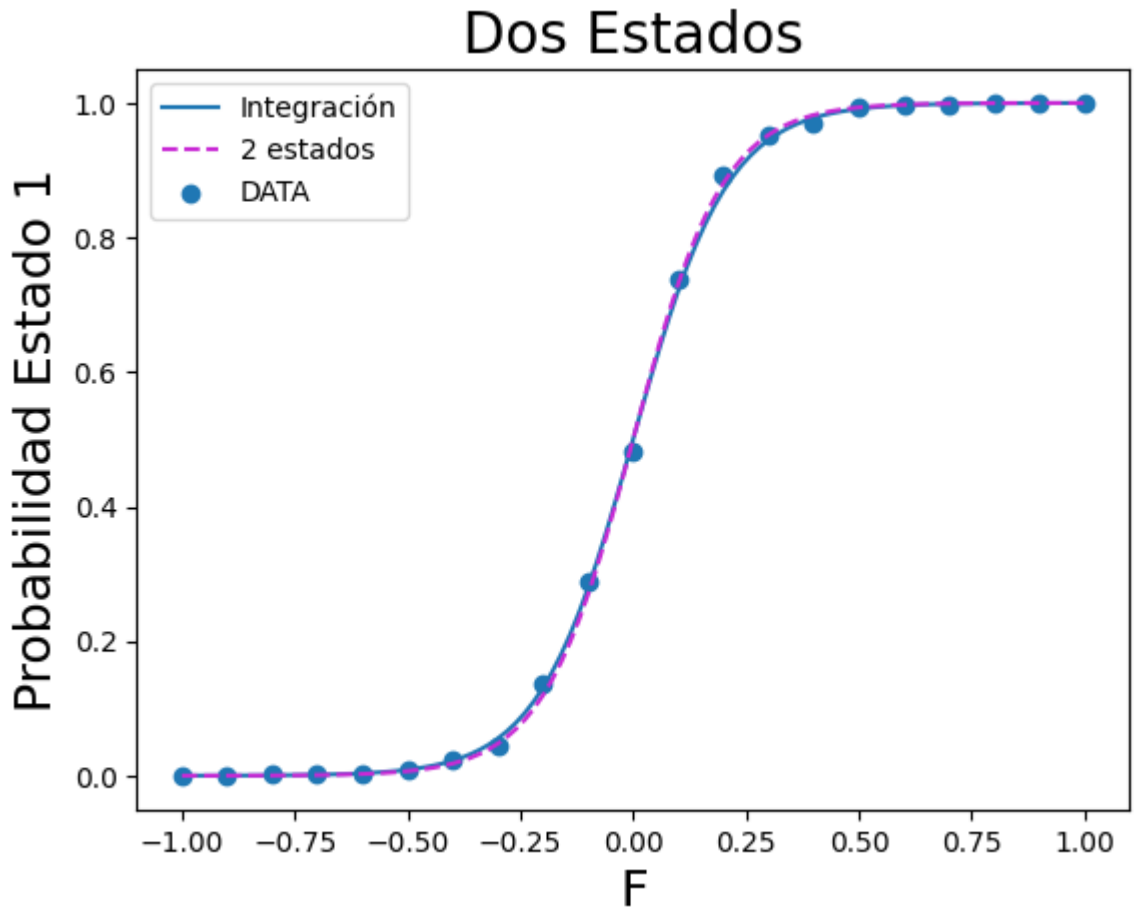
Aproximación modelo de dos estados:

$$P_{2estados}(1) = \frac{e^{F/k_B T}}{e^{F/k_B T} + e^{-F/k_B T}}$$

DOBLE POZO: Fuerza constante



DOBLE POZO: Fuerza constante



$$P_{2estados}(1) = \frac{e^{F/k_BT}}{e^{F/k_BT} + e^{-F/k_BT}}$$

Ajuste al modelo de dos estados:

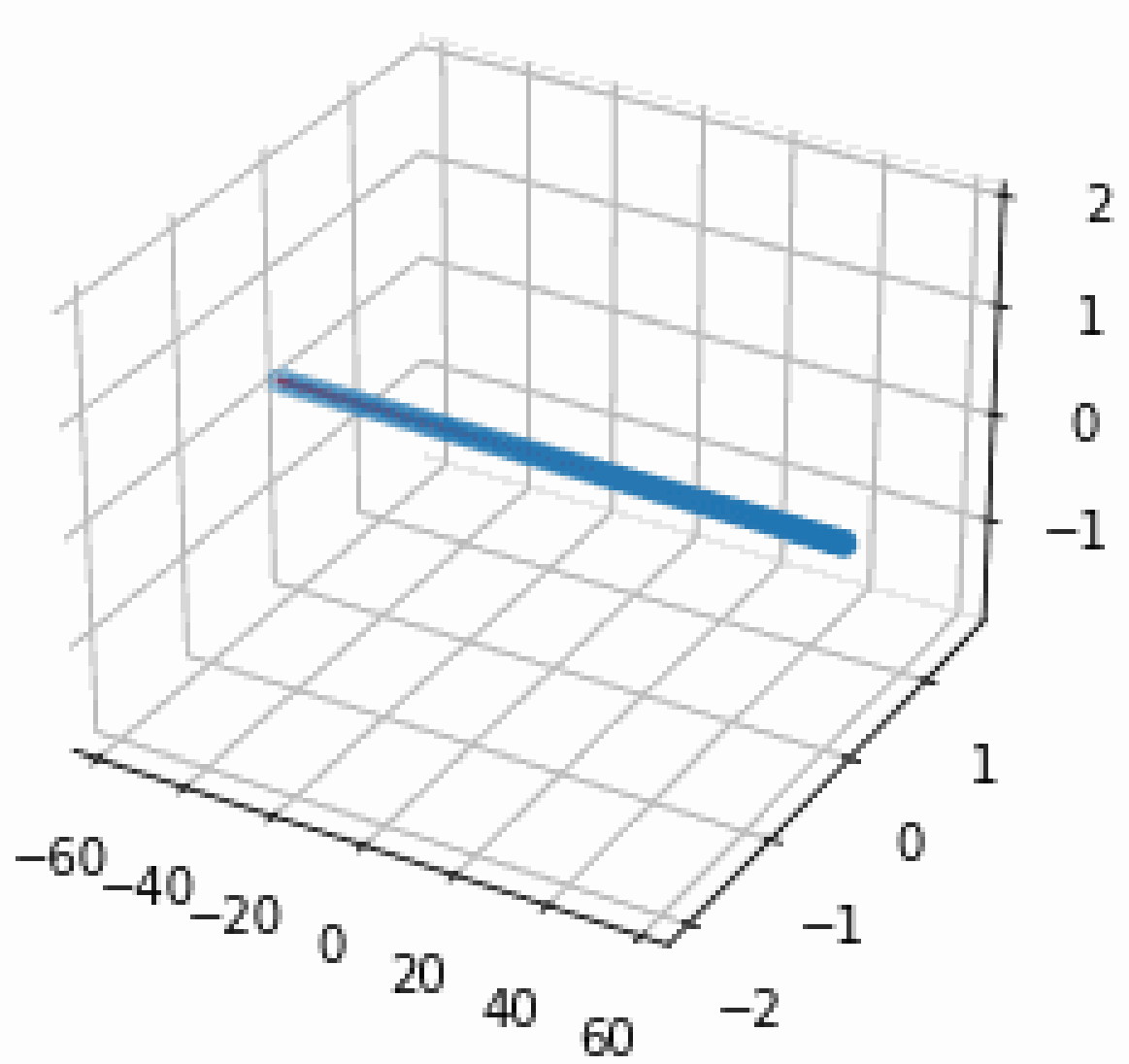
$$k_BT = 0.205 \pm 0.004$$

$$k_BT_{teo} = 0.2$$

INTRODUCCIÓN MOLÉCULA

$$V(\vec{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2} k_e (|\vec{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

b ⟶ longitud del enlace (= 1)

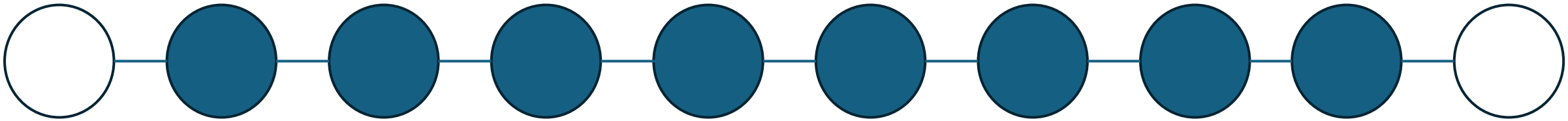


$$V(\overrightarrow{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2} k_e (|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

Puntos intermedios

$$\mathrm{F}_x = -\frac{dV}{dx} = -k_e (|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b) \frac{(x_i - x_{i+1})}{|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}|} + k_e (|\overrightarrow{r_{i-1}} - \overrightarrow{r_i}| - b) \frac{(x_{i-1} - x_i)}{|\overrightarrow{r_{i-1}} - \overrightarrow{r_i}|}$$

$$\forall x \epsilon \{x, y, z\}$$

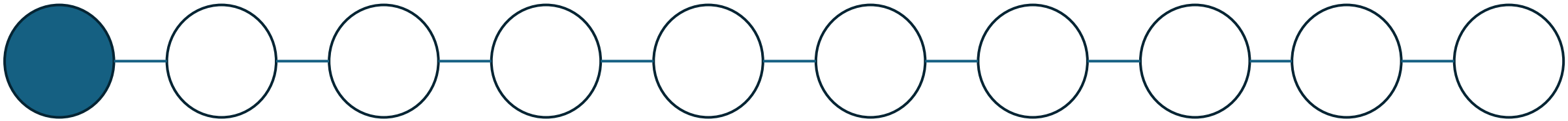


$$V(\overrightarrow{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2} k_e (|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

Puntos extremos Primero

$$\mathrm{F}_x = -\frac{dV}{dx} = -k_e (|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b) \frac{(x_i - x_{i+1})}{|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}|}$$

$$\forall x \epsilon \{x, y, z\}$$



$$V(\overrightarrow{r_i}, \overrightarrow{r_{i+1}}) = \frac{1}{2} k_e (|\overrightarrow{r_i} - \overrightarrow{r_{i+1}}| - b)^2$$

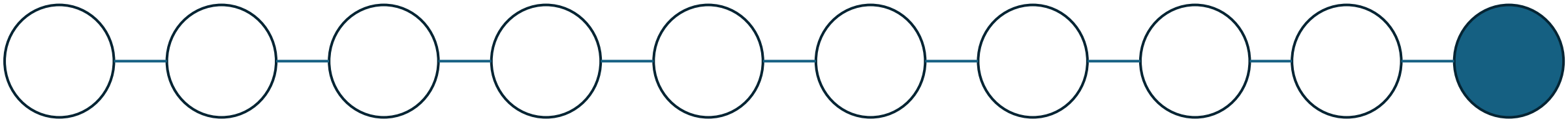
Puntos extremos

Último

$$\mathrm{F}_x = -\frac{dV}{dx} =$$

$$k_e (|\overrightarrow{r_{i-1}} - \overrightarrow{r_i}| - b) \frac{(x_{i-1} - x_i)}{|\overrightarrow{r_{i-1}} - \overrightarrow{r_i}|}$$

$$\forall x \epsilon \{x, y, z\}$$

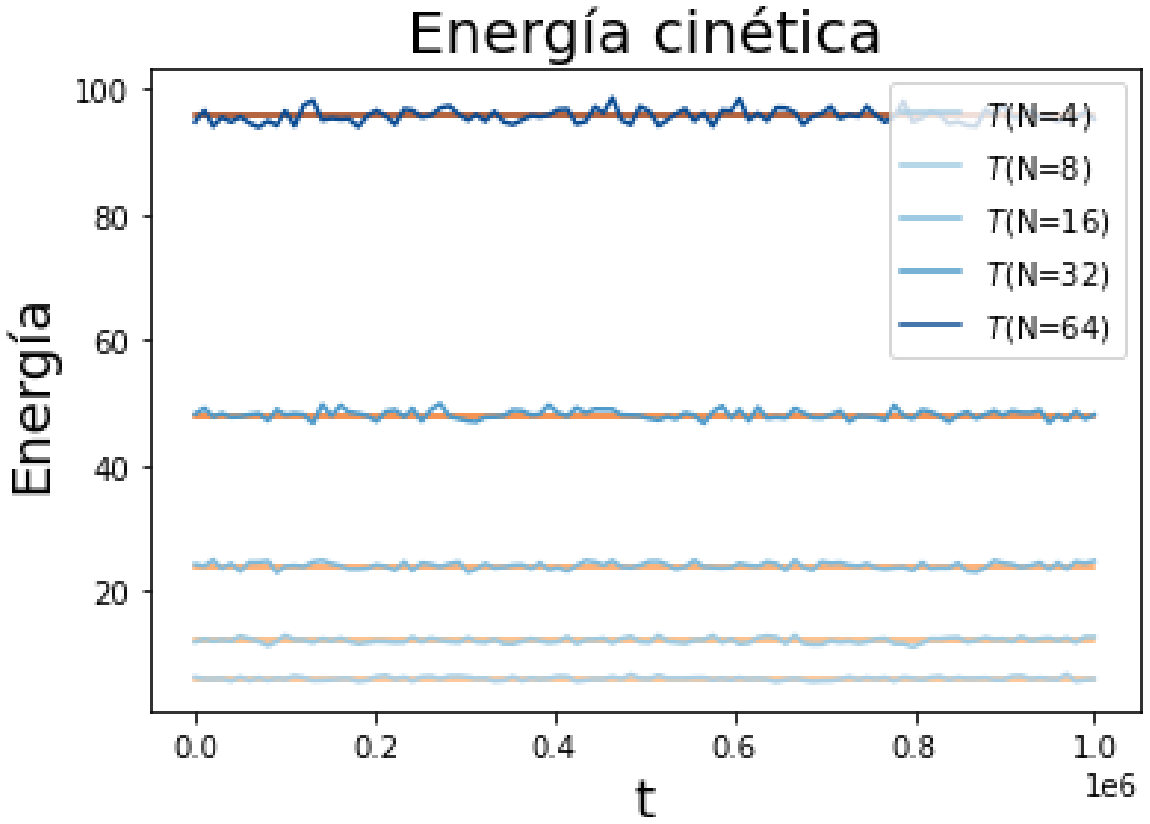


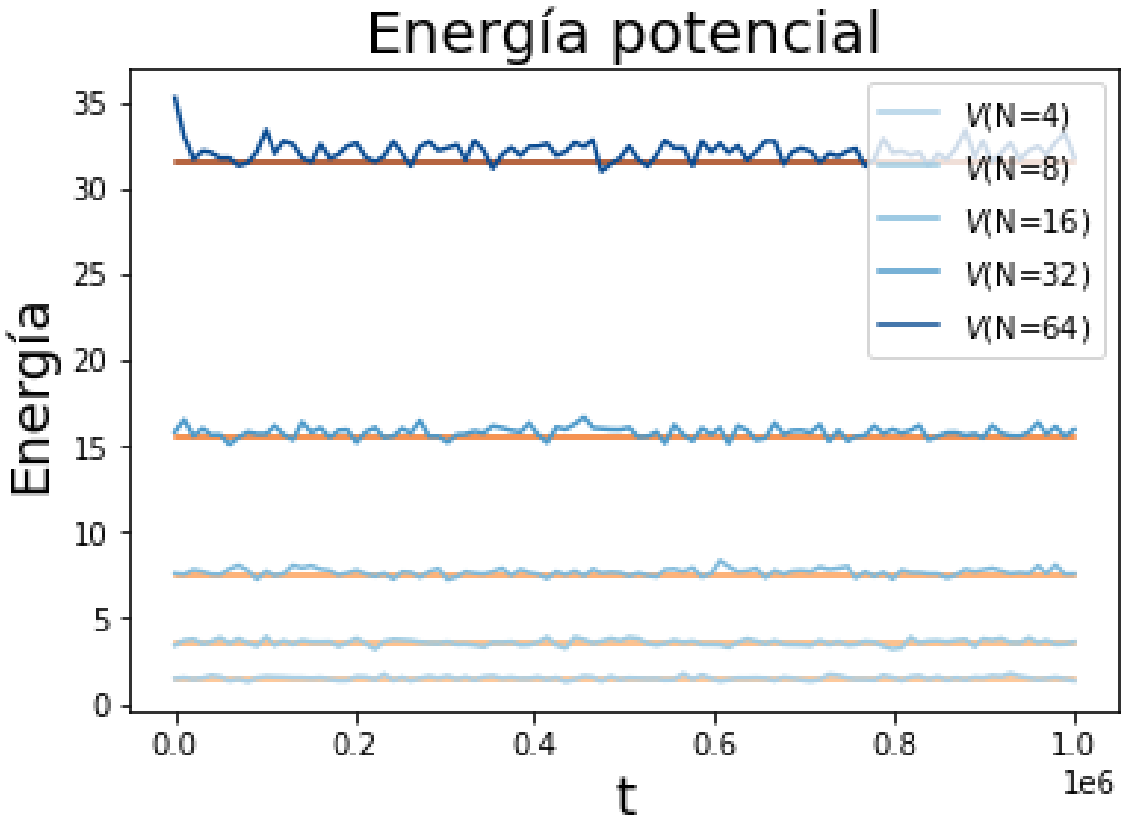
MOLÉCULA: Equipartición de la energía

- Potencial cuadrático: Se equiparte la energía
- Al haber más de una partícula y estar en 3D:

$$\langle T \rangle = \frac{3}{2} N k_B T \qquad \langle V \rangle = \frac{1}{2} (N-1) k_B T$$

MOLÉCULA: Equipartición de la energía





MOLÉCULA: Ley de escala del radio giro

$$R_G^2 = \frac{1}{N}\sum_{i=1}^{N}\left|\overrightarrow{R_i} - \overrightarrow{R_{CM}}\right|^2$$

MOLÉCULA: Ley de escala del radio giro

$$R_G^2 = \frac{1}{N}\sum_{i=1}^{N}\left|\overrightarrow{R_i} - \overrightarrow{R_{CM}}\right|^2 \quad \Rightarrow \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(\overrightarrow{x_i} - \vec{\mu})^2 = \langle x^2\rangle - \langle x\rangle^2$$

RANDOM WALK 1D:

$$x(n) = x(n-1) \pm \Delta\mathrm{x} \quad \rightarrow \quad \langle x(n)\rangle = \langle x(n-1)\rangle = (\dots) = \langle x(0)\rangle$$

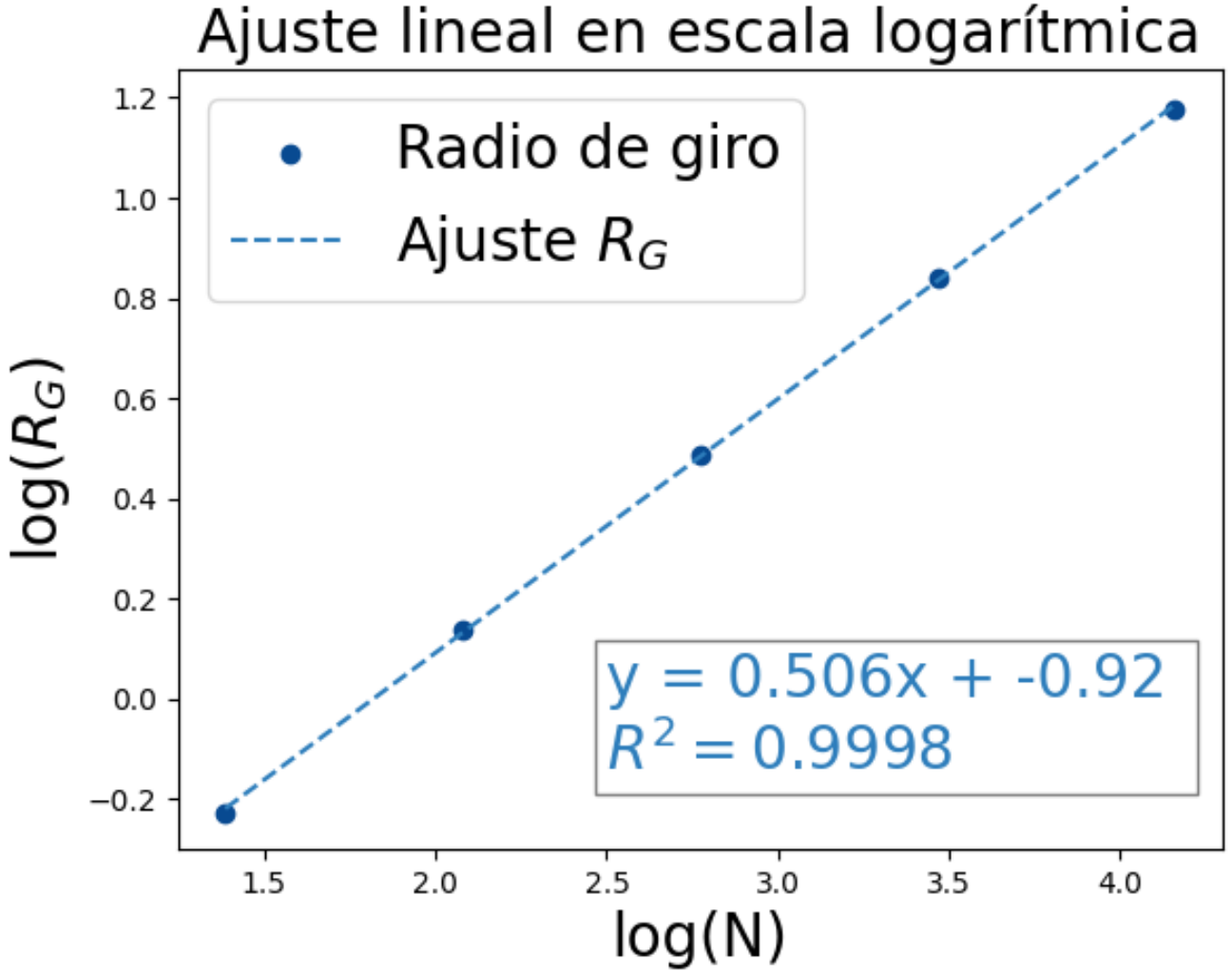
$$\langle x^2(n)\rangle = \langle x^2(n-1) \pm 2\Delta x x(n-1) + (\Delta x)^2\rangle = \langle x^2(n-1)\rangle + (\Delta x)^2$$

$$\langle x^2(n-1)\rangle = (n-1)(\Delta\mathrm{x})^2$$

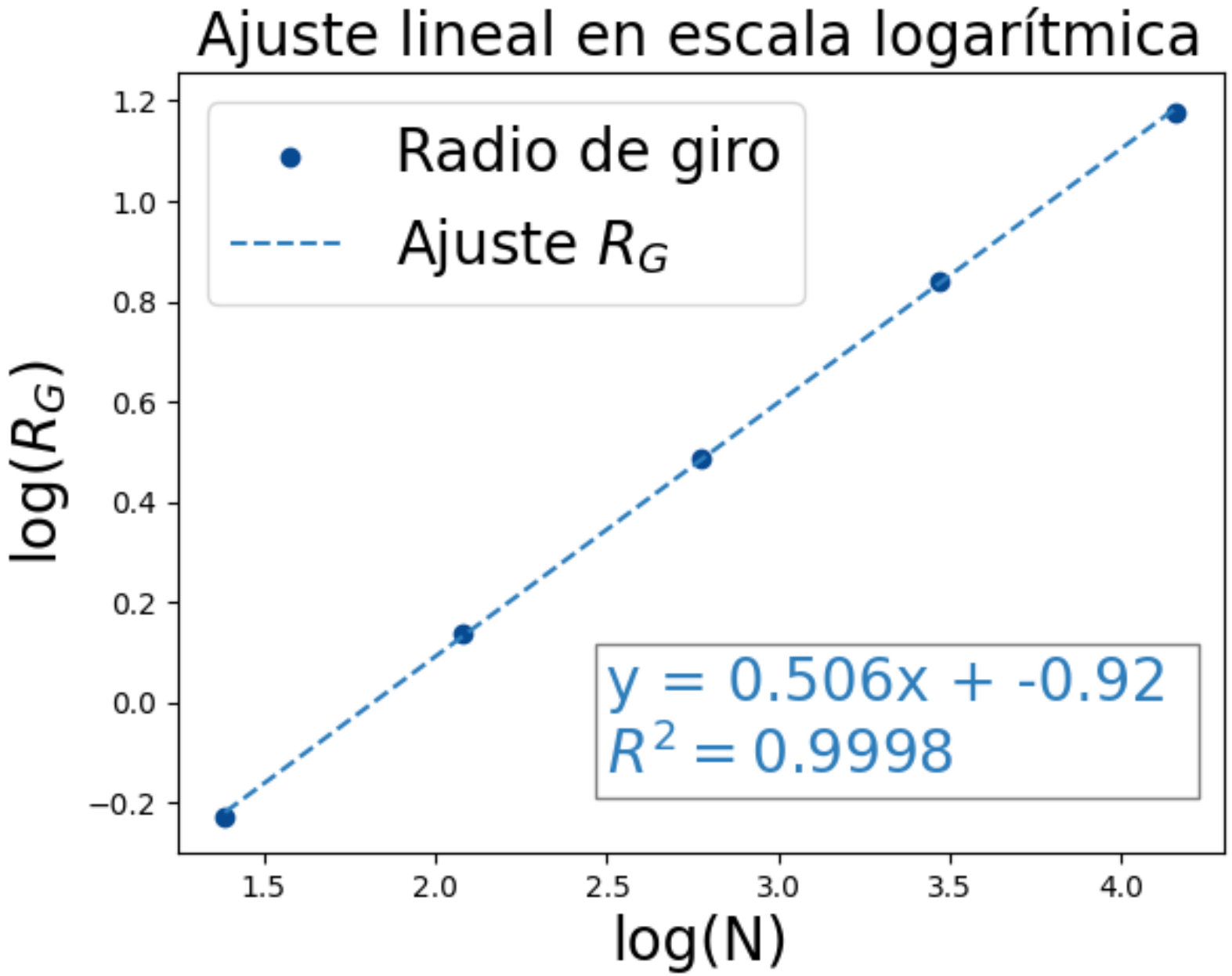
$$\rightarrow R_G^2 \propto n$$

MOLÉCULA: Ley de escala del radio giro

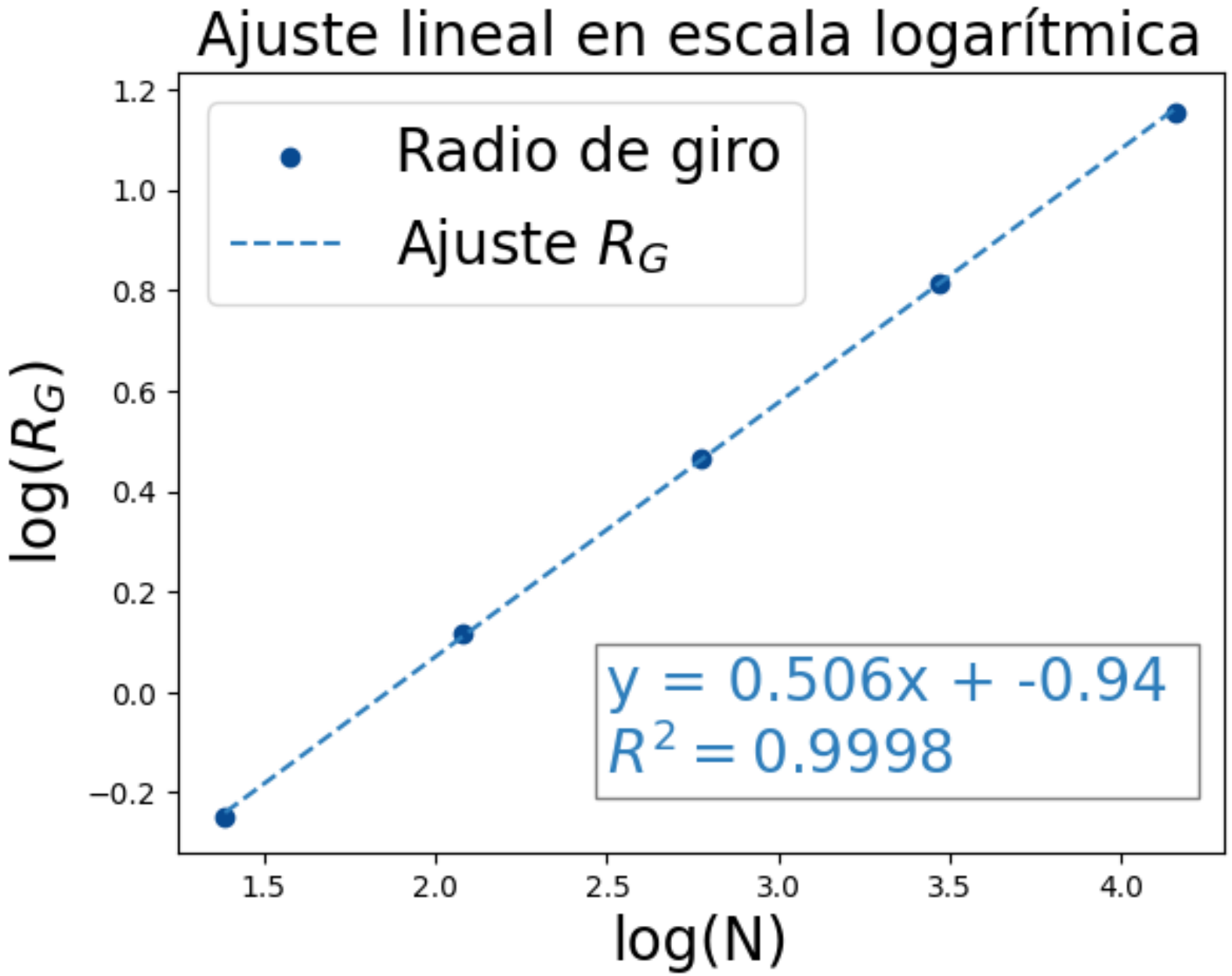
$$R_G^2 = \frac{1}{N}\sum_{i=1}^{N}\left|\overrightarrow{R_i} - \overrightarrow{R_{CM}}\right|^2$$

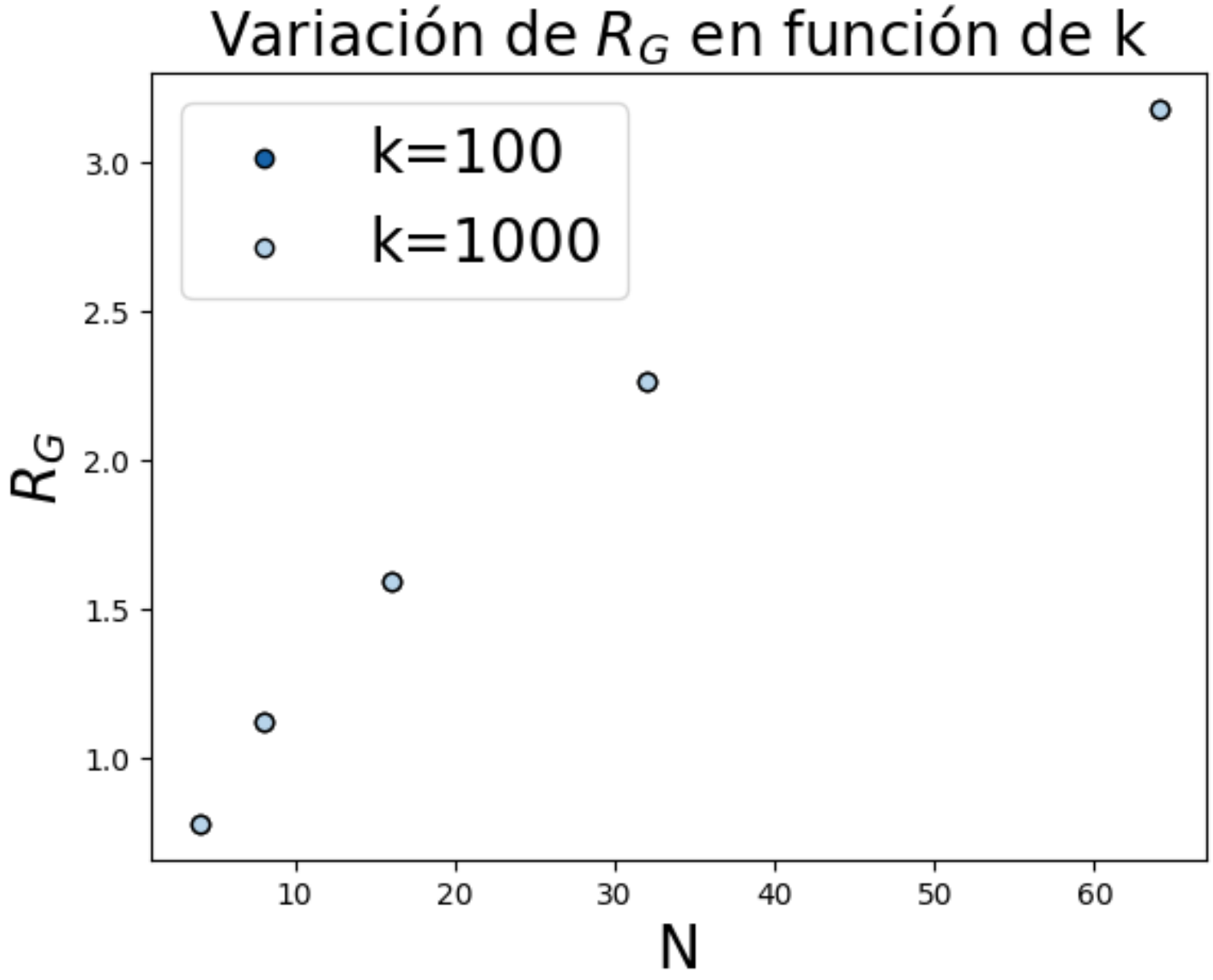


k=100


Ajuste lineal en escala logarítmica
Radio de giro
Ajuste R_G
y = 0.506x + -0.92
$R^2 = 0.9998$
log(R_G)
log(N)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
1.5
2.0
2.5
3.0
3.5
4.0

k=1000


Ajuste lineal en escala logarítmica
Radio de giro
Ajuste R_G
$y = 0.506x + -0.94$
$R^2 = 0.9998$
$\log(R_G)$
log(N)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
1.5
2.0
2.5
3.0
3.5
4.0



discrepancia

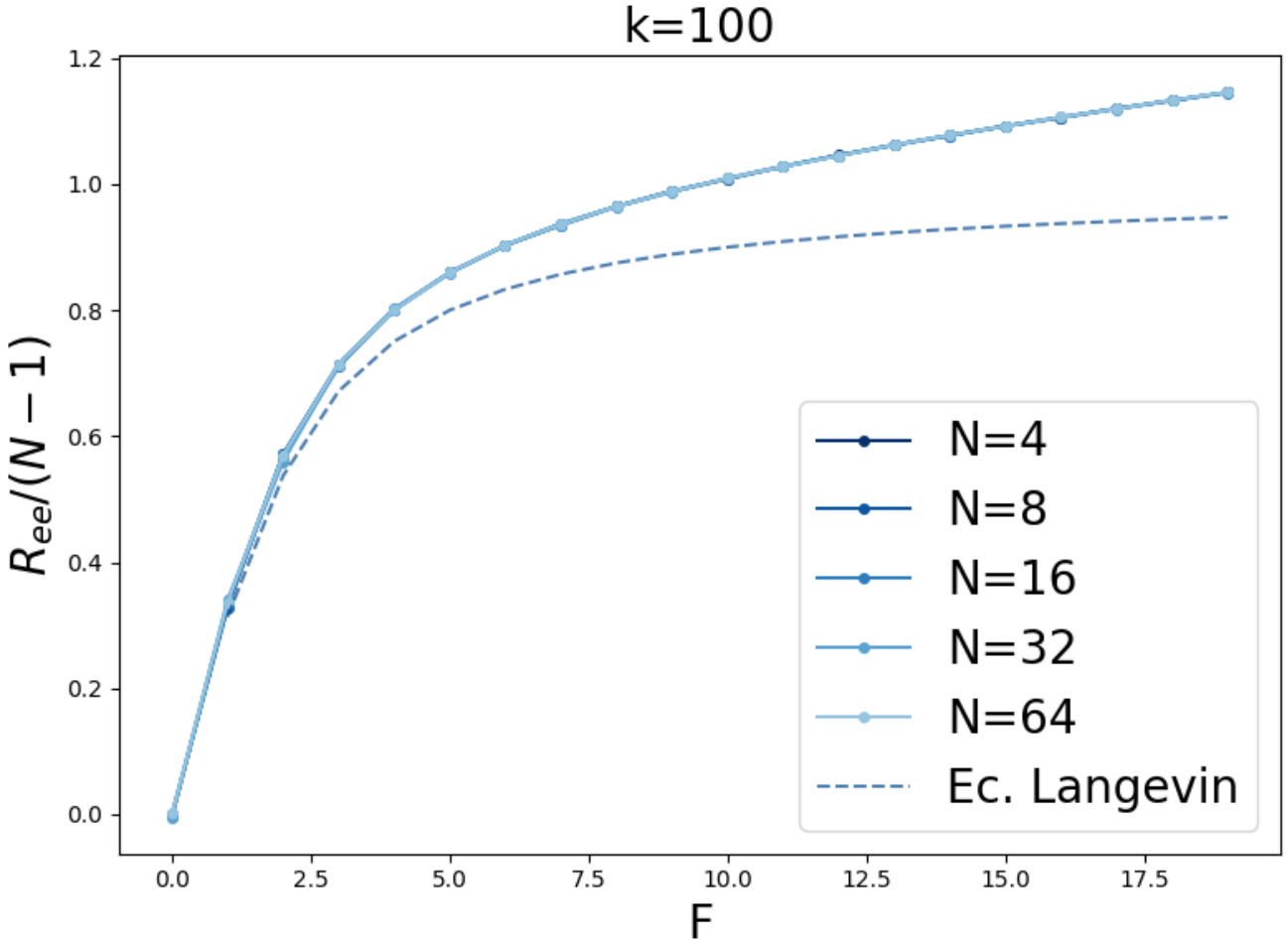
2,13%

MOLÉCULA: Fuerza cte y REE

$$\frac{R_{ee}}{N-1} = b\left(\coth\left(\frac{Fb}{k_B T}\right) - \frac{k_B T}{Fb}\right)$$

$$\overrightarrow{R_{ee}} = \overrightarrow{R_n} - \overrightarrow{R_1}$$

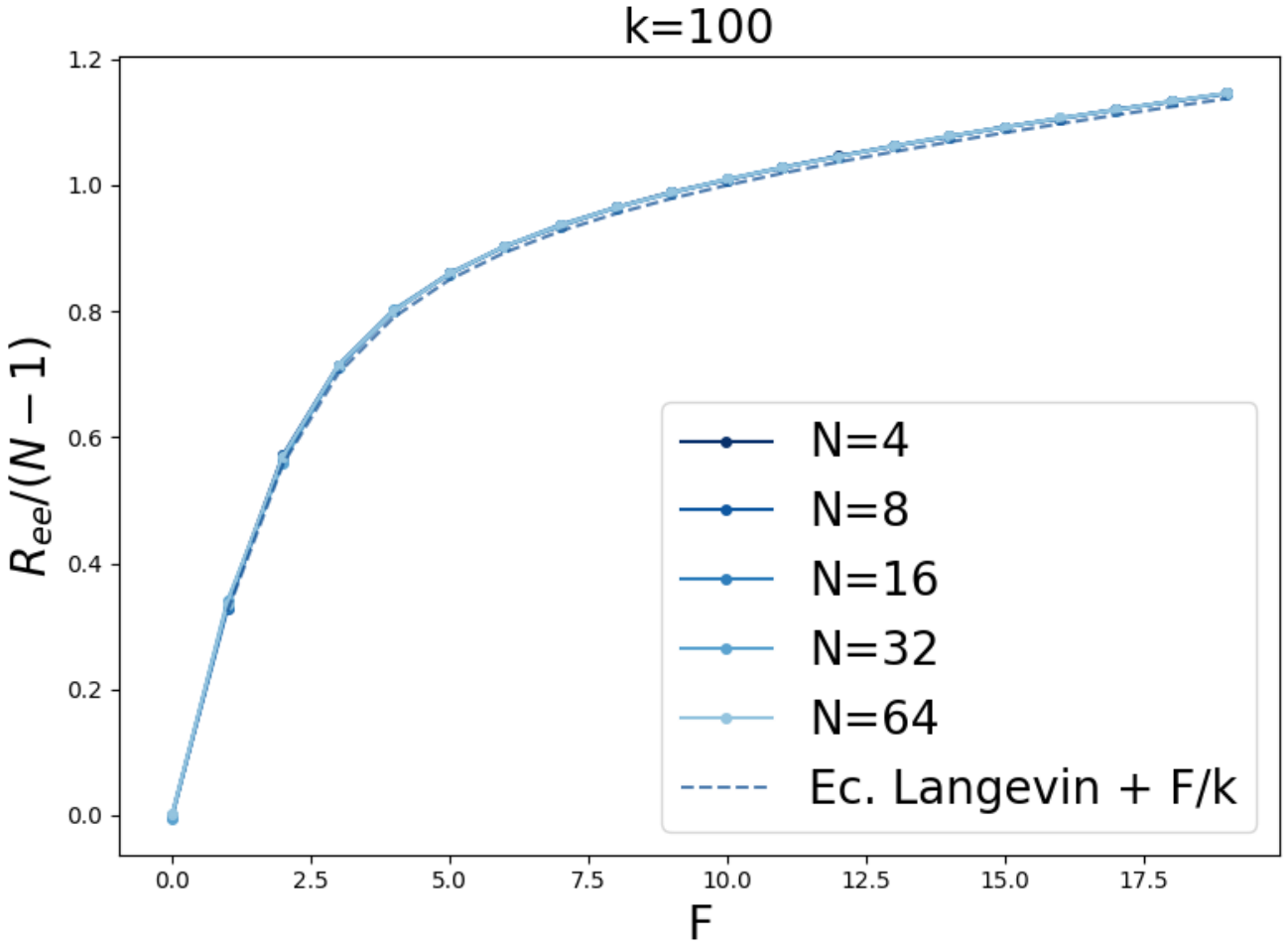
¡Langevin asume que los muelles son rígidos!

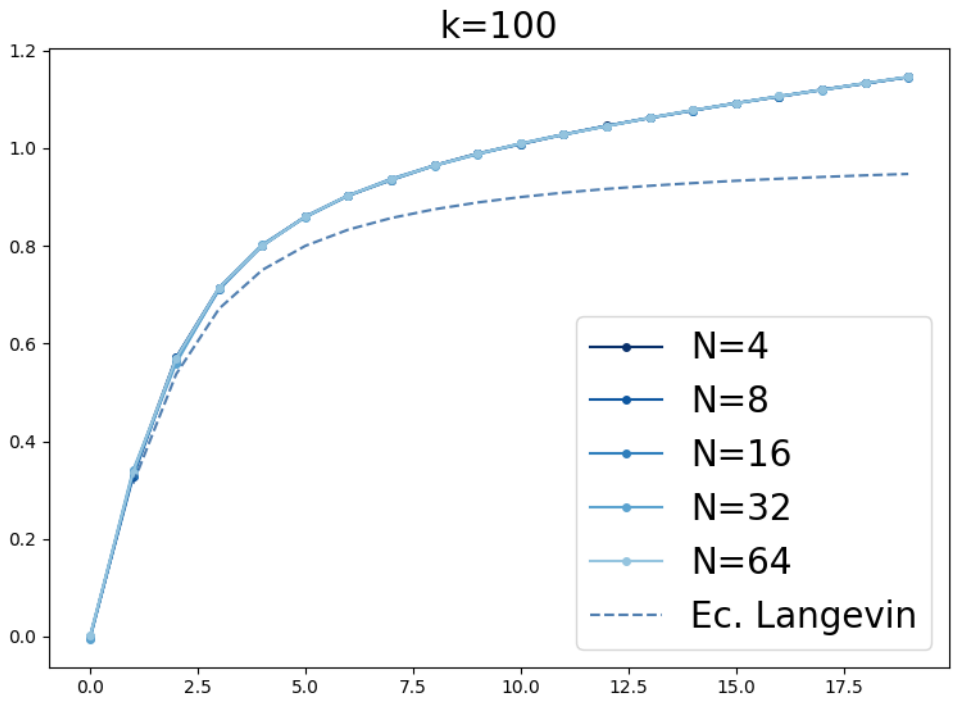


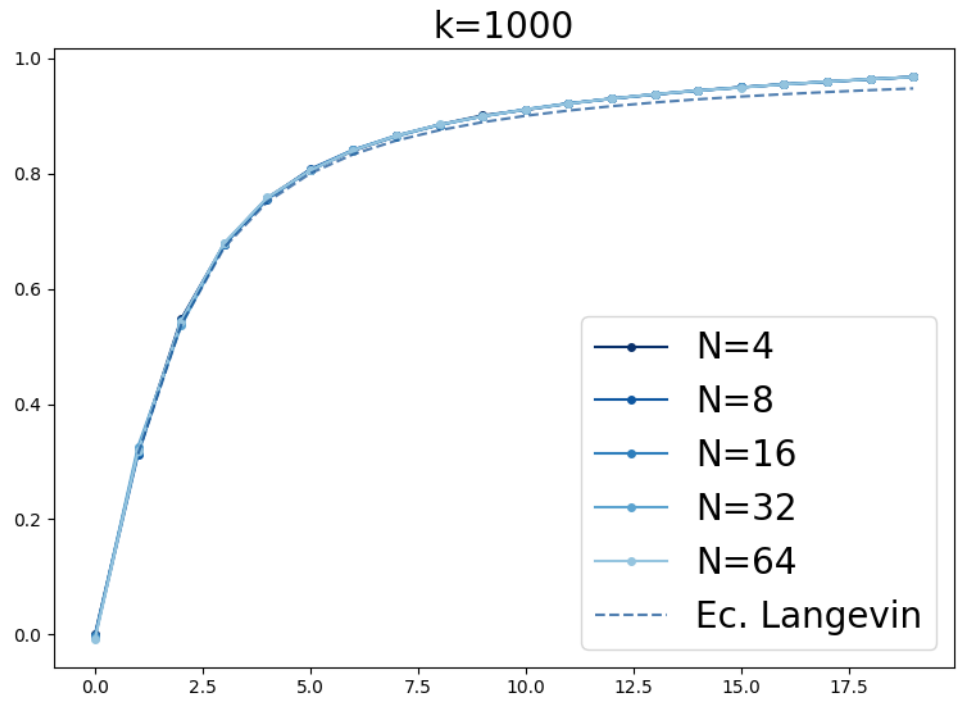
MOLÉCULA: Fuerza cte y REE

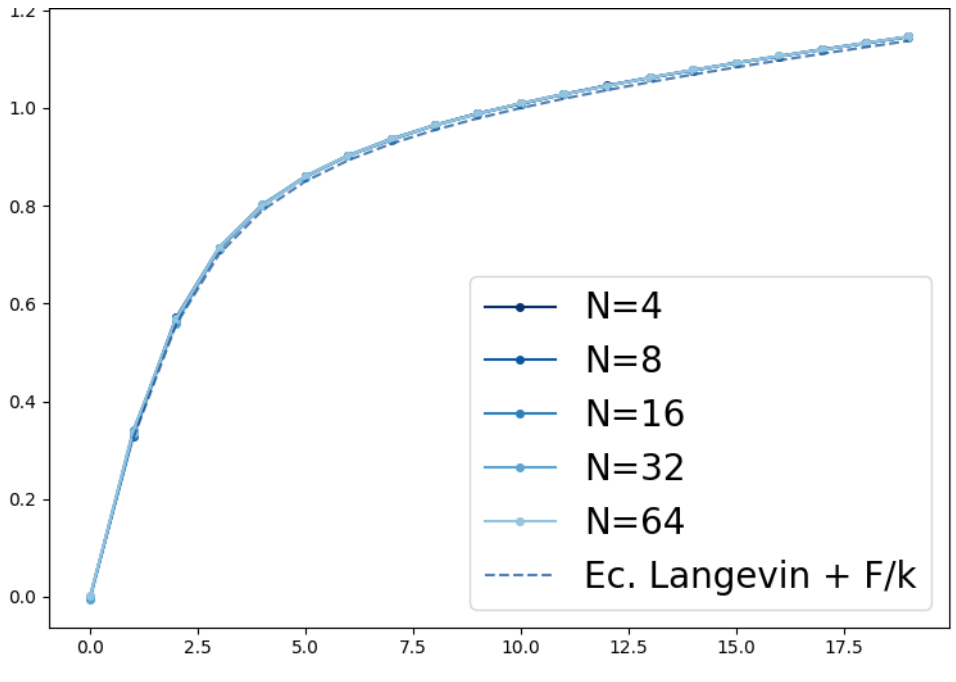
$$F = k\Delta x \Rightarrow \Delta x = \frac{F}{k}$$

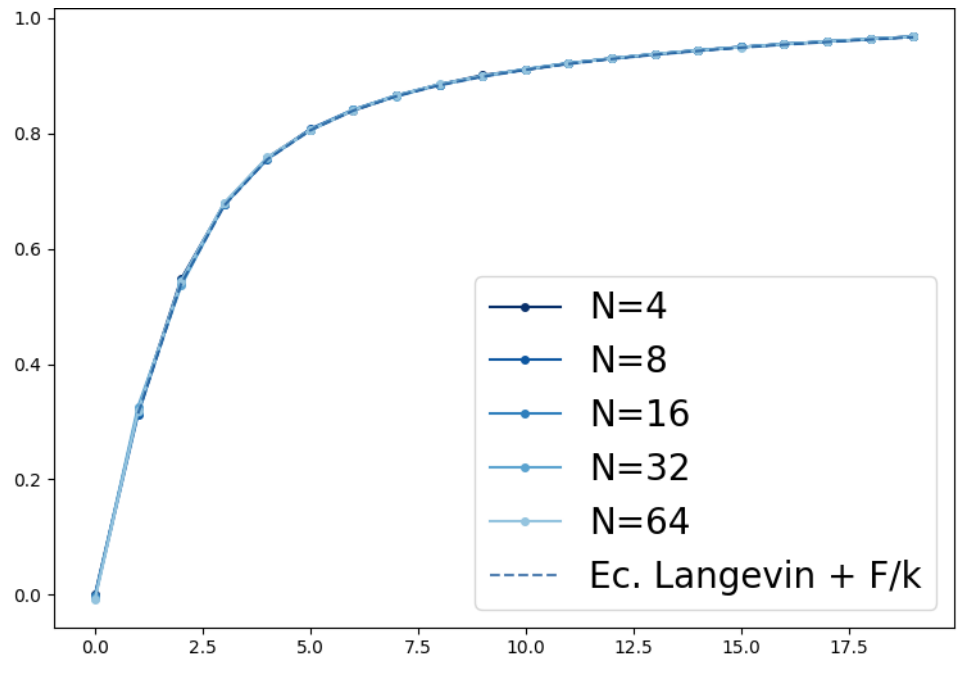
$$\frac{R_{ee}}{N-1} = b\left(\coth\left(\frac{Fb}{k_B T}\right) - \frac{k_B T}{Fb}\right) + \frac{F}{k}$$




k=100
N=4
N=8
N=16
N=32
N=64
Ec. Langevin


k=1000
N=4
N=8
N=16
N=32
N=64
Ec. Langevin

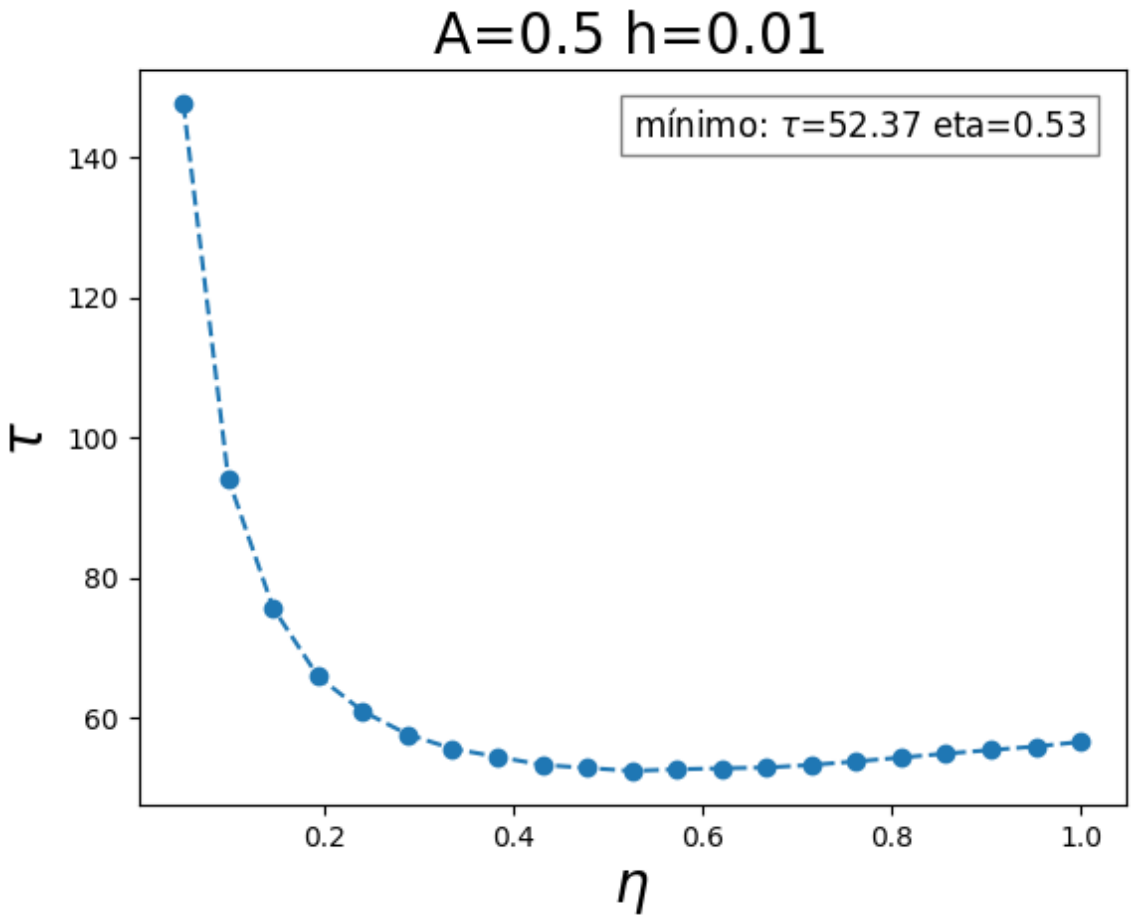

N=4
N=8
N=16
N=32
N=64
Ec. Langevin + F/k


N=4
N=8
N=16
N=32
N=64
Ec. Langevin + F/k

CONCLUSIONES

Marco teórico:

Efectos de la dinámica de Langevin y del parámetro η

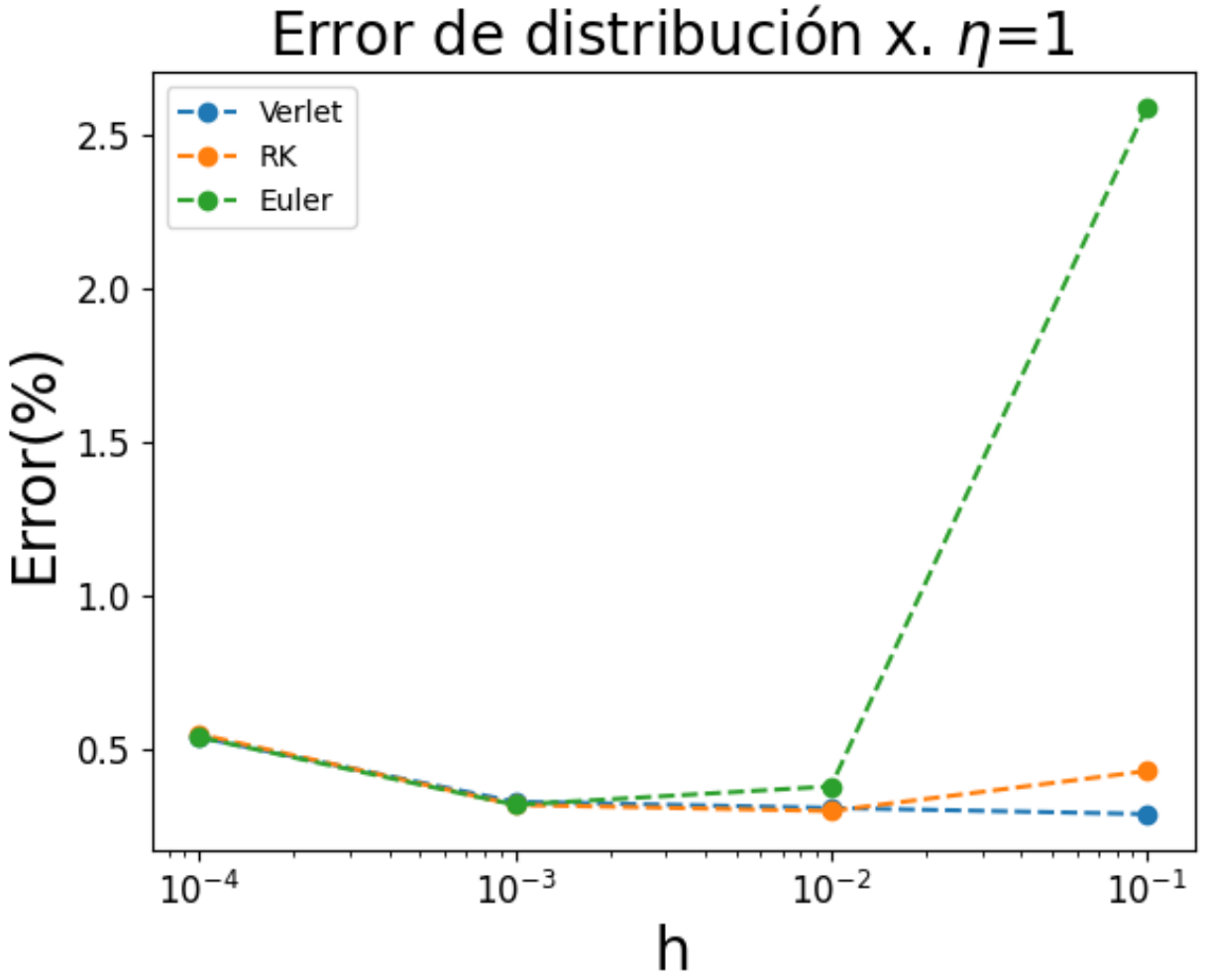


CONCLUSIONES

Implementación:

Estudiar los sistemas en los rangos de los parámetros de simulación

Decidir por el algoritmo más apropiado



CONCLUSIONES

Sistemas físicos:

Oscilador: elección del método

Doble Pozo: componente aleatoria

Molécula: aplicación real a la simulación

